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I, robot... you, unemployed: How workers are responding to AI fears

July 1, 2026

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While media headlines this year have been filled with stories of AI-driven layoffs, less attention has been paid to how workers are responding to the new AI-centric world they find themselves in. Are people so fearful of displacement by AI that they are pulling back on spending and increasing savings? Or are they embracing its potential to enhance their careers and lifestyles?

This research report uses our proprietary dbDataInsights household survey [1] to examine how workers are responding to fears that they will lose their jobs to AI. The analysis shows that, contrary to the idea that worker's fear could slow down the economy as they spend less to boost precautionary savings, individuals most concerned about AI-driven job loss are not retreating. Instead, they are taking some proactive steps.

— **Proactive upskilling:** Workers who are more concerned about losing their job to AI are more likely to undertake AI-related training – potentially in an attempt to complement, rather than be substituted by, AI.

— **Increasing wage demand:** They are more likely than other workers to seek out a pay rise.

— **Confident spending:** They are more likely to be currently spending more, in direct contrast to the idea that greater concern about job security will act as a drag on spending.

The survey also reveals who these concerned individuals are. They tend to be younger, more educated, have a better understanding of AI, and are already using AI in their professional and personal lives.

The findings suggest that workers are not resigned to the fear of losing their jobs to AI. They are attempting to adapt to the new AI world. Supporting this trend – both at the level of individual businesses and broader public policy – could help reduce the social cost of AI adoption that some fear.

[1] For more background on the dbDataInsights household survey see the Annex at the end of this report.

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Survey background

Our dbDataInsights household survey asks a wide range of questions each month to nationally representative samples in Germany, France, Italy, Spain, the United Kingdom, and the United States. The analysis in this report draws exclusively from the survey. Respondents to the survey are asked directly about their fears of losing their job to AI:

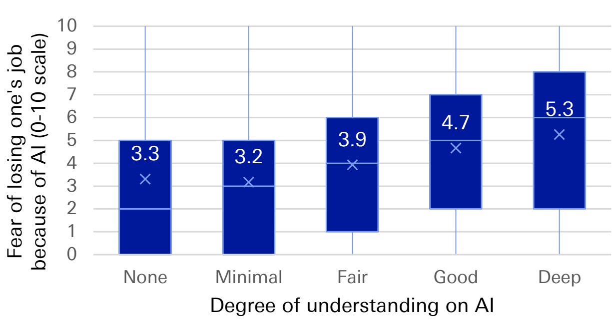
Using a scale of 0 to 10, where 0 is 'Not at all concerned' and 10 is 'Extremely concerned', how concerned are you that you will need to look for a new job within the following time frames because of AI?

This report will focus specifically on the one-year ahead horizon, although results are similar if we consider the two- or five-year ahead horizon instead. We began asking this question in our June 2025 survey round, and the level of concern has been broadly stable since then, both over time and similar across countries. Therefore, this analysis aggregates data from all monthly surveys and countries to maximise statistical power

The fear of losing one's job to AI is not born out of ignorance or irrational fear about AI – it is driven by informed concern. In our survey, those that have a greater understanding of AI show higher levels of concern about losing their jobs.

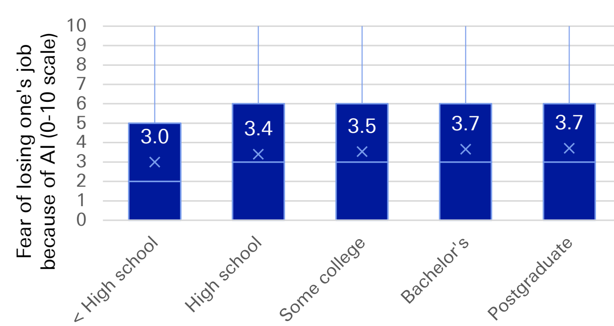
When asked to describe their own knowledge level of AI, respondents who reported higher levels of understanding generally had a higher concern about losing their job to AI than those with less self-reported knowledge (Figure 1). This relationship is not likely to be driven by bias in people's self confidence. **On average, those with higher levels of education also showed higher levels of fear** (Figure 2), and **those who have actively used AI in the past three months either at home or at work – and thus should have greater knowledge about the latest AI developments – also showed greater concern on average about losing their job to AI** (Figure 3).

Figure 1: Those with a higher self-report understanding of AI are also those with the highest level of fear about losing their jobs over the next year



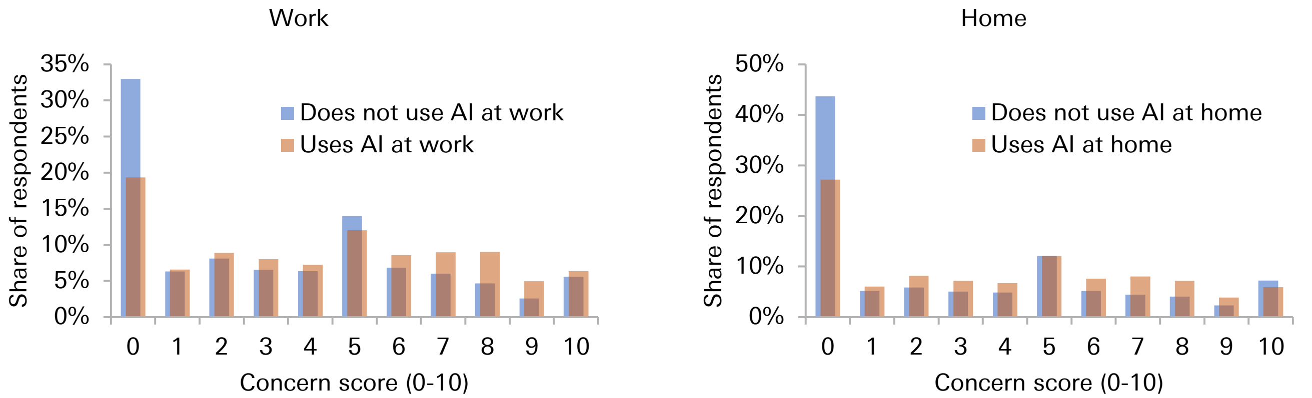
Source : Deutsche Bank Research, dbDataInsights
Notes: Numerical values denote the average fear (on a 0-10 scale) of each group

Figure 2: The more educated respondents show more concern about losing their job to AI over the next year



Source : Deutsche Bank Research, dbDataInsights
Notes: Numerical values denote the average fear (on a 0-10 scale) of each group

Figure 3: The distribution of concern about losing one's job to AI over the next year is higher for those respondents who have used AI at work or at home within the last three months



Source : Deutsche Bank Research, dbDataInsights

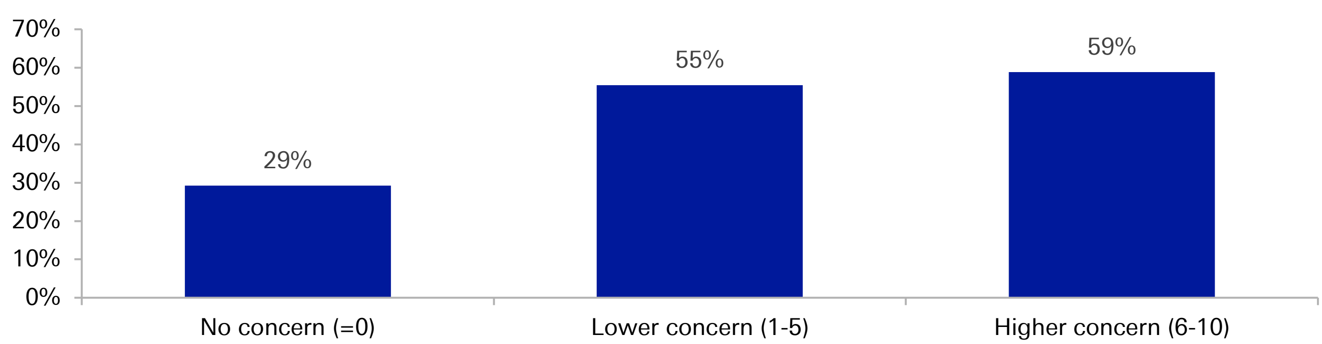
The proactive worker: Upskilling to become a complement to AI rather than a substitute

Worker's fear of losing their job to AI over the next year translates into direct action by the workers. They are not taking the transition to a new AI world lying down. One way in which they are doing this is by upskilling – adapting to become a complement to AI investment rather than a substitute for it.

Up until December 2025, our survey asked respondents what AI learning activities they may have been engaged in: (1) Watching videos to learn about AI; (2) Completing a course to learn more about AI; (3) Reading articles to learn more about AI; or (4) none of these. If the respondent has done any of these types of training (1-3) then they are classified as having undertaken training.

As Figure 4 shows, **the proportion of respondents undertaking some form or AI upskilling/training is increasing with the level of concern about losing one's job to AI over the next year.**

Figure 4: The proportion of respondents undertaking AI training/upskilling is higher for those with higher levels of concern about losing their jobs to AI over the next year



Source : Deutsche Bank Research, dbDataInsights

To work out if the higher likelihood of undertaking training is associated with the level of concern, and not driven by differences in the types of people who are more likely to be concerned, a logit regression analysis is carried out. The regression includes controls for a range of other factors of respondents in an attempt to isolate the effect of concern about job loss on the likelihood of undertaking AI training/ upskilling.[2] The results are shown in the table of Figure 5.

Figure 5: Logit model: Impact of concern about losing one's job to AI on the likelihood of doing AI training

	Share upskilling	Average marginal effect
Full sample	55.3%	1.67pp (SE 0.12)
Only those concerend (concern>0)	61.6%	0.51pp (SE 0.17)

Source : Deutsche Bank Research, dbDataInsights.
Notes: pp = percentage points

The regression is run on the full sample, and then again on the subset who say they have some level of concern (concern >0 on the 0-10 scale). The reason for this is that The change in behaviour from going from "not at all concerned" (a score of 0) to a score of 1 could be quite different from the change in behaviour of someone who is already concerned becoming slightly more concerned (say moving from a 4 to a 5).

The regression analysis shows that a one-unit increase in AI concern (moving one point higher on the 0-10 scale) correlates with a statistically significant average marginal effect of around 1.7 percentage points (pp) for the full sample, and 0.51pp for the concerned subsample. **This means a one-point rise in an individual's concern about losing their job to AI over the next year is, on average, associated with a 0.5-1.7pp increase in the probability of that person undertaking some form of AI training.**

To put that into context, the average level of concern across our survey has risen slightly over half a point over the last year. Therefore, the average respondent has become 0.25-0.9pp more likely to be undertaking some form of AI because of their rising concern about job loss.

This finding is a positive development. Workers are not waiting to be replaced by AI. They are actively working to become AI complements rather than substitutes. If AI does lead to the productivity gains business leaders hope for, tomorrow's labour force are more likely to have the skills and knowledge needed to take advantage of the AI, thereby further boosting the productivity gains from AI.

The pay-rise paradox: Demanding more in the face of disruption

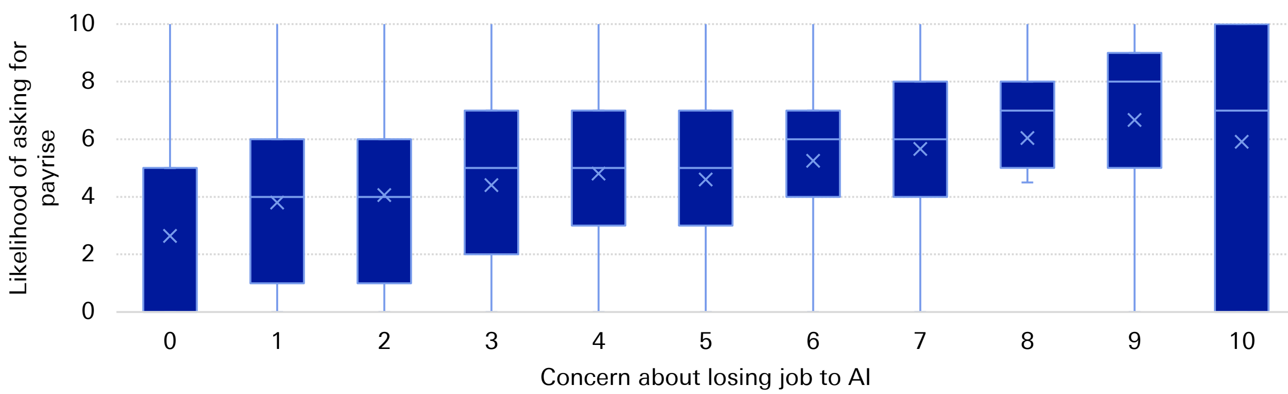
Conventional economic theory would suggest that the because AI is a technology that threatens to replace workers, the adoption of AI by businesses should weaken the bargaining power of workers, making them less likely to push for pay rises.

2 For more on the regression framework used in this report, see Annex 2.

However, the data from our dbDataInsights survey shows the opposite: workers who are more concerned about losing their job to AI are, in fact, more likely to ask for higher wages.

Figure 6 illustrates this paradoxical finding. In our survey, we ask respondents how likely (on a scale of "0 – not at all likely" to "10 – extremely likely") they are to ask their employer for a pay rise in the next three months. Looking at the distribution of responses to this question for each different level of concern about losing one's job to AI over the next year there is a clear pattern of **respondents with higher levels of concern being the most likely to ask for a pay rise in the next three months.**

Figure 6: Paradoxically, workers who report higher levels of concern about losing their jobs to AI are also saying they are more likely to ask for a pay rise in the next three months



Source : Deutsche Bank Research, dbDataInsights

To explore this relationship further, two different forms of regression analysis are used. The first is a panel OLS regression. The results are presented in the table of Figure 7. This approach makes the simplifying assumption that the likelihood of asking for a pay rise is a continuous variable (rather than the discrete 0-10 buckets). The advantage of making this assumption is the coefficient from the regression has a rather simple and intuitive interpretation.

Figure 7: Panel OLS: The impact of concern about AI on the likelihood of asking for a pay rise

	Likelihood of asking for a wage increase (0-10)		Coefficient
	Mean	Median	
Full sample	4.22	5	0.29 (SE 0.01)
Only those concerend (concern>0)	5.01	5	0.30 (SE 0.02)

Source : Deutsche Bank Research, dbDataInsights

The second approach is an ordered logit regression. The results of this are presented in the table of Figure 8. This approach accounts for the discrete buckets for the likelihood of asking for a wage rise, and only requires that the buckets be in ascending order (and not necessarily equally spaced apart).

Figure 8: Ordered Logit: The impact of concern about AI on the likelihood of asking for a pay rise

	Likelihood of asking for a wage increase (0-10)		Odds ratio for a 1pt increase
	Mean	Median	
Full sample	4.22	5	1.21 (95% CI: 1.20, 1.21)
Only those concerend (concern>0)	5.01	5	1.22 (95% CI: 1.20, 1.22)

Source : Deutsche Bank Research, dbDataInsights

Both regression approaches point to the same conclusion. **An increase in the concern about losing one's job in the next year is associated with a higher likelihood of asking for a wage rise in the next three months, even after we control for the characteristics of respondents that are likely to influence their behaviour.**

The panel OLS regression coefficient indicates that a one-point increase in concern about losing one's job to AI (on our 0-10 scale) is associated, on average, with a 0.3 point increase in the likelihood of asking for a pay rise.

So, that would suggest that if an individual's concern about losing their job rose by 3 points on the 0-10 scale, then the likelihood of them asking for a pay rise in the next three months would increase by nearly one point on its 0-10 scale.

The ordered logit gives us an odds ratio of around 1.21-1.22. **That tells us that each one-point increase in the concern of losing one's job to AI is associated with a 21-22% higher likelihood that the individual reporting a higher likelihood of asking for a pay rise in the next three months (than someone with a lower level of concern).**

Why do we find such a counter-intuitive result? It is difficult to say with any certainty. But there are a few theories that could explain the pattern:

- **Omitted Variable Bias:** The regression analysis includes a range of other variables (like the respondents age, whether they are using AI, country-time fixed effects), that attempt to control for factors that could also influence the respondents willingness to ask for a wage rise, and could thus confuse our estimate of the effect from AI job loss concern. But it is possible that the regression does not control for every possibility.

  It's possible that unobserved respondent characteristics influence both concern about losing one's job to AI and the propensity to ask for a wage increase. For instance, ambitious or career-driven individuals might be more inclined to seek pay rises while at the same time are more attuned to both the threats (and opportunities) presented by AI, leading to higher concern as well.

- **Strategic Preparation for Future Redundancy:** Although not reported here, the level of concern about losing one's job to AI over the next year is not too dissimilar to the concern over the next five years. Perhaps workers expect that the time horizon for being replaced by AI is quite long. This could lead them to strategically maximize their current earnings, including seeking higher wages today, in an attempt to bolster future severance payouts or raise their incomes today to build up their savings in anticipation of a future job loss.

The confident consumer: Spending and saving in the AI era

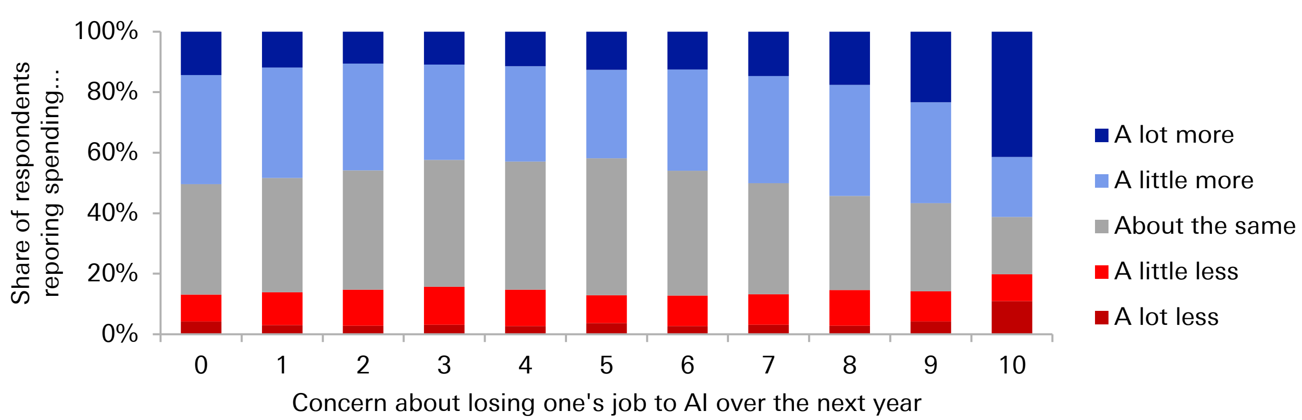
Consumption smoothing behaviour tells us that if workers become more concerned about losing their job in the future, they should reduce their consumption today and save more, thereby having more money available for when they are unemployed.

Thus while spending on AI data centres can boost GDP and economic activity in the near term, at the same time worries about job security can act as a drag on spending, thereby lowering GDP and economic activity.

The results from our household survey present mixed results regarding this behavioural prediction of households. **While workers concerned about losing their job to AI state an intention to save more (a prudent response to uncertainty), they are also more likely to report spending more than this time last year (something that goes against consumption smoothing).**

In the survey, we ask respondents if there are current spending: a lot more; a little more; about the same; a little less; or a lot less, relative to a year ago. Figure 9 shows how the distribution of responses varies by the level of concern about losing one's job to AI. While all groups are more likely to report spending a little or a lot more than last year (a consequence of higher inflation rates), those with a greater level of concern about losing their job are also the ones more likely to report spending more than this time last year.

Figure 9: Respondents who are more concerned about losing their job to AI are also more likely to report spending more than this time last year.



Source : Deutsche Bank Research, dbDataInsights

Figure 10 reports the results from the ordered logit regression of spending behaviour onto the fear of job loss (and other controls).[3] **The results suggest that each one-point increase in the concern of losing one's job to AI is associated with a 3-6% higher likelihood that the individual reports a higher degree of spending relative to last year.** The result is statistically significant.

Figure 10: Ordered logit: The impact of concern about AI on spending growth

	Odds ratio for a 1pt increase
Full sample	1.032 (95% CI: 1.02, 1.04)
Only those concerend (concern>0)	1.060 (95% CI: 1.05, 1.07)

Source : Deutsche Bank Research, dbDataInsights

This positive correlation between AI concern and increased spending contradicts the initial hypothesis that AI-driven job loss concerns would dampen consumer spending. Spending less would imply a negative correlation (i.e., an odds ratio below 1).

Savings and consuming are two sides of the same coin. It is difficult to accurately measure savings behaviour in household surveys. However, we do ask respondents about their intentions to save. Specifically, respondents are asked: *'Thinking 12 months ahead, do you anticipate your current savings amount(s) will increase, decrease, or remain about the same?' Response options in the survey included: 'Increase significantly,' 'increase slightly,' 'stay about the same,' 'decrease slightly,' or 'decrease significantly.'*

The ordered logit regression analysis was repeated on the savings intension of the survey respondents. The results are presented in the table of Figure 11.

The results suggest that each one-point increase in concern about losing one's job to AI over the next year is associated with a 6-9% higher likelihood that the individual reports a higher intention to save more over the next year.

Figure 11: Ordered logit: The impact of concern about losing one's job to AI on savings intentions

	Odds ratio for a 1pt increase
Full sample	1.06 (95% CI: 1.05, 1.07)
Only those concerend (concern>0)	1.09 (95% CI: 1.08, 1.10)

Source : Deutsche Bank Research, dbDataInsights

3 Inflation expectations is included as an additional control here as it could impact on the decision to bring forward the timing of consumption spending. We also include a control related to the question "are you better or worse off financially than a year ago" as being better off financially than a year ago could explain why the respondent is spending more today than last year.

Thus, a higher concern for losing one's job to AI is, on average, directly associated with an intention to increase savings in the near future. This positive correlation aligns with the concept of precautionary savings, where the perceived heightened risk of AI-induced job in the future loss motivates consumers to save more today.

Thus while the reported spending behaviour of respondents contradicts to predictions that concerned households should be spending less and saving more, those respondents also report a greater intention to save more going forward. But whether they actually follow through on these intentions is not something we can observe. Therefore, the weight of the evidence from our survey suggests that the uncertainty shock from AI fear is not (yet) dampening consumption.

Conclusions and implications

The standard economic theory tells us that workers/households should fear AI. As a substitute for their labour, it weakens their bargaining power and makes it more likely they will be laid off.

But concerns about AI-driven job loss are not creating a passive, fearful workforce poised to retreat from the economy. Instead, our dbDataInsights household survey reveals a powerful paradox: the individuals who are most anxious are also the most proactive, turning their concern into decisive economic action.

- Fear of job-loss from AI is not born out of ignorance. Our survey shows that those who are most informed about AI show the greatest fear of losing their job.
- Workers who are fearful of AI are responding by investing in their own skills and trying to learn more about AI, turning themselves from a type of labour AI can substitute for, to a type of labour that is a complement to AI.
- Rather than weakening worker's bargaining power, fear of AI seems to be encouraging workers to ask for higher wages.
- The confidence also extends to their role as consumers. Higher fear of losing one's job to AI is not associated with workers spending less (savings for the future when they are unemployed). Instead they seem to be spending more.

The key insight from the findings here is that fear of AI is not (yet) acting as a drag on households as standard theory would predict. While businesses are spending big on AI models and data centres, workers are actively trying to adapt to the new world we are moving into. Proactive policies can support workers adapting to AI and lead to productivity gains that go well beyond simply replacing labour with AI tools.

But it must be kept in mind that these findings present only a snapshot of a highly dynamic situation. For now, the 'AI Anxiety Paradox' shows a workforce that is resilient and adaptive. However, these behaviours are rooted in the anticipation of change. Should widespread, large-scale AI-driven layoffs materialise, this dynamic could fundamentally alter, potentially shifting today's proactive ambition back towards a precautionary retreat.

The author would like to thank Anthony Chaimowitz from the dbDataInsights team for his invaluable assistance in administering the survey and collating the data.

Annex 1: dbDataInsights household survey

The dbDataInsights household survey provides valuable insights into household/ consumer attitudes and expectations regarding personal finances, labour market conditions, and buying conditions. The survey is conducted by Deutsche Bank and is used by economists and sector analysis in their research on the macroeconomy and specific sectors.

The survey covers households/consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States. For each country, a national-representative sample of 550 households is surveyed each month (650 households in the US). Results are collected throughout the month to provide a high-frequency look at how expectations and attitudes are evolving in response to the latest developments in the economy.

The survey has been running since 2018 and is continuously evolving to address relevant issues. The survey is conducted online. Although the individual respondents change each month, the panel remains nationally representative each month.

Annex 2: Regression analysis framework

This Annex describes in more detail the regression analysis framework used throughout the report.

To assess the correlation between AI concern and behavior, a series of panel regressions is run using micro-level survey data. In these regressions, the variable of interest (e.g., savings intentions, spending) is regressed against the self-reported concern level, incorporating fixed effects to control for individual respondent characteristics.

Standard fixed effects in most of the analysis account for:

- **Age:** Categorized by ranges (18-24, 25-34, 35-44, 45-55, 55-64, 65+)
- **Using AI:** Binary indicator of whether the respondent has used AI in the last 3 months, either at work or at home.
- **Income Brackets:** Low, medium, and high.
- **Country-Month Interactions:** To capture broader macroeconomic trends (e.g., economic shifts, political events).

Other fixed effects are added where appropriate. For example when examining worker's savings intentions, inflation expectations of the respondents are also included.

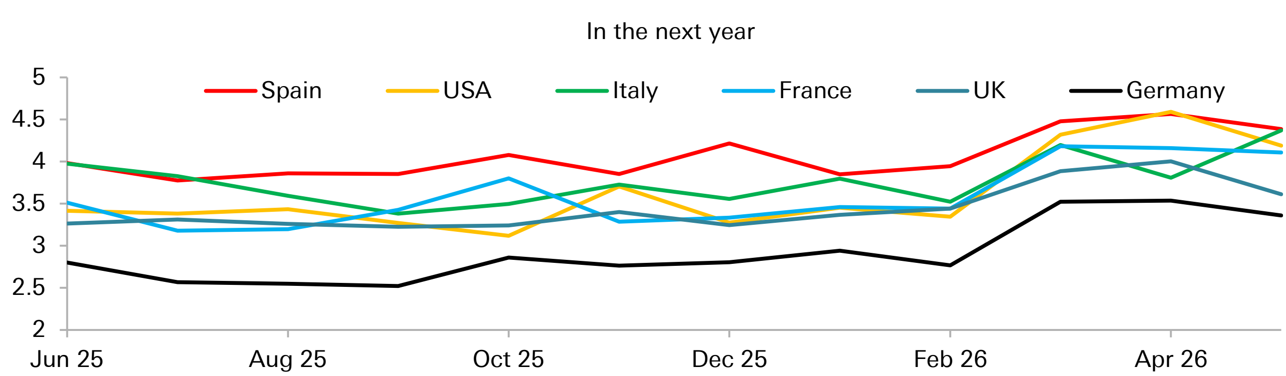
All results in the paper are based on concern of losing one's job to AI over the next year. However, using their concern of losing their job over the next two and five years produce similar results.

Annex 3: Further results about what type of respondents are worried about AI job loss

Figure 12 plots average level of concern in each country we survey (Germany, France, Italy, Spain, the UK, and the US) for each month. A few patterns stand out here:

- **Up until March, the average level of concern was quite stable.** Between June 2025 and February 2026 the average level of concern in all countries remained broadly stable. This period we saw release of some major new LLM models.[4] But the stability of concern shown in Figure 12 suggests the improvement of these models did little to affect the average concern of workers, either because the improvements did not change the opinion of respondents, or respondents may have been ignorant of these latest developments.[5]
- **A shift occurred in March, with average concern rising significantly in all countries.** This increase may be linked to broader geopolitical anxieties, such as the Iran war, which could heighten general economic and financial well-being concerns, making respondents more sensitive to AI-related job loss risks.
- **Across all surveyed nations, Germany consistently records the lowest average level of concern, while Spain registers the highest or near highest.** France, Italy, and the UK show broadly similar levels of concern, though a recent surge in the US has brought its average closer to Spain's level.

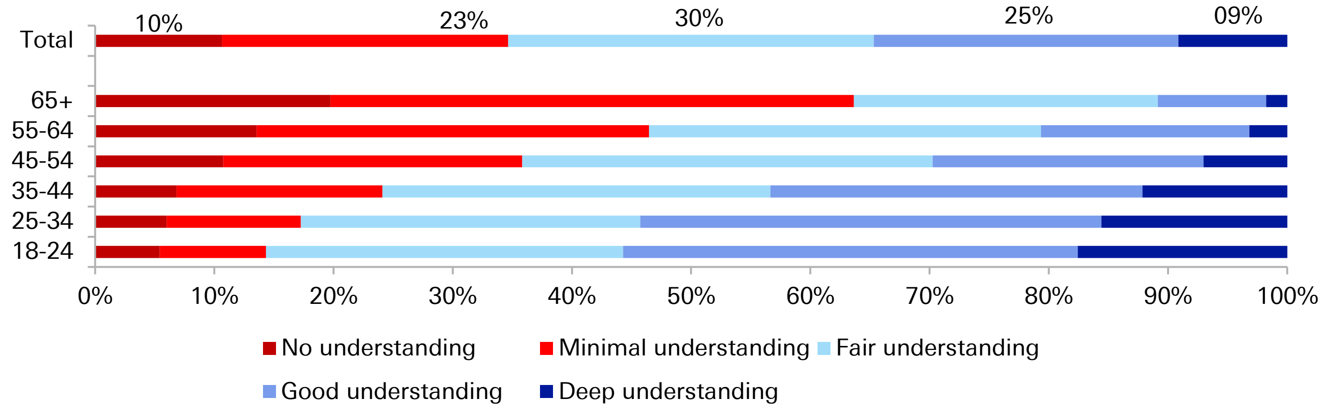
Figure 12: Average concern (scale 0-10) of losing your job to AI



Source : Deutsche Bank Research, dbDataInsights

4 Including OpenAI's GPT-5 (August 2025), Google's Gemini 3.1 Pro (February 2026), etc.
5 However, as shown in Figure 13, a majority of respondents report at least "a fair understanding of AI", suggesting that the new model developments would have likely to have been known by a significant proportion of respondents.

Figure 13: Self reported understanding of AI by age group – The young claim to understand AI better



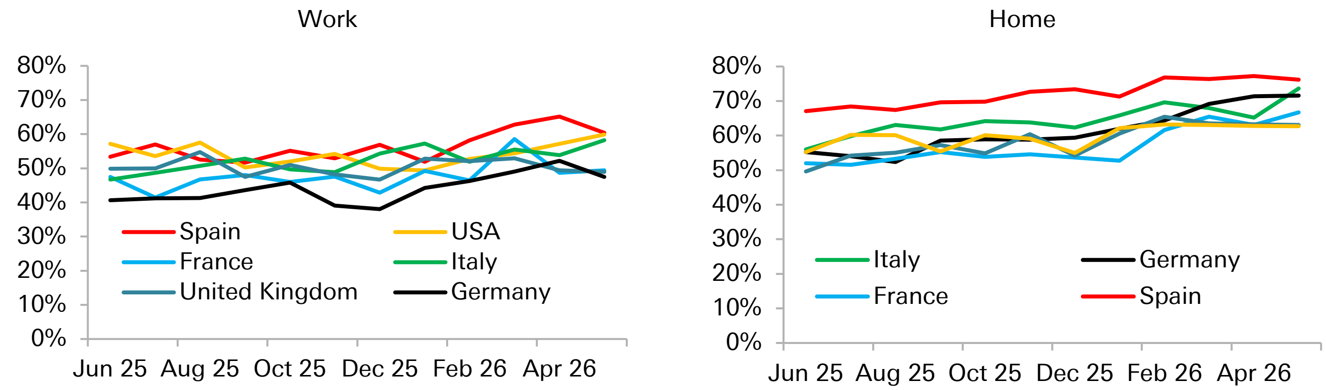
Source : Deutsche Bank, dbDataInsights

Younger individuals also report a higher self-assessed understanding of AI (Figure 13). Younger individuals tend to be more tech-savvy. Their deeper understanding of AI's capabilities likely influences their heightened concern about job loss.

Our survey indicates a strong association between AI usage and elevated concerns about losing one's job to AI. Figure 14 shows the percentage of respondents who have used AI in the last three months, both at work and at home. A few interesting patterns stand out in Figure 14.

- **The proportion of respondents using AI at work is broadly comparable between the US and the European countries we surveyed.** This finding offers some reassurance amidst European political concerns about falling behind in the 'AI race,' suggesting that even without developing as many cutting-edge models of their own, European businesses are effectively leveraging existing AI tools.
- **While US respondents show similar rates of AI usage at work and home, in Europe, home usage generally surpasses work usage.** This could signal a greater appetite among European workers for integrating AI into their professional lives, provided they are granted more autonomy with the tools.
- **AI adoption, both at work and at home, is steadily increasing.** Across all surveyed countries, the percentage of respondents using AI in the past three months shows a consistent upwards trend.

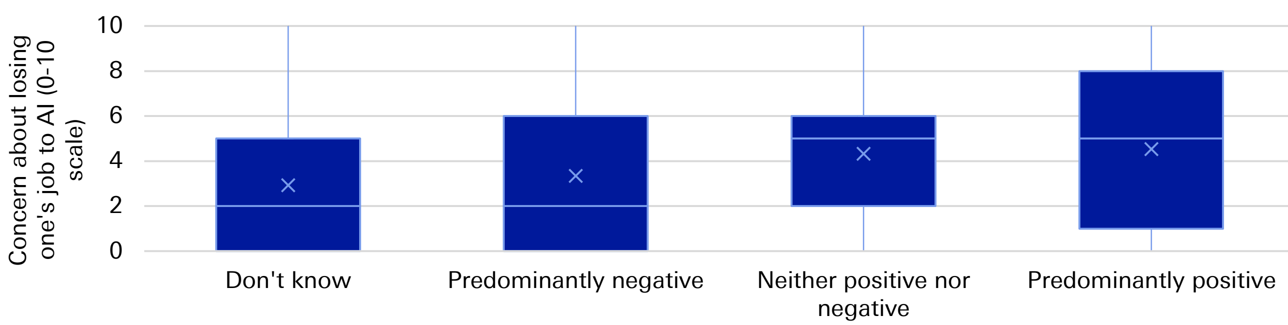
Figure 14: Share of respondents who have used AI in the last 3 months at…



Source : Deutsche Bank Research, dbDataInsights

Those with a more positive outlook on AI's societal impact also demonstrate the highest concern about losing their own jobs (Figure 15). This pattern suggests that techno-optimism/pessimism may not be the primary determinant of individual job security concerns.

Figure 15: Those thinking AI will have a more positive social impact are the most concerned about losing their jobs to AI



Source : Deutsche Bank Research, dbDataInsights

Appendix 1

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This Month in Geopolitics: July 2026

Key events, themes and trends to watch in the month ahead

July 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in **July 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.

US/Mexico/Canada:
- **1 July:** Beginning of the mandatory six-year joint review process of the US-Mexico-Canada (USMCA) trade agreement

United Kingdom
- **9-17 July:** Should no other Labour MP mount a leadership challenge, it is likely that **Andy Burnham will become the next Prime Minister** by mid-July

Europe:
- On 1 July, the EU introduced **new levies on low-cost shipments** and **rules to protect the EU steel market** – both implicitly aimed at China. At the same time, the EU and China have agreed to enter three months of talks to try to avoid a trade war.

India:
- **1-3 July: Japan's** Prime Minister Takaichi is expected to visit India
- **24 July:** India and the **US** could seek to finalise talks on an interim trade agreement by this date

China:
- **Late July:** Politburo meeting is expected to set economic priorities for 2026-H2

NATO Summit: (Ankara)
- **7-8 July: NATO 3.0: Redefining Burden-Sharing : US** pressure on European allies to carry more of the security burden will likely headline the summit.

Switzerland/UN
- **6-10 July:** Geneva will host three UN-backed summits focused on AI / digital governance and cooperation

Australia/Indonesia/New Zealand:
- **6-11 July:** India's Prime Minister Modi could reportedly pay an official visit to the three countries

United States:
- **13 July:** The US could introduce higher tariffs, price floors or other measures on some critical mineral imports
- **24 July:** The US global 10% import tariff is set to expire, and is expected to be replaced by new Section 301 levies

US-Iran:
- **All July:** Peace negotiations are set to continue. Watchpoints include implementation disagreements, Gulf states' requirements for security guarantees, and domestic political opposition from key parties

Source: Deutsche Bank

In the spotlight for July 2026: **US-Iran** peace talks and **China-EU** trade tensions

US-Iran peace talks (all of July)

The US and Iran are keen to press ahead with the framework agreement even if this doesn't lead to a full-scale peace agreement. This process is unlikely to be without friction. There are many moving parts in negotiations and we should watch closely for disagreements over implementation, Gulf states' requirements for security guarantees, and domestic political opposition from key parties. A critical factor to watch will be the potential for Israeli aspirations in Lebanon to upend the deal. We suspect that there is significant US pressure on President Netanyahu to cease operations and accept the accrued gains to prevent a derailing of the deal.

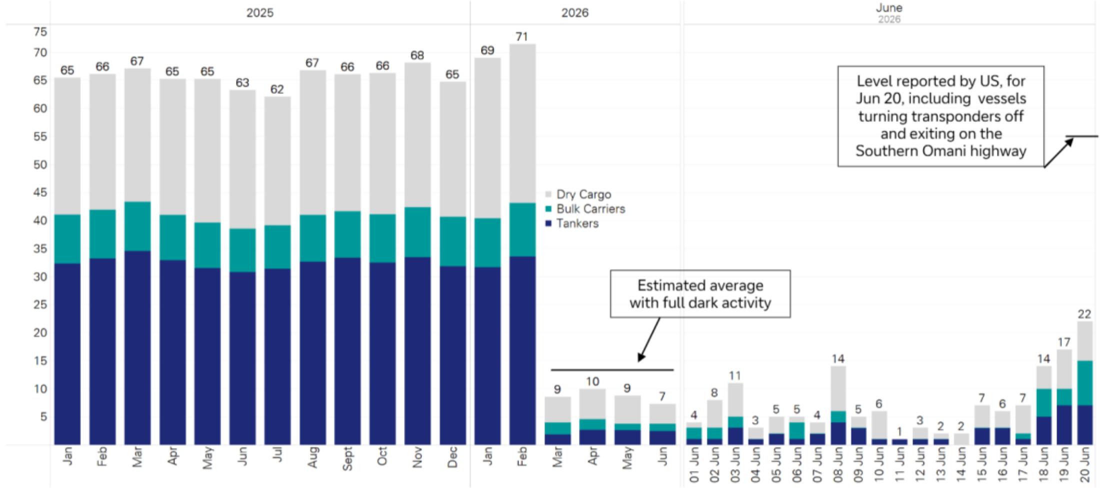
Shipping levels through the Strait of Hormuz have increased (see top chart), but **traffic will likely take time to recover to pre-war levels** given the challenges of de-mining, a lack of trust between the US and Iran, and Iranian threats. Infrastructure repair to ports, terminals and energy infrastructure, as well as the provision of security guarantees are other factors to watch. While crude oil prices have recovered to pre-war levels, full operational normalization, involving infrastructure repairs, is not expected until 2027 and beyond.

China-EU trade tensions (all of July)

While efforts have been initiated to avoid a trade war, the EU continues to implement new measures against China. On 29 JUN, the EU and China agreed to a 3-month dialogue with a commitment to reconvene in October on four "workstreams" to make progress on trade disagreements, including export controls, trade and investment balancing, and a joint monitoring mechanism to improve transparency and manage trade frictions.

At the same time, the EU Council **has reportedly tasked the European Commission with developing new trade defence tools** implicitly aimed at addressing trade imbalances with China. The trade imbalance, along with perceptions that China's currency is undervalued, is a growing political concern for the bloc (see bottom chart). **Two new EU measures implicitly aimed at China entered into force on 1 JUL.** The first is a temporary €3 (US$3.4) **customs duty per item levied on low-cost shipments entering the bloc**. The measure intends to address imports from Chinese e-commerce companies and will remain in force until 1 JUL 2028. **New rules intended to protect the EU steel market from global overcapacity** entered into force, which include a revised quota system. China is the world's largest steel producer, accounting for around 50% of global output.

Blue-water commercial cargo tankers exitin the Strait of Hormuz (from 2025) version count

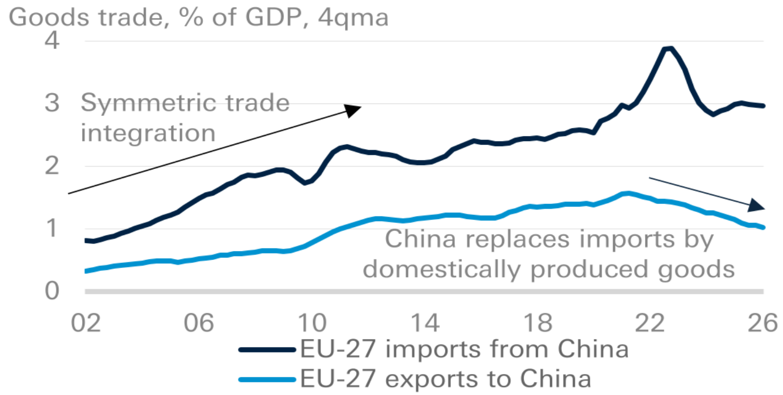


Source: dbDataInsights

Note: Pre-crisis figures do not account for 'dark activity' (estimated 3-6 vessels per day with transponders off, mainly Iranian tankers). Following the crisis, we implemented a methodology to estimate partial dark activity by cross-referencing Persian Gulf port departures with confirmed arrival records outside the SoH, and media reports. Based on this, our recorded average daily vessel crossings from March to mid-June were nine. However, if we were to triangulate all estimated dark activity, average daily crossings during this period would likely have been in the low-teens. Dark activity appears to have picked up significantly since the MoU was signed as the U.S. Central Command said 55 commercial ships passed through the strait on 20 June.

For a detailed analysis of the impact of the Iran conflict and the outlook for the Strait of Hormuz, please see Twenty Miles That Shook the World

Structural trend of a widening EU trade deficit with China



Source: Eurostat, Deutsche Bank. For more detail see: Germany Blog: Stepping up trade defences against China

Deutsche Bank Research Institute

Key events, issues and themes to watch in **July 2026**

Listed in chronological order

US trade policy (all of July)

July could see the introduction of new US tariffs against dozens of trading partners. The expiry of US 10% global Section 122 tariffs **on 24 July**, and their expected replacement by Section 301 levies, should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. Countries with existing framework agreements with the US could see the rates of new levies capped at previously agreed levels. **New US Section 301 tariffs on China** could increase bilateral tensions, although we assess that a return to significant trade escalation remains unlikely as long as both leaders continue to signal a commitment to strategic stability.

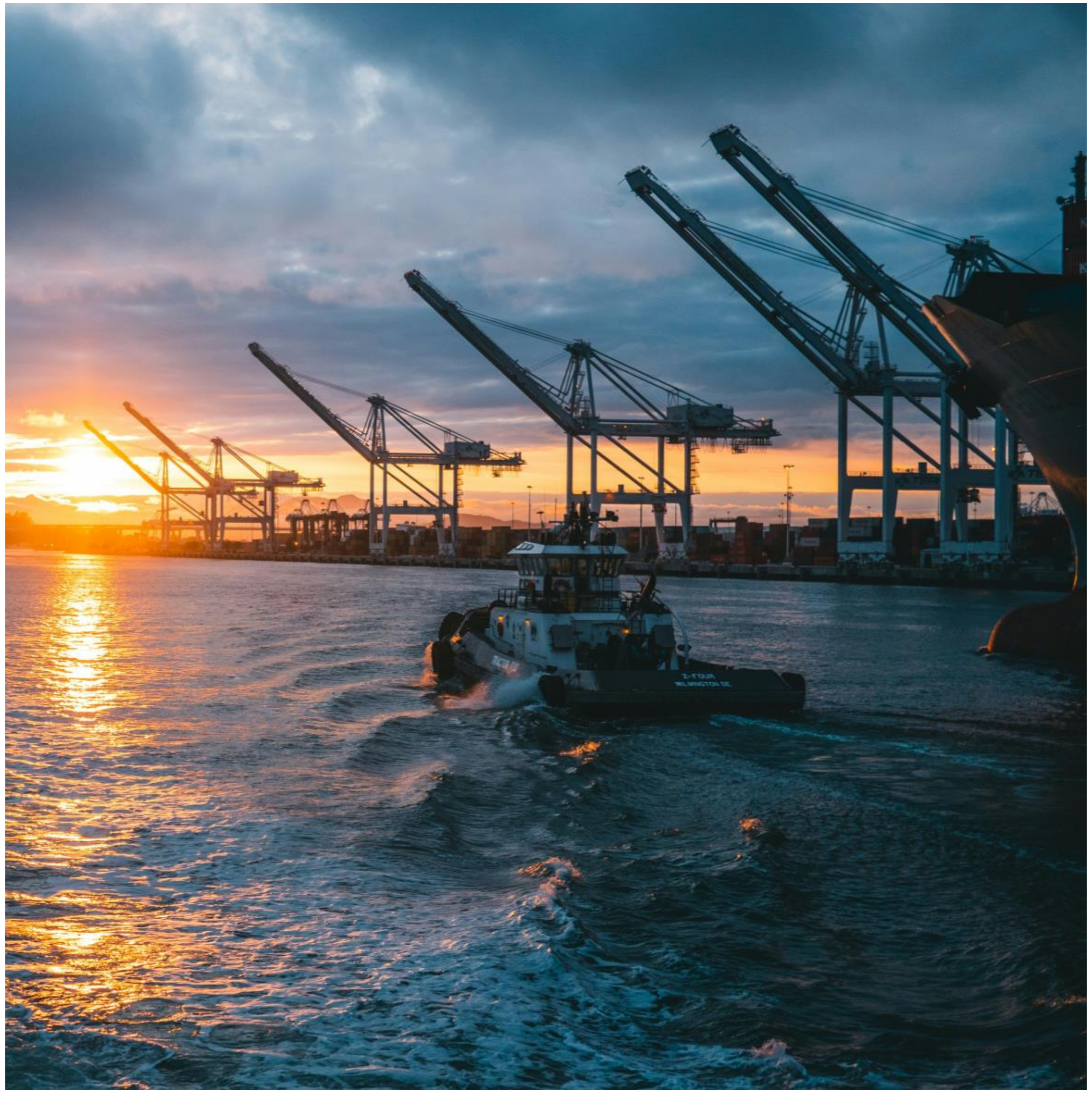
Under a separate Section 232 investigation **into critical minerals**, after 13 July, the US may choose to impose tariffs, price floors or other trade measures in response to minerals agreements judged to be incomplete or ineffective.

US-Mexico-Canada trade talks (1 July onward)

The renewal talks are expected to be challenging and potentially protracted. 1 July marks the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process. The three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode if consensus is not reached. **Officials have signaled that the complex negotiations may continue beyond the initial review date**. President Trump's statements in June expressing skepticism over the deal and its value to the US could add another layer of uncertainty to the talks.

Japanese Prime Minister Takaichi's expected to visit India (1-3 July)

The visit will likely aim to strengthen bilateral trade and investment ties. Prime Minister Takaichi's first official visit to India will likely include a bilateral meeting with Prime Minister Modi. Areas of focus in talks could include increased cooperation in semiconductors, strengthening supply chain resilience, as well as security cooperation. Both countries are members of the Quad initiative, and bilateral ties have strengthened in recent years as New Delhi and Tokyo diversify their international ties amid rising geopolitical uncertainty. **Japan is India's fifth-largest foreign investor**, particularly in sectors such as automotive manufacturing, digital technology, clean energy and infrastructure.



Source: Ronan Furuta on Unsplash

Key events, issues and themes to watch in **July 2026**

Listed in chronological order

AI/digital summits in Geneva (6 -10 July)

Switzerland will host three UN-backed summits during July. The UN will hold its inaugural **Global Dialogue on AI Governance** (6-7 July), intended to establish shared principles on AI governance and improve interoperability across emerging national regulatory frameworks. The summit takes place alongside two meetings organized by the International Telecommunication Union (ITU). The annual **AI for Good Global Summit** (7-10 July) aims to translate AI policy discussions into technical standards, public-private partnerships and scaled deployment of AI solutions. The **World Summit on the Information Society** (WSIS; 6-10 July) centres on promoting global digital cooperation.

NATO leaders' summit in Ankara (7-8 July)

Pressure on European allies to carry more of the security burden will likely intensify as US stockpile and resource constraints bite after the Iran war. President Trump's demands will likely headline the NATO Leaders Summit. **It is slated to assess the credibility of national defence spending plans** and advance the emerging "NATO 3.0" agenda under which European NATO members take on more responsibility for their own conventional defence.

Ahead of the meeting, **the US has announced some military pullbacks from Europe** and reportedly notified allies of further planned reductions to its NATO Force Model contributions. To slow the withdrawal of hard-to-replace US capabilities (such as air defence, long-range weaponry and strategic airlift), European leaders will stress their commitment to NATO's new 3.5%-of-GDP target for "core" defence spending and their readiness to help clear the Strait of Hormuz.

India's Prime Minister Modi could visit Indonesia, Australia and New Zealand (6 -11 July)

Media reports indicate that Modi could visit the three countries during July, even though no plans have been officially confirmed. The prime minister's itinerary could reportedly include talks with Indonesian President Prabowo Subianto, Australian Prime Minister Albanese and New Zealand's Prime Minister Luxon. If confirmed, the visit would likely reflect New Delhi's **intensifying economic and security engagement with regional middle powers** amid rising global geopolitical risk and global trade volatility.

Source: Jannik on Unsplash

Key events, issues and themes to watch in **July 2026**

Listed in chronological order

Appointment of new UK Prime Minister (mid-July)

UK will see the appointment of its 7th Prime Minister since 2016. Prime Minister Keir Starmer announced his resignation on 22 JUNE. The Labour leadership nomination process opens on 9 JUL. At the time of publication, Any Burnham is the only declared candidate. Assuming no other Labour MP mounts a leadership challenge, Burnham could become the Prime Minister by mid-July, pending the completion of the Labour leadership process and formal appointment by the King.

Burnham's key priorities appear to be significant devolution of power, regional economic growth, fiscal discipline, and political reform.

US-India trade negotiations (24 July as a soft deadline)

Trade talks are reportedly nearing completion, reaffirming strategic stability despite recurring bouts of friction. The Modi-Trump meeting on the sidelines of the June G7 summit has reportedly injected renewed momentum into bilateral trade talks, which are now believed to be nearing completion. India's Trade Minister Piyush Goyal has expressed **optimism that an interim deal could be concluded by 24 July**, when the US' global 10% tariffs expire.

But even if negotiations extend beyond this deadline, we maintain our view that a convergence of strategic interests – including long-term competition with China – should keep bilateral ties broadly on track. **Nevertheless, this does not preclude the possibility of recurring bouts of friction over trade, tariffs and security issues**.

Politburo meeting in Beijing (late July)

The July meeting of the Politburo – China's top policy-making body – is one of the more important sessions in this year's political calendar. The meeting will most likely assess the state of the Chinese economy during the first half of 2026 **and set the economic policy direction for the rest of the year**. Post-meeting readouts could provide insights into priorities around industrial policy, support for the property sector, and the implementation of the 15th Five-Year Plan adopted in March 2026.

Source: 典 方 on Unsplash

Appendix 1

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Research Institute

AI IPOs 101: why they'll change the boom forever

July 01, 2026

Four reasons why the expected IPOs of OpenAI and Anthropic matter so much

The expected wave of initial public offerings of the world's leading AI companies will change the AI boom forever.

It will bring unprecedented **transparency**, **funding** and **accountability** to the industry – as well as finally giving investors a **pure-play benchmark** to value the likely winners and losers of a revolutionary technology.

The news at the end of last week that OpenAI is considering delaying its own share sale, originally expected for later this year, indicates just how much is at stake. The company, which lit the fuse for the AI frenzy with its launch of ChatGPT in November 2022, as well as its now slightly larger rival Anthropic, which makes Claude, confidentially filed for IPOs in early June. Those share sales would reportedly aim to raise about $60bn apiece and value each company at more than $1 trillion.

They and their peers need to **strike while the iron is hot** to secure computing power, distribution and capital. But there is no guarantee that there will be enough investor appetite to absorb hundreds of billions of dollars in rapid-fire demand.

The AI boom is fast approaching a **moment of truth**, as rapid growth and soaring valuations collide with ballooning capital expenditure, a public backlash and the challenges of real-life adoption. Challenges abound, including Chinese open-source models that promise similar performance for a fraction of the price. Whoever comes out on top could define the era.

In this report, we look at:

- Why have an IPO?
- Which AI-related companies are likely to have IPOs?
- OpenAI and Anthropic in a nutshell
- Four ways that IPOs will change the AI boom forever

1. Greater transparency for financials – and business models
2. More diverse funding sources – and more spending
3. Greater accountability for their actions – and the industry
4. New era for comparing valuations to peers – and the past

Authors

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Research Analyst

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Why have an IPO?

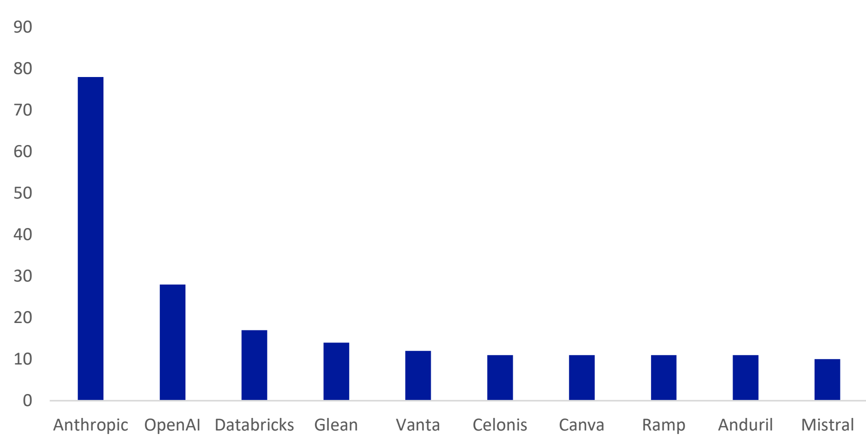
1. **Access new sources of funding** to invest, reduce debt and improve financial flexibility.
2. **Create liquidity** to give existing investors and employees a way to cash out and enable acquisitions with publicly traded shares.
3. **Improve public credibility and profile**, lending additional weight with consumers, businesses and lawmakers.

Which AI-related companies are likely to have IPOs?

The two leading contenders to sell shares to the public this year are Anthropic, given a 78 percent chance, according to bets on Polymarket, and OpenAI, now down to 28 percent.

Other AI-related companies that may go public include data analytics platform Databricks, enterprise data platform Glean, and compliance tool Vanta, according to Polymarket bets. Others include business process platform Celonis; graphic design tool Canva; finance operations platform Ramp; defence company Anduril; and Europe's only notable frontier-model maker Mistral.

Figure 1: Implied probability of select AI-related IPOs by 2027



Source : Polymarket, Deutsche Bank Research

OpenAI in a nutshell

- San Francisco-based OpenAI was **founded in 2015** as a non-profit research laboratory led by **Co-Chairs Elon Musk and now-CEO Sam Altman**, the former head of Y-Combinator. It was backed by some of the biggest investors in Silicon Valley, including Peter Thiel and Reid Hoffman. (Musk severed ties with OpenAI in 2018, triggering a legal dispute that concluded in May in OpenAI's favour.)
- Its **mission** was "to ensure that artificial general intelligence – AI systems that are generally smarter than humans – benefits all of humanity".
- On November 30, 2022, OpenAI launched **ChatGPT**, a generative AI chatbot based on its GPT-3.5 large language model, which became a

runaway success with 100 million users in three months. GPT-4, released in March 2023, was so far ahead of the competition that it took a year or more for serious challengers to gain a foothold. However, GPT-5, launched last August, was quickly matched amid stiffer competition.

- The non-profit board voted to remove **Altman** as CEO in November 2023. He was reinstated four days later, after a revolt by investors and employees.
- Its **most important strategic relationship has been with Microsoft**. When OpenAI transitioned into a "capped-profit" entity in 2019, it was able to obtain capital investment including $1bn from Microsoft, which later added another $10bn in early 2023. Their exclusive partnership and revenue-sharing agreements enabled Microsoft to be an early leader in AI and secured funding and computing power for OpenAI.
- The companies **scaled back their partnership** in late April 2026, though Microsoft retains a 27 percent stake in OpenAI. (OpenAI's for-profit LLC became a public benefit corporation controlled by the non-profit in May 2025.)
- Today, **more than one in ten people on the planet use ChatGPT every week**. The name is to chatbots what PostIts are to sticky labels. It reportedly surpassed $25bn in annualised revenue in March and has indicated to investors that it will achieve its first annual profit in 2030.
- Although the market is increasingly crowded, OpenAI is still producing **cutting-edge models**, with GPT-5.5 excelling at agentic coding. Brand new GPT-5.6 won the same badge of honour as the latest Claude model last Friday by being restricted to a small group of users over US government concerns that it was so powerful that it could pose a security risk. It also has a strong developer ecosystem building on top of its Application Programming Interface (API).

Anthropic in a nutshell

- San Francisco-based Anthropic was **founded in 2021** by seven former employees of OpenAI, including **Dario Amodei, its CEO**, and his sister Daniela Amodei, its president. Anthropic is a public benefit corporation whose mission is "to ensure that the world safely makes the transition through transformative AI".
- Led by its **Claude models**, the company has focused heavily on the enterprise market, positioning itself as a bastion of high standards of safety and reliability.
- **Google parent Alphabet and Amazon** have invested billions of dollars in Anthropic. Like OpenAI, it has struck complex agreements that commit it to use chips and data centres from its investors in return, including a commitment reported by The Information to spend about $200bn with Google Cloud over five years.
- This has been a **breakout year** for Anthropic. The release of its Cowork and Claude Code agent tools in January and February prompted a "SaaSpocalypse" slump in stocks of software-as-a-service companies, amid concerns that their utility-like business models could be largely disintermediated.
- The **Claude website had almost one billion visits in May**, still behind ChatGPT at 5.6 billion and Google's Gemini at nearly three billion, but way up from 200 million in January, according to data from Similarweb. Meanwhile, Claude Code now accounts for slightly more than half of the business-focused coding market.
- **Anthropic's revenue** reached an annual recurring rate of $47bn in May,

overtaking OpenAI. CEO Amodei said that an 80-fold increase in usage in the first quarter on an annualised basis was "just crazy" and "too hard to handle," forcing it to limit access for some users and sign up new sources of compute to keep up with demand.

- A **disagreement with the US Administration** came out into the open in February when the Department of Defense announced that it was a supply-chain risk. The company had demanded restrictions on the use of its technology for domestic surveillance and autonomous weapons.
- Then two weeks ago, Anthropic abruptly **switched off its most powerful Fable 5 and Mythos 5 models** worldwide after the US Department of Commerce ordered it to restrict access by foreigners, due to concerns that it could be weaponised by bad actors. It won approval to restore some access at the end of last week.

Four ways that upcoming IPOs will change the AI boom forever

1. Greater transparency for financials and business models

Nothing sharpens an entrepreneur's mind – and business model – like the prospect of an IPO.

- It is a decision that they may take with some trepidation, given the obligation to open the doors and explain their finances, business model and outlook in detail.

To prepare for an IPO, AI companies will have to produce public-company-grade reporting, credible performance indicators, risk analyses and forecasts.

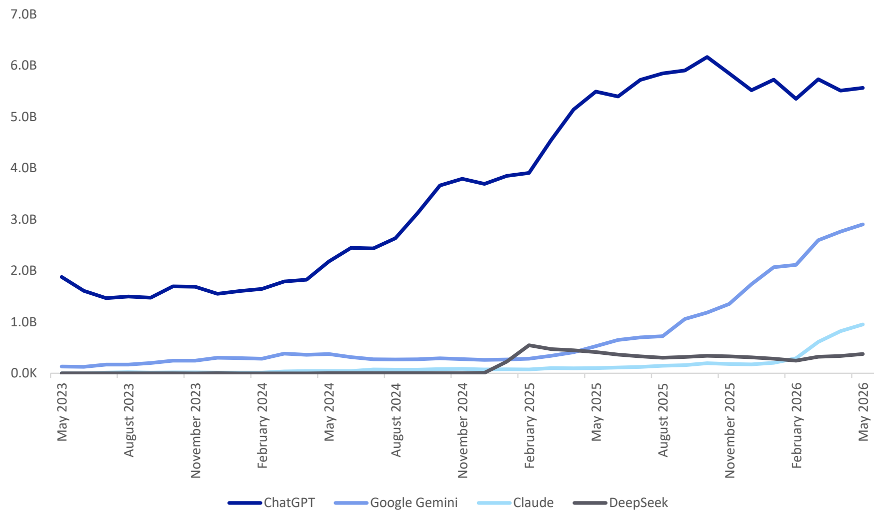
- They will also have to make any relevant disclosures about a growing number of lawsuits over copyright, alleged negligence regarding safety, and corporate governance.
- Remuneration may also have to be revamped, with public metrics around pay for performance for management and provisions for employees to vest stock.

Greater transparency may help investors take a view on how much they consider currently lossmaking AI models to be winner-takes-all, like Amazon or Meta, where network effects create a virtuous circle, or commodities, where one model is viewed as interchangeable with another.

- While OpenAI has the leading chatbot by far with around 900 million monthly users, it has been rapidly pivoting its business model this year to try to attract enterprise users who will pay for it.
- That includes creating the OpenAI Deployment Company in May with 19 private-equity groups led by TPG to distribute its enterprise products, spearheaded by so-called "forward deployed engineers". It said previously that enterprise revenue was on track to match consumer revenue by the end of the year.
- Since CEO Sam Altman declared a "code red" in December, OpenAI has scrapped what it called "side quests" such as its video generating tool Sora and is working on a "unified superapp" that brings its chatbot, coding, browsing and agentic capabilities together, aiming to convert satisfied home users to a similar business offering.

- Anthropic overtook it in business subscriptions in May, according to data from Ramp. And Similarweb data shows monthly visits to ChatGPT fell below a majority of the generative AI market for the first time in May, suggesting consumers are increasingly willing to switch between models.

Figure 2: Top four generative AI platforms: monthly visits worldwide (desktop and mobile web)



Source : Similarweb, Deutsche Bank Research

- OpenAI has indicated that it is generating about $25bn to $33bn in annualised revenue, up from a $4bn annual rate at the end of 2024. By comparison, Anthropic's annualised run rate shot up from $1bn a year at the end of 2024 to $47bn in late May. The company previously said that its first profitable full year will be in 2029, one year earlier than OpenAI.

Greater disclosure around IPOs will finally illuminate the business models of the frontier model makers.

- While chips and data centres are easy to measure, actual monetisation of AI is opaque, given that private model makers make only limited disclosures and public hyperscalers rely on AI models for just a small part of their AI revenues.
- The key questions are how intense and rapid adoption will be; is the return on investment justified once businesses have to make hard choices to scale beyond pilots; and will premium all-singing, all-dancing frontier models win out over cheap, stripped-down, task-specific alternatives?

The stakes for pure-play model makers are higher now than ever before, with pressure from above and below.

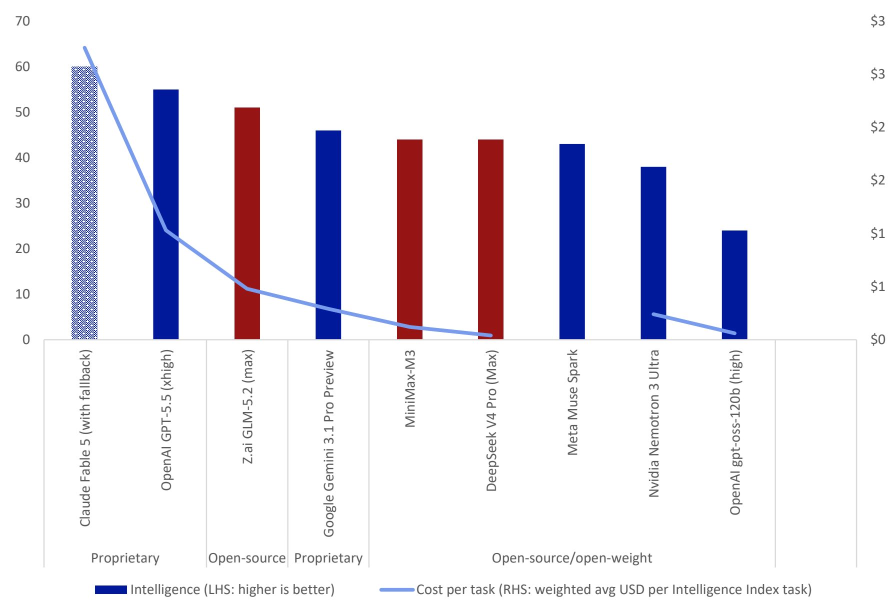
- On the one hand, the hyperscalers have unsurpassed access to data, distribution and compute, as well as the ability to offer competing models on their platforms and still take a cut.
- On the other hand, ever-smarter open-source models may be good enough for most tasks. GLM 5.2, an open-source model from China's Z.ai, has become a viral hit in the US after rocketing up the leaderboards over the past couple of weeks. It scores 51 in Artificial Analysis's Intelligence Index, not

far behind Claude's (currently restricted) Fable 5 and OpenAI's GPT-5.5, yet costs just 48 cents per task in the index, versus $2.75 and $1.03 respectively.

Price is becoming a battleground as enterprise customers begin to balk at the true costs of rolling AI out at scale.

- Yes, cost per token, the building block of AI, is falling by between nine and 900 times a year, depending on the task, according to Epoch AI. But that is being offset by ever more compute-hungry agentic applications – as well as by well-meaning initiatives to encourage employees to be creative that have incentivised budget-busting "token-maxxing".
- The economics of AI suddenly look less attractive once token usage is no longer being subsidised by investors. Anthropic has changed from a flat fee to a usage-based model, and OpenAI's Altman said last month that cost had come out of the blue to become a "huge issue" for customers this year. Both are reported to be considering price cuts that could erode their profit margins.

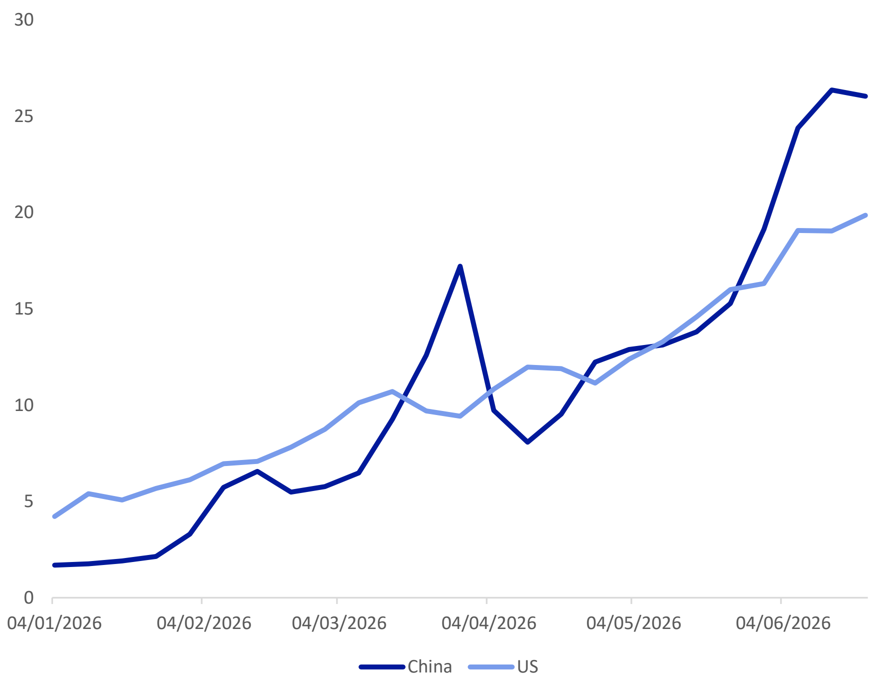
Figure 3: Artificial Analysis Intelligence Index of selected AI models vs cost per tasks



Source : Artificial Analysis Intelligence Index. "Fallback" mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

- OpenRouter data showing weekly token usage of the top models on its platform indicates that the number of tokens processed by US models has risen fivefold this year to 20 trillion, convincingly outpaced by a thirteenfold increase by Chinese models to 26 trillion. The platform allows customers to access hundreds of LLMs from different providers through a single account.

Figure 4: The number of tokens processed by OpenRouter using Chinese models has overtaken tokens processed using US models this year (trillion)



Source : OpenRouter, Deutsche Bank Research

- Amazon, Walmart and Cisco are among companies that have reined in token use. Uber burned through its AI budget for the year in four months and has now capped spending, with its Chief Operating Officer saying that "the link is not there yet" between the use of Claude Code and customer innovation.
- Demand is growing for new model architectures that can divert queries to the most efficient model for any given prompt, such as OpenRouter's new Fusion plugin.

2. More diverse funding sources – and more spending

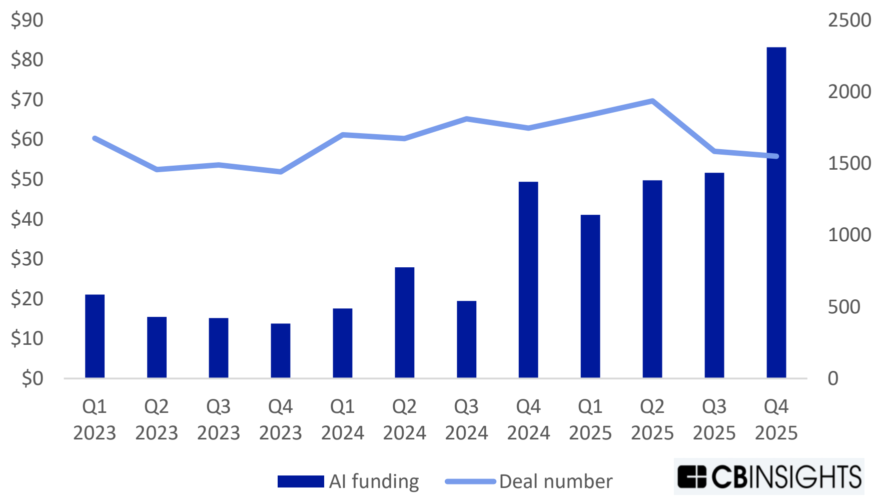
AI is an arms race for compute, pitting startups against established hyperscalers.

- Specialist model-makers are on the back foot compared with a company like Alphabet parent Google that has ample cloud compute facilities and is generating roughly $50bn in cash from operations every quarter that it can use to buy even more.
- Private companies cannot tap the same pool of investors forever. Abundant public market funding would break the cycle of relying on the piecemeal multi-billion-dollar investments and complicated financing arrangements among the leading firms of the past few months.

To be sure, this has not been a bad time to be raising private money.

- Three lean years for venture investment reversed last year as investors injected $425bn, an increase of almost a third from the previous year, into more than 24,000 companies, Crunchbase data show. AI funding was particularly healthy, exceeding $225bn globally, albeit concentrated in few companies, according to CB Insights.

Figure 5: Global AI funding surged in recent quarters, albeit concentrated in fewer companies: quarterly equity funding ($bn) and deal count



Source : CB Insights, Deutsche Bank Research

- Anthropic said at the end of May that it raised $65bn at a post-money valuation of $965bn, more than double its valuation in February. In one leap, it overtook OpenAI, which raised $122bn at an $852bn post-money valuation in March. If either achieved the mooted market value of $1trn, it would be comfortably within the world's top 20 most valuable companies.

But growth does not come cheap.

- OpenAI told investors in February that it was targeting some $600bn in infrastructure investments by 2030, less than half of its earlier estimate of $1.4tn amid concerns about whether future revenue could justify the outlay. Its emphasis has changed from building its own data centres to just renting them.
- OpenAI's funding round in March was anchored by long-term partners Amazon, Nvidia and SoftBank. Amazon will invest $50bn in OpenAI in return for OpenAI using 2GW of its Trainium chip capacity and cooperating on new joint products.
- Long-term shareholder and partner Microsoft still took part even though the two have grown more distant and unwound some exclusive arrangements in April. The startup also took investments from at least 19 institutional investors and funds, as well as some retail investors via banks and exchange-traded funds, and expanded a revolving credit facility to about $4.7bn.
- Meanwhile, Anthropic's latest round looped in at least two dozen investors and included $15bn of previously committed investments from hyperscalers including $5bn from Amazon – after the startup agreed to buy up to 5GW of compute capacity from it and a similar amount from Google and Broadcom.
- In a sign of just how complicated the AI funding org chart is becoming, Google, which already reportedly owned a 14 percent stake in Anthropic

worth $3bn, pledged to invest another $40bn in it in April and more recently was said by Bloomberg to have agreed to backstop lease payments worth $35bn on five data centres.

IPOs would not just unlock new equity finance.

- Public listings would give the companies and their backers leverage capacity: the ability to raise debt against liquid, transparently valued collateral. Those funds could then be recycled into further expenditure on data centres, acquisitions and other ventures.
- A publicly funded OpenAI or Anthropic would likely not just build bigger models but also seek to become the default foundational platform layer similar to what Windows was for PCs and to Amazon's AWS for the cloud. They could use larger war chests to expand their API services, developer tools and enterprise partnerships, commoditising the layers where rivals (and perhaps some customers) operate.

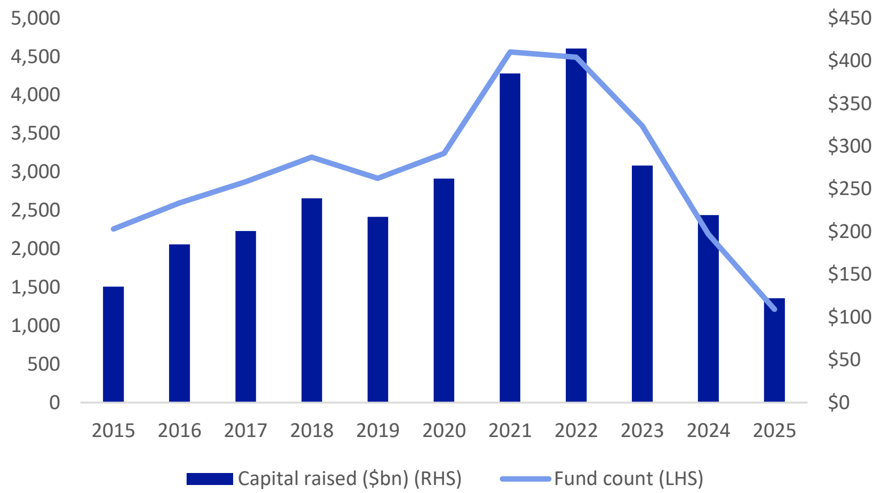
Financial flexibility matters for all of the players in the land grab for capacity in the AI boom.

- Capital expenditure by hyperscalers looks set to rise by two-thirds to more than $750bn this year. Even cash-rich Google turned to the debt markets this year, raising almost $32bn in less than 24 hours in February, then raising $84.75bn in the biggest equity capital markets transaction ever on June 1. Nvidia sold $25bn of high-grade bonds in mid-June.
- And smaller companies are turning to the public markets, most recently with South Korean memory chip maker SK Hynix announcing that it is planning to raise more than 45 trillion won, or almost $30bn, in a landmark US listing.

An opening of the IPO floodgates after a five-year drought would give companies the currency they need to embark on a spending spree.

- It could even break the inertia of an upstream drought in fundraising for early investments by unblocking trapped liquidity. Global venture capital fundraising fell last year to $122 billion, little more than a quarter of the peak in 2021 and 2022, as investors shied away from illiquid investments in favour of less risky strategies, PitchBook data show.

Figure 6: Global private-equity fundraising dropped significantly after 2022, trapping upstream liquidity



Source : PitchBook, Deutsche Bank Research. Data as of Dec 31, 2025

- That could mean a surge in mergers and acquisitions, turning other private companies into targets and giving an exit route to impatient investors. This year has already seen more than 5,000 technology acquisitions or investments pending or completed with a value of about $850bn, more than double the level for the same period last year, Bloomberg data show.
- Elevated valuations and regulatory hurdles for the biggest companies would put them out of reach of all but a tiny number of strategic buyers (the hyperscalers) or sovereigns. An alternative approach has been "acquihires", pioneered by Microsoft two years ago when it hired most of the staff of Inflection, including co-founder Mustafa Suleyman, and licensed the use of its AI models for $650 million.
- But smaller companies have a much greater chance of being acquired.
- Most recently, US chipmaker Onemi agreed to buy Synaptics, which makes semiconductors for smart devices, for about $6.2bn in stock late last week. Indeed, Bending Spoons, which buys up struggling software companies, including AOL and Vimeo, itself filed for an IPO last week. The Italian company aims to raise as much as $1.6bn at a $19bn valuation early this month.

Last but not least, IPOs will give companies a potent new weapon in the battle for talent: liquid, publicly traded stock options.

- For top-tier researchers and engineers, this would be a far more tangible and immediate form of wealth than illiquid shares in a private rival that might seek to lure them away.
- Such a liquidity shock could force private competitors into responding with ever larger funding rounds at higher valuations on the promise of "jam tomorrow" for star employees – or into going public themselves. They would also have to be on high alert to prevent their existing core teams from being poached by a newly public OpenAI or Anthropic armed with a trillion-dollar market capitalisation.
- Meta's Mark Zuckerberg sparked a bidding war for top AI researchers last

year, offering compensation packages worth hundreds of millions of dollars in some cases. OpenAI and Anthropic were the most recent players to fish in the small pond of genuine AI experts a couple of weeks ago, hooking two legends of the industry. Shares of Google, where they worked, fell more than 5 percent on the day.

3. Greater accountability for their actions – and for the industry

The AI industry has lost its lustre in recent months.

- Warm initial perceptions of it as a slightly magical composer of wedding speeches have faded and may vanish once Mom and Pop own shares in pure-play AI companies for the first time. Those companies risk becoming a lightning rod for broader fears about the impact of AI on trust, jobs and vulnerable industries.

At the most basic level, public companies are held to higher standards around consistency and accountability.

- For example, OpenAI has attracted criticism for reversing its earlier opposition to advertising in an attempt to create a more sustainable business model. Critics have said conversations could be biased by sales imperatives.

Beyond that, OpenAI and Anthropic have already had a taste of the political challenges in store as public companies.

- Anthropic was blacklisted by the US Department of Defense in February for refusing to let Claude be used for autonomous weapons and mass surveillance. The dispute cost Anthropic a $200m contract and is now in the courts. OpenAI was then criticised from the other side after it struck a deal to replace Anthropic. Altman later described the move as being "complex but the right decision with extremely difficult brand consequences and very negative PR for us in the short term".
- More recently, the US Department of Commerce issued an export control directive late on June 12 to suspend all access by foreign nationals to Anthropic's Claude Mythos 5 and less powerful Fable 5 models because of security concerns.
- Although it had to temporarily disable access to all customers to ensure compliance, its reputation for having the most powerful models will have benefitted (see our report AI twist in the tale for Anthropic's Fable and Mythos). The silver lining may eventually eclipse the cloud amid a branding boost that money can't buy. However, alternatives to US models are also likely to benefit from concerns that access could be switched off without warning.

Most broadly, the makers of the best-known chatbots may pay a price for being seen as proxies for all of AI when controversy hits the industry.

- Hot topics include cybersecurity. The leaders of the Five Eyes cybersecurity agencies of the US, UK, Australia, Canada and New Zealand warned in a joint statement last week that frontier models are likely to supercharge cyber threats: "The timeline is not years, it is months".

The biggest underlying popular concern is what AI will mean for jobs.

- The leaders of both Anthropic and OpenAI have moderated their predictions of severe disruption in recent weeks, with Amodei pivoting to say automation would dramatically increase productivity and Altman saying the impact on jobs has been less than he expected.

- One challenge is that AI has become a scapegoat for job cuts at companies ranging from Block (4,000 employees, or nearly half the workforce) to Atlassian (10 percent of roles) and even British American Tobacco (with AI cited as one of a number of factors in a planned one-fifth reduction in its workforce).
- However, the tide has begun to turn on predictions of catastrophic job losses. A slump in entry-level graduate jobs may have been more due to rising interest rates and remote work than AI – and may be reversing. And business leaders like Amazon Web Services' CEO Matt Garman have said eliminating the pipeline of talent makes no sense.

Nowhere has the AI backlash become more focused than over data centres.

- As fast as these "AI factories" are mushrooming around the US and the world, so are concerns – both justified and unjustified – about them overwhelming the electricity grid, forcing up power prices, and sucking rivers dry for water to cool their chips.
- How the leading model makers, hyperscalers and infrastructure companies handle the issue could determine whether it remains contained or unleashes moratoriums, regulation or swingeing taxes that put the entire boom at risk.
- In the US, home to half of the world's data centres, opposition to "AI factories" is becoming a rare area of bipartisan consensus, uniting prominent Democratic Representative Alexandria Ocasio-Cortez and independent Senator Bernie Sanders on the left, and populist strategist Steve Bannon on the right.
- Maine recently passed a moratorium on larger data centres while, across the Atlantic, data-centre hub Ireland has introduced a "Bring Your Own Power" policy that requires new facilities to generate their own electricity or secure renewable energy contracts.

Data centers are – literally – close to home for voters.

- That makes them ripe fodder for targeted lobbying and grassroots campaigns in a way that vague fears about long-term job losses are not. The US midterm elections in November will be an important litmus test of the political consequences of local pressure.
- A Gallup poll in May found that 71 percent of Americans would oppose a data centre being built in their area. At least 75 projects worth a record $130bn were blocked or delayed in the first quarter, according to Data Center Watch.
- Tech companies have done their bit to respond. Microsoft, for example, said in its most recent data centre announcement last week that it would create thousands of jobs at the facility in Pecos, Texas; pay to ensure it does not cause a rise in electricity prices; and replenish more water than it uses. "We are approaching this project as a new neighbour," it said.

The IPOs of the biggest AI companies could also have repercussions in the geopolitical contest for technological supremacy, raising their status to that of strategic national assets.

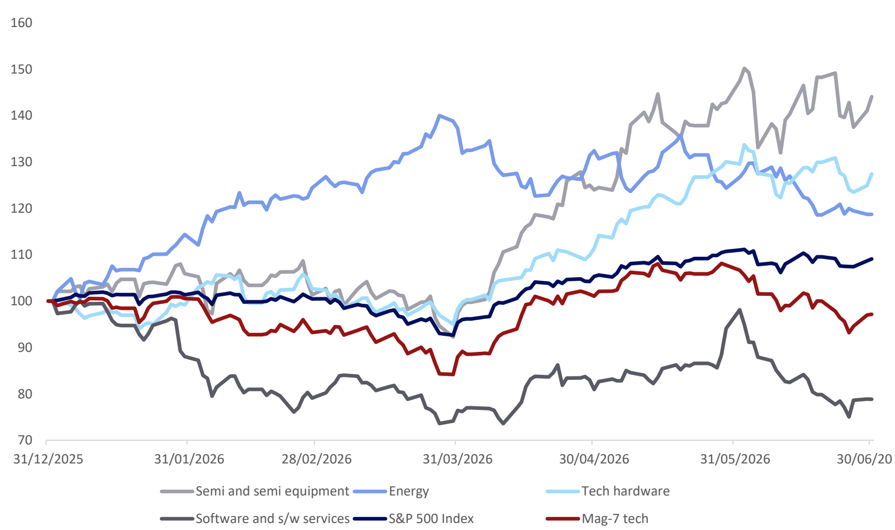
- That could encourage other powers to accelerate support for their own national champions, like China's "AI tigers" and France's Mistral, to avoid becoming dependent on US-controlled foundation models. Other countries may find themselves having to choose to align with one centre of power or the other.

4. New era for comparing valuations to peers – and the past

Part of the anxiety about an AI bubble is the lack of pure-play, public, investable companies to act as a benchmark. A listed OpenAI or Anthropic would fill a huge gap.

- Sentiment about AI has driven much of the market since the launch of ChatGPT, with anticipated gains from investment and productivity helping to explain the two-thirds rise in the S&P 500 Index during that period.
- That rise has been bumpy, soaring and swooning from week to week, as one area after another, such as software, semiconductors or infrastructure companies, has come into favour and then fallen out again. Investors have crowded into trades on a concentrated and shifting list of potential AI enablers and adopters, and away from prospective casualties.
- To give one example, the Magnificent Seven biggest technology stocks have whipsawed over the past month amid concerns about how much of their AI capital expenditure will pay off, shedding as much as 12 percent by late last week. They have tripled over the past three-and-a-half years to account for almost a third of the S&P 500.

Figure 7: S&P 500 Index, Magnificent Seven tech stocks and other select AI-related sectors performance YTD to June 30 (indexed to 100)



Source : Bloomberg Finance LP, Deutsche Bank Research

- That makes investing in AI right now feel a bit like going to a horse race and having to bet on the jockeys and the condition of the track because the horses themselves are unavailable.

Of course, there is no guarantee that any filing ends up in a successful IPO.

- The San Francisco-based co-working company WeWork was valued at

$47bn in early 2019 but failed to gain traction when it filed its prospectus in August that year. It eventually went public in 2021 at a heavily reduced valuation and ended up filing for bankruptcy in late 2023.

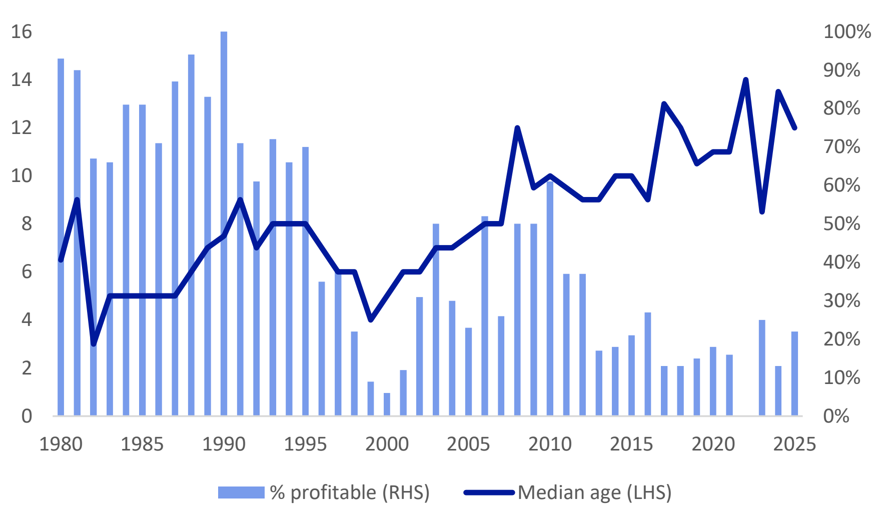
Another way that major IPOs could remove some distortions is by acting as a release valve for private markets, enabling long-awaited exits from investors.

- Private-equity companies are sitting on a record backlog of unsold companies, thanks in part to geopolitical disruption and fears of a "SaaSpocalypse", according to data from Pitchbook and others. The median age of still-held assets continued to trend upwards from three years in 2022 to four years in 2025 as exit activity stalled.

Despite concerns that the boom turns to bust, there are differences in the IPO markets from the height of the dot-com bubble.

- Overall valuations are clearly elevated by historical standards, with the Shiller Cyclically Adjusted Price-Earnings (CAPE) ratio for the S&P 500 recently surpassing 40, nearing the record 44 it reached at the peak of the dot-com bubble.
- However, the IPOs coming to market are more mature and more profitable.
- Back in 1999 to 2000, the median age of venture-capital-backed tech startups going public shrank to about 4.5 years, according to University of Florida data. Since then, the median age has been relatively stable at ten-and-a-half years and was 12 years last year.
- Only 6 percent of those tech startups going public at the height of the dot-com bubble were profitable, compared with 22 percent since 2010, and they had a median opening price-to-sales ratio of over 40, four times higher than the average since 2010.

Figure 8: Median age and profitability of technology IPOs (1980-2025)



Source : University of Florida, Jay Ritter, Deutsche Bank Research

Perhaps the greatest insurance against a bubble is the fixation of today's senior investment managers on the dot-com bust that scarred them in their formative years in the market.

- For them, mega IPOs bring back memories of the IPO of Netscape in August 1995, widely regarded as the start of the dot-com bubble.However, there is just as important a lesson in how long it took for that bubble to go bad despite prophesies of doom and gloom right from day one.
- Bloomberg published an article two days after the IPO on August 9, 1995, entitled "Why investors say Netscape shows a market poised for a fall" and warns "Be worried, very worried". "Netscape was a little insane," says a trader quoted in the article, after the stock doubled on its first day of trading, giving the lossmaking company a $2.9bn market value. "It certainly hints at more frothiness in the market and that's usually the sign of the beginning of the end of a bull market".
- Yet in the end, the Nasdaq shot up from 1,005 on the day of Netscape's IPO, not peaking until almost five years later, at over 5,000 on March 10, 2000. It was only then that it dramatically fell back, reaching a trough of about 1,100 in October 2002.
- As for Netscape, it was bought by America Online (AOL) for $10bn in 1999, and then vanished in the fallout from AOL's record-breaking $124bn purchase of Time Warner in 2001.

Yet Amazon, another star of the times, shows the power of patience – and picking a winner.

- Adjusting for stock splits, the shares debuted at 7.5 cents in the company's $54m IPO on May 15, 1997. After the roller coaster of the dot-com bubble and crash, the shares took a decade to recapture their early peak – and then continued to rise, reaching almost $275 a month ago.
- Yet on the day of its IPO, when the stock rose 31 percent, a Pennsylvania-based fund manager told Bloomberg: "I personally am going to stay away from it. I don't think it's been proven that one can make a lot of money over the web."

Recent relevant research

Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh (June 9, 2026)

ChatIPO: Quick take on OpenAI's record-breaking IPO plans (May 21, 2026)

Will Claude's AI finance agents change your life? (May 6, 2026)

AI, Iran and the Gulf 101 (April 13, 2026)

What AI says about AI eating itself and the world… (Feb 16, 2026)

AI Q&A: Why have market vibes suddenly shifted on AI? (Feb 16, 2026)

Three AI themes for 2026: the honeymoon is over (Jan 20, 2026)

Appendix 1

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Deutsche Bank
Research

Q2 Tech Performance Review: Memory Wins, Chips Dominate

July 1, 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

Deutsche Bank
Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.


Apple increases MacBook and iPad prices by 20%
Micron Soars After AI-Fueled Forecast Shatters Estimates
Wall Street ends lower on semiconductor selloff as AI spending concerns mount
Chinese AI, chip firms are driving an onshore IPO rebound
Alibaba sues Pentagon over inclusion on Chinese military blacklist
Anthropic Halts Access to Top AI Models After U.S. Ban on Foreign Use
Apple seeks to buy memory chips from blacklisted Chinese company

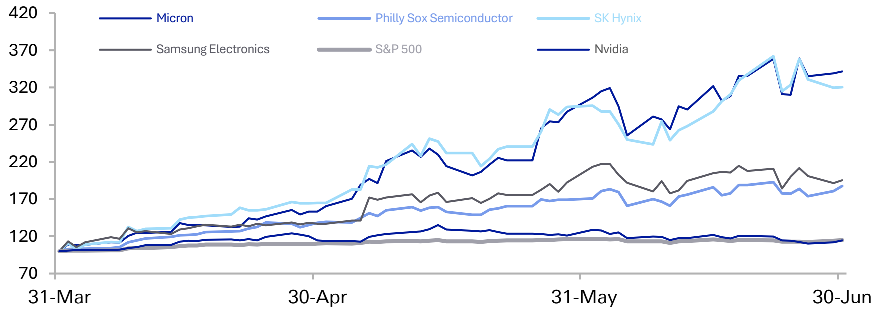
SUMMARY: Memory wins, chips dominate

In Q2, equities ran largely uncorrelated from events in Iran, driven by a tech rally despite ongoing concerns over the war. Strong earnings growth fuelled the Mag 7's rise in April (+15%) and May (+7%). Then last month, risk assets sold off despite the June 15 US-Iran peace deal, as worries over rising Fed hawkishness, the memory shortage's inflationary impact, and doubts over AI demand sustainability triggered a sell-off in AI stocks. Still, despite the meltdown over the last few weeks, tech delivered a remarkable run this quarter, with AI excitement significantly outweighing macro and geopolitical concerns around the war. The Mag 7 ended up +12% on the quarter.

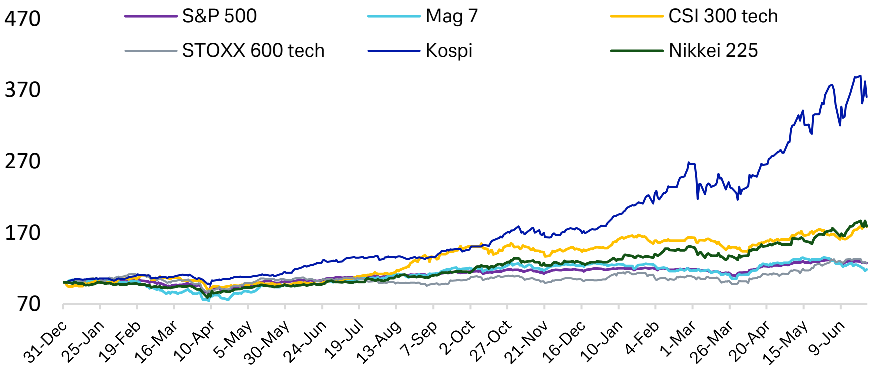
The spotlight has been on memory chipmakers in particular, which drove most of tech's outperformance – and its June selloff. The ongoing memory shortage has led to rising returns for Micron (+242% q/q), SK Hynix (+228% q/q) and Samsung Electronics (+100% q/q), all of which have now joined the $trn dollar market-cap club. Asian equities followed, with the Nikkei 225 (+37% q/q) and Kospi (+67% q/q) outperforming their Western counterparts.

What will Q3 bring? Micron's results were probably enough to restore confidence in AI-driven demand, particularly in data-centre memory. But volatility under a more hawkish Fed will likely persist, and BIS has warned of AI exuberance risk ending in a lengthy investment bust. In crypto, payments and fintech, expect continued pressure as investors rotate into AI. Tech should remain the underlying volatility driver, with the US most exposed.

Memory chipmakers have had geometric rises…



… pushing up Asian equities in particular

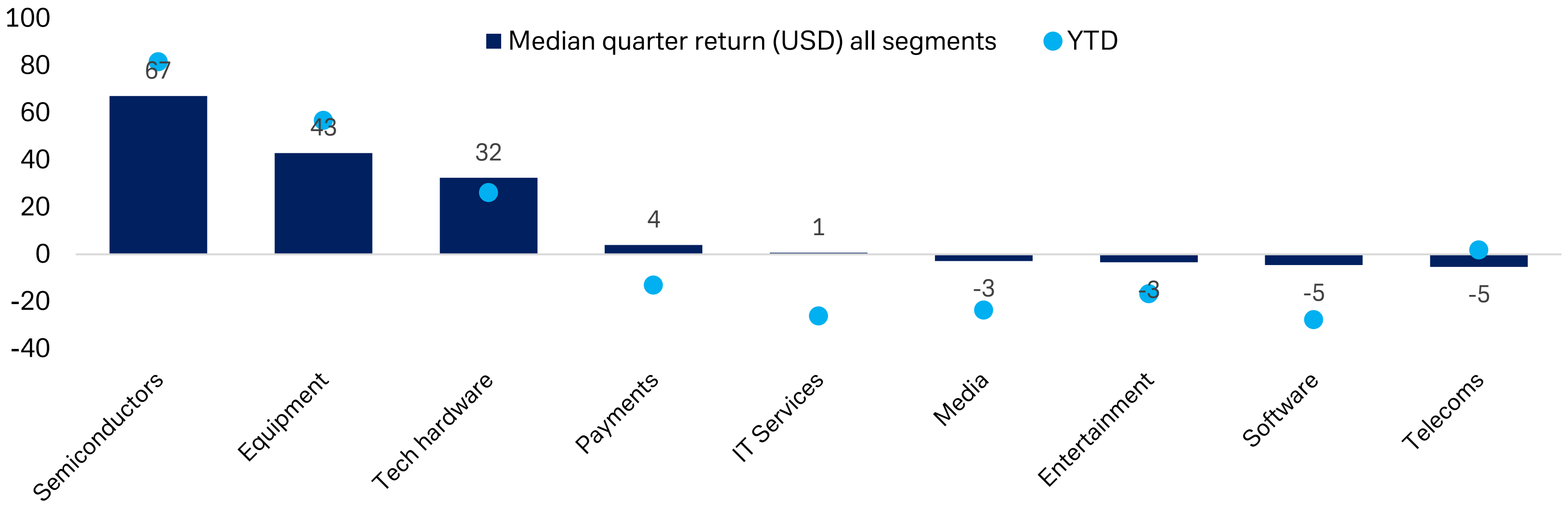


Sources: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026. Returns in local currency.

1. GLOBAL INDUSTRY BREAKDOWN

Semiconductors and chip-related stocks are tech's clear group-stage winners in Q2 and YTD, led by Chinese and other Asian companies like GigaDevices (+248% q/q) and ACM Research (+210% q/q). Outside of semis, equipment and tech hardware, the rest of the field struggled. Payments and IT services continue to underperform in both AI risk-on and risk-off environments, with this quarter also weighed down by Accenture's (-37% q/q) weak FQ326 earnings in June. Software remains the worst performer, as AI disruption continues to hit companies including Intuit (-39% q/q), Horizon Robotics (-38% q/q) and Sensetime (-27% q/q).

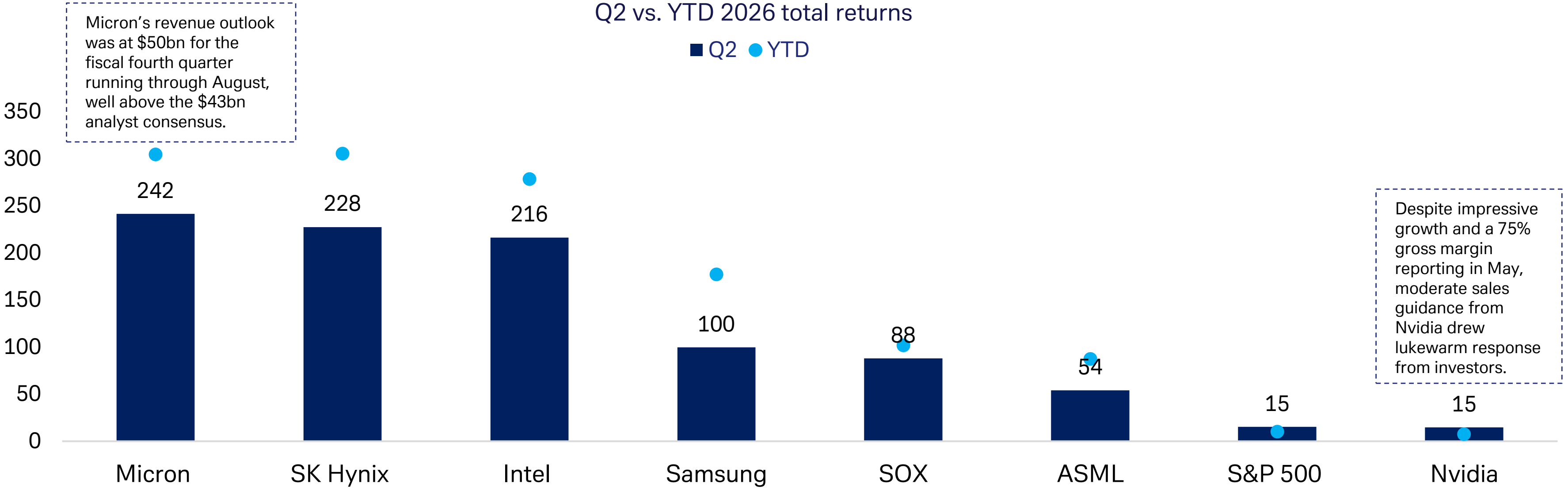
Median total returns for tech sectors over Q2 vs. YTD returns



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

2. SEMICONDUCTORS

Memory chipmakers will continue to benefit from persistent industry imbalances, with Micron and SK Hynix the standout performers. Micron's earnings beat ignited hopes about AI-fuelled growth and helped push back against bubble fears. SK Hynix has also confirmed it will seek over $29bn via the issue of ADRs on the Nasdaq. Intel (+216% q/q) surged on a strong earnings beat and accelerating foundry momentum, with reports that Apple is evaluating Intel's 18A process and Google has reportedly placed orders for more than 3mn TPUs for 2028. Elsewhere, the Philadelphia Semiconductor Index (+88% q/q) posted its best quarterly performance since the index began in the early 1990s.

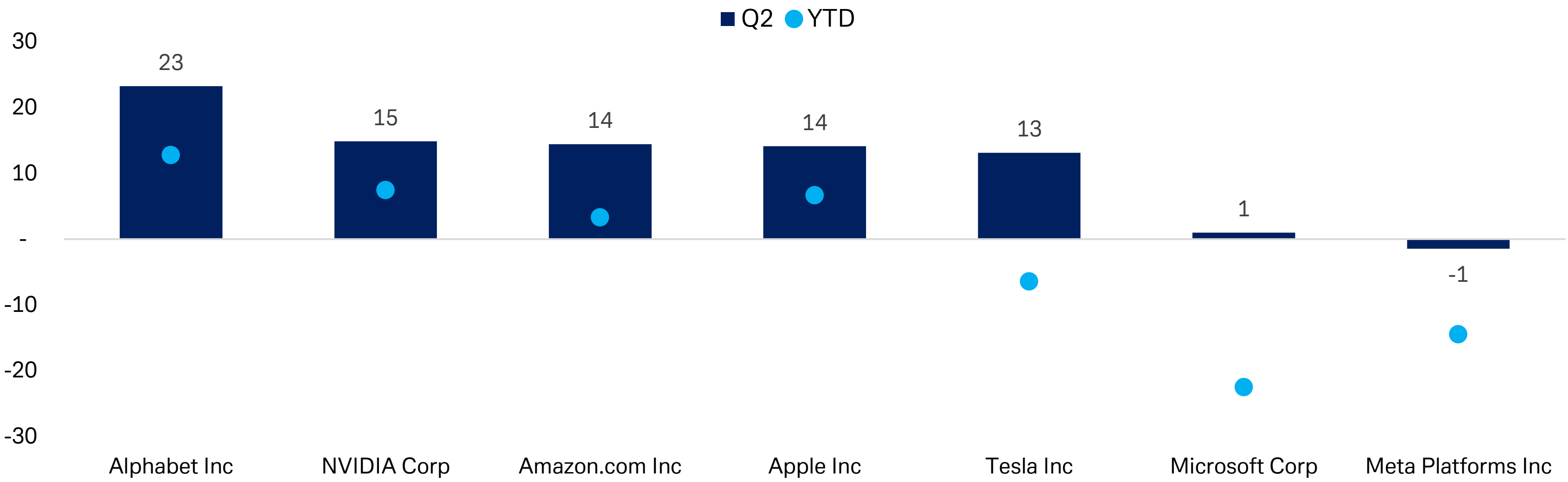


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026. Returns in local currency.

3. MAGNIFICENT 7

The Mag 7 started Q2 on a high, driven by strong earnings growth against a backdrop of hyperscaler capex expected to exceed $700bn this year. In June, elevated large-cap tech positioning combined with AI concerns temporarily dragged the index into correction territory, although the group recovered by quarter-end (+12% q/q). Microsoft (+1% q/q) continues to lag YTD, as dual concerns over its ability to compete on AI spending and AI disruption risk persist.

Mag 7 total returns

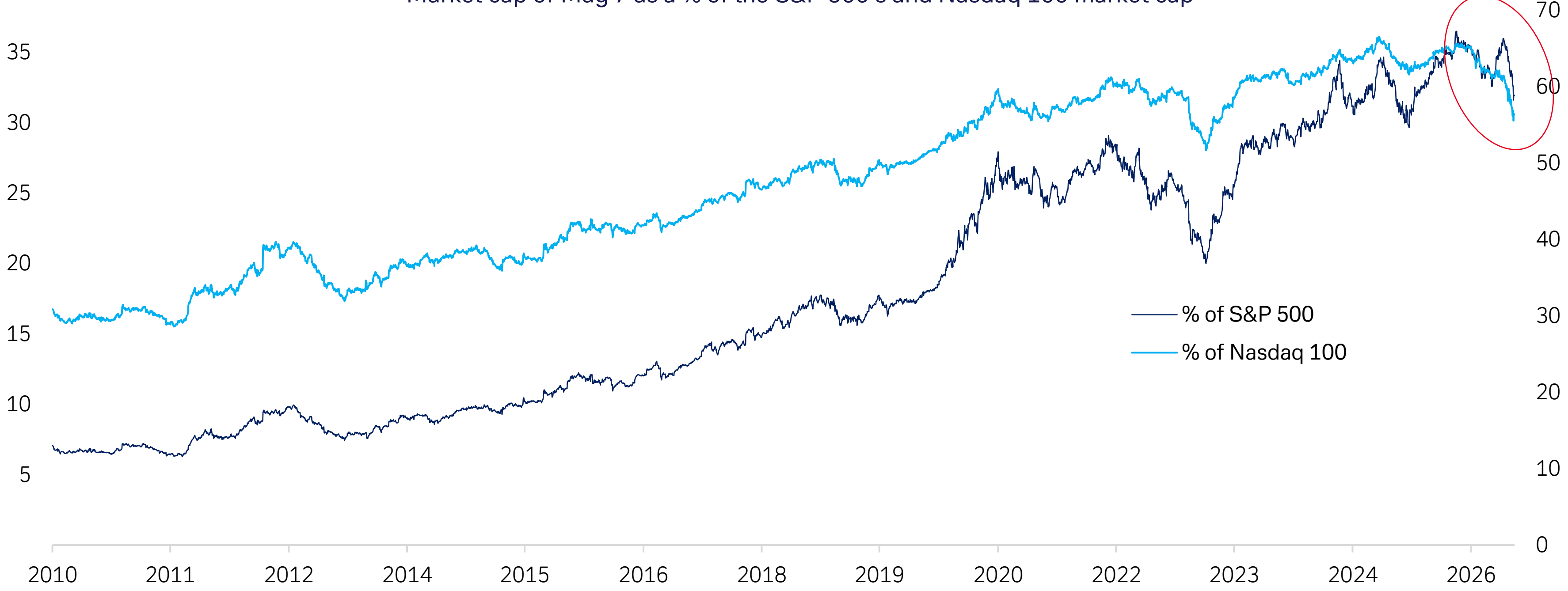


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

3.1 MAGNIFICENT 7

Large-cap tech stocks are down significantly, dragging the S&P 500 down modestly, even as two-thirds of the overall stocks and industries are up, alongside small- and mid-caps. Still, according to our strategists, there is an ongoing rotation away from large-cap tech and into other sectors and regions that have lagged.

Market cap of Mag 7 as a % of the S&P 500's and Nasdaq 100 market cap

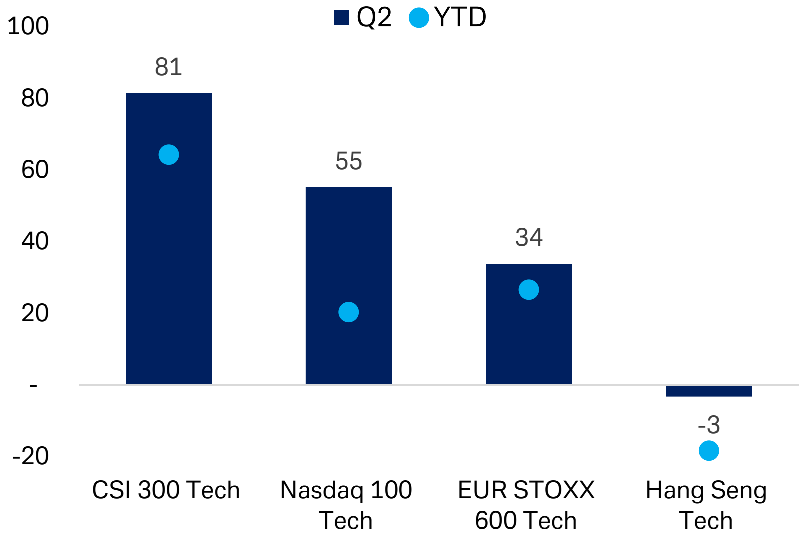


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

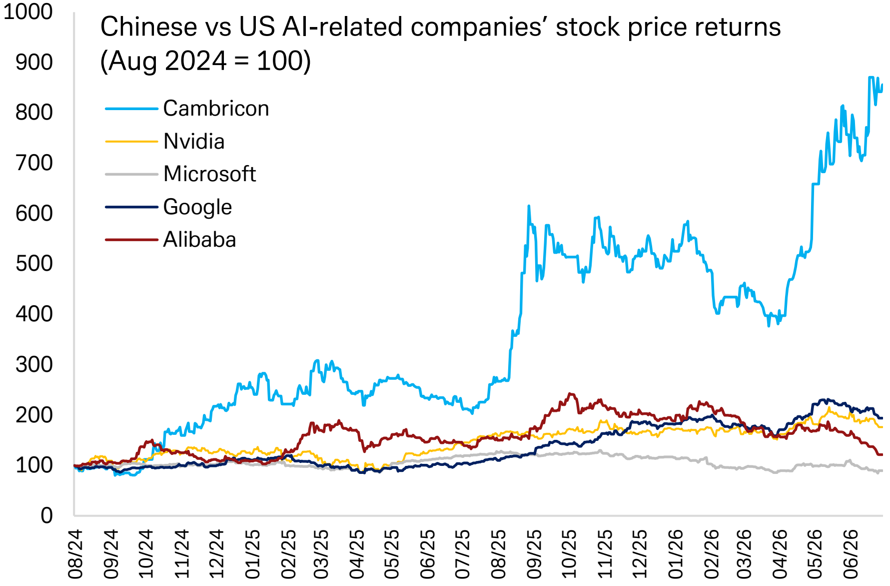
4. REGIONAL BREAKDOWN

Chinese stocks are leading YTD and quarterly gains, buoyed by local memory chipmakers, while off-shore HK-listed tech continues to suffer from old-guard heavyweights like Xiaomi (-32% q/q), Alibaba (-21% q/q), and Tencent (-10% q/q). Elsewhere, European tech equities saw some growth but suffered substantial outflows during the war, with software companies particularly affected over the last six months.

Top global tech indices – Q2 vs. YTD total returns



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated June 30, 2026.

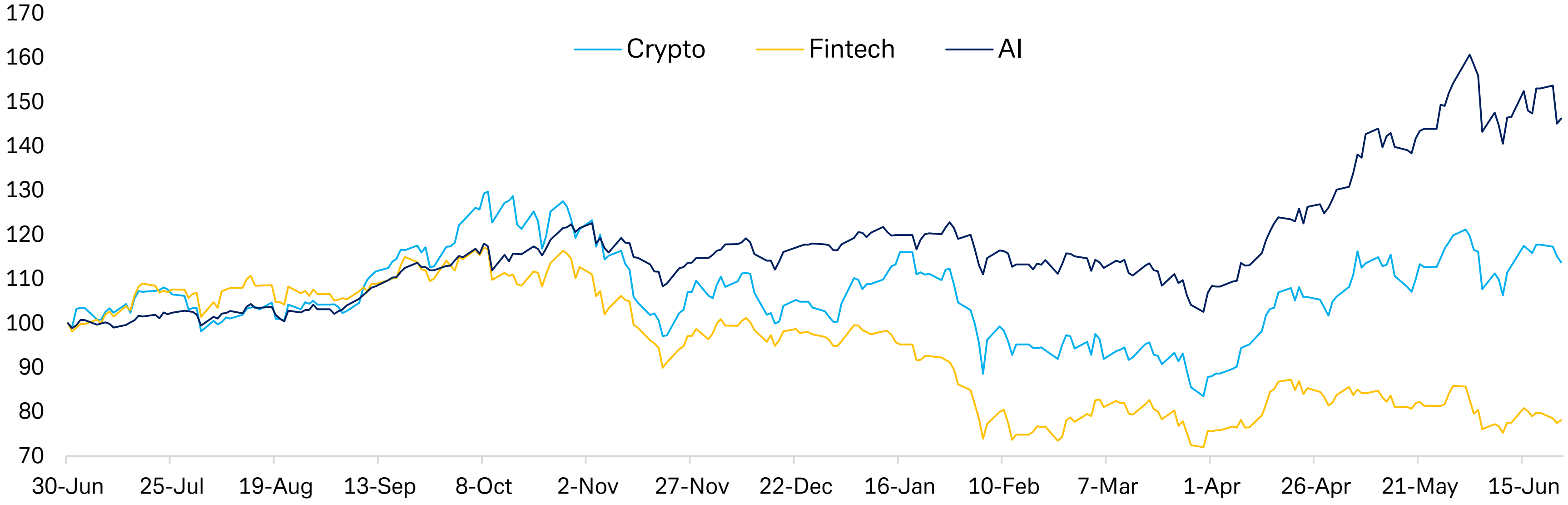


Source: Deutsche Bank Research, Bloomberg Finance LP. Indexed to 100. Updated June 30, 2026.

5. CRYPTO, AI & FINTECH

Crypto extended its struggles in Q2, falling below $60k in June as a renewed sell-off was driven by a more hawkish Fed outlook, record institutional ETF outflows, uncertainty over the CLARITY ACT, and rotation of risk capital into AI (see here). Our Fintech and blockchain ETF remains the worst-performing segment YTD (-17%), weighed down by ongoing pain at companies like Canton Strategic Holdings (-14%) and Circle (-34%). Fintech sentiment remains weak more broadly, with the steady demand deterioration coupled with long-term fears around AI disintermediation.

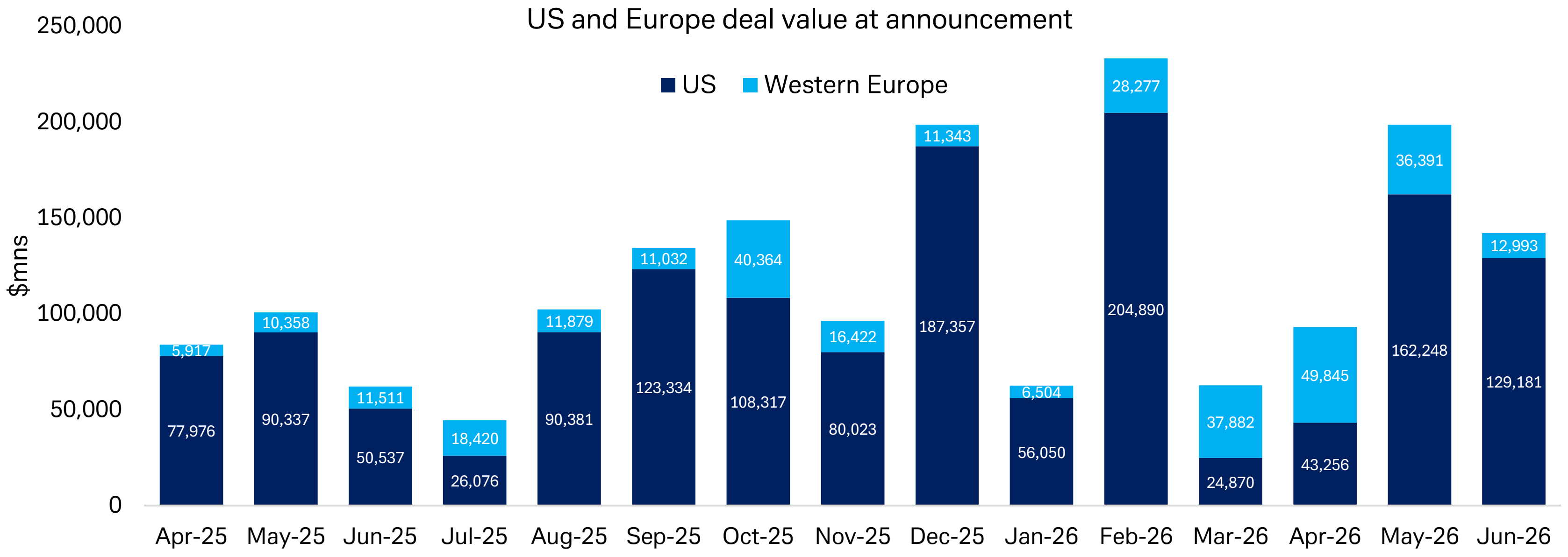
Crypto, AI & Fintech performance rebased to June 2025



Source: Deutsche Bank Research, Bloomberg Finance LP. Data updated June 30. The following are based on AIQ ETF, BLOK US Equity ETF, and ARKF US Equity ETF.

6. M&A

After a halt in April, Q2 was a blockbuster quarter for mega-cap AI IPOs, with Anthropic ($65bn) and OpenAI ($852bn) filings. Beyond IPOs, semiconductors M&A led the way, including Analog Devices' proposal to acquire Empower Semiconductor for $1.5bn in May and ONSemi's $6.8bn acquisition of Synaptics Inc, announced on 25 June.



Source: Deutsche Bank Research, Dealogic. Values up to June 29.

DB select forecasts

Sector	Company	Reporting Currency	Listing Currency	Current Price	Target Price	Rec.	M. Cap	P/E (x) 2026E	P/E (x) 2027E	EV/EBITDA (x) 2026E	EV/EBITDA (x) 2027E
IDM	NXP	USD	USD	277.0	315.0	Buy	61.4	18.3x	15.0x	12.6x	10.6x
	Intel	USD	USD	128.3	100.0	Hold	565.9	111.7x	77.7x	34.5x	27.3x
	ST Microelectronics	USD	EUR	62.6	75.0	Buy	57.0	51.1x	24.1x	18.5x	11.8x
	Infineon	EUR	EUR	78.3	90.0	Buy	102.3	40.3x	30.9x	18.6x	14.8x
	Texas Instruments	USD	USD	285.4	240.0	Hold	227.9	35.7x	29.1x	24.2x	20.2x
	Renesas*	JPY	JPY	4,799.0	NA	NR	48.7	18.6x	17.5x	17.6x	15.0x
	Analog Devices	USD	USD	386.9	365.0	Hold	165.4	30.2x	26.5x	25.5x	22.1x
	onsemi	USD	USD	90.7	110.0	Buy	31.2	28.8x	16.8x	18.3x	12.4x
	Microchip Technology*	USD	USD	87.9	NA	NR	41.8	32.0x	23.2x	23.0x	17.1x
Semi Equip	ASML	EUR	EUR	1,578.2	1600.0	Buy	612.6	48.2x	32.6x	37.9x	25.9x
	Applied Materials	USD	USD	626.8	550.0	Buy	436.7	48.2x	35.3x	40.8x	30.1x
	KLA	USD	USD	248.6	1750.0	Hold	285.0	56.9x	46.3x	46.7x	38.6x
	Lam Research	USD	USD	379.1	325.0	Buy	416.0	55.5x	42.5x	45.8x	35.4x
	TEL*	JPY	JPY	74,000	NA	NR	187.7	49.9x	38.6x	36.0x	27.4x
	Screen*	JPY	JPY	16,665	NA	NR	17.2	28.9x	23.2x	18.2x	14.9x
	ASM Int'l	EUR	EUR	948	925.0	Buy	46.8	44.5x	36.7x	29.3x	24.0x
	VAT Group	CHF	CHF	669.8	550.0	Hold	21.8	64.8x	46.0x	46.1x	33.7x
	BE Semiconductor	EUR	EUR	282.7	275.0	Buy	22.9	66.4x	45.1x	55.1x	38.4x
	Technoprobe	EUR	EUR	33.0	42.0	Buy	21.6	74.0x	44.4x	45.4x	26.6x
Fabless	AMD	USD	USD	521.6	365.0	Hold	746.3	70.1x	43.5x	56.8x	34.9x
	MediaTek	TWD	TWD	3,910.0	2850.0	Buy	172.5	57.8x	34.7x	44.8x	26.8x
	Nvidia	USD	USD	192.5	255.0	Hold	4,088.5	21.3x	15.8x	17.8x	12.3x
	Marvell	USD	USD	266.8	240.0	Buy	204.8	67.5x	43.8x	50.6x	33.1x
	Broadcom	USD	USD	365.0	515.0	Buy	1,523.9	27.9x	17.0x	22.0x	13.2x
	Qualcomm	USD	USD	189.4	160.0	Hold	175.2	18.0x	17.7x	14.4x	14.4x
Fab	TSMC	TWD	TWD	2,370.0	2500.0	Buy	1,690.5	24.1x	18.8x	15.8x	12.2x
	UMC*	TWD	TWD	164.0	NA	NR	56.7	35.1x	28.4x	15.9x	12.9x
	SMIC*	USD	HKD	82.3	NA	NR	104.4	81.5x	59.7x	24.6x	20.1x
	Tower Semiconductor*	USD	USD	249.9	NA	NR	24.7	70.8x	41.1x	38.4x	25.6x
IP/EDA Comp	arm	USD	USD	334.3	205.0	Hold	313.3	162.2x	123.3x	133.1x	105.0x
	Synopsys	USD	USD	454.3	590.0	Buy	76.3	29.8x	25.2x	17.4x	15.8x
	Cadence	USD	USD	377.3	375.0	Buy	91.3	47.3x	39.1x	36.2x	30.6x
IT Hardware	Logitech	USD	CHF	78.8	80	Hold	13.7	17.3x	15.6x	14.3x	13.5x
	Apple*	USD	USD	283.8	NA	NR	3,657.4	31.6x	28.6x	23.6x	21.6x
	HP Inc*	USD	USD	22.9	NA	NR	18.4	7.6x	7.5x	5.6x	5.5x
Memory	Samsung Electronics*	KRW	KRW	327,500	NA	NR	1,092.6	7.2x	5.2x	4.0x	2.4x
	SK Hynix*	KRW	KRW	2,643,000	NA	NR	1,072.8	8.5x	5.8x	6.1x	3.5x
	Micron	USD	USD	1,132.3	1550	Buy	1,122.2	11.2x	6.8x	8.5x	5.0x
Telecom Equipment	Cisco Systems*	USD	USD	113.8	NA	NR	393.5	25.3x	22.8x	18.8x	17.0x
	Ericsson	SEK	SEK	106.7	100.0	Hold	32.5	14.4x	12.9x	7.5x	6.8x
	Motorola Solutions*	USD	USD	402.9	NA	NR	58.7	23.7x	21.7x	17.2x	15.7x
	Nokia	EUR	EUR	11.4	13.50	Buy	65.4	35.2x	29.6x	20.5x	17.0x
	Ciena Corp*	USD	USD	479.5	NA	NR	59.6	67.6x	45.7x	46.6x	33.5x
	F5 Networks*	USD	USD	394.5	NA	NR	19.5	23.6x	21.8x	16.2x	14.9x
	Keysight*	USD	USD	328.7	NA	NR	49.3	31.4x	27.2x	24.0x	21.2x
	ZTE Corp*	CNY	CNY	35.3	NA	NR	20.3	23.5x	20.7x	16.6x	15.3x
Software Comps	Adobe	USD	USD	202.7	260.0	Hold	77.2	8.2x	7.0x	5.9x	5.1x
	Amadeus	EUR	EUR	52.1	58.0	Hold	25.8	15.3x	13.8x	9.4x	8.3x
	Autodesk	USD	USD	196.3	330.0	Buy	39.9	15.8x	13.6x	12.1x	10.3x
	Cadence Design	USD	USD	377.3	375.0	Buy	103.1	47.3x	39.1x	34.8x	29.5x
	Dassault Systemes	EUR	EUR	17.9	21.0	Hold	26.7	13.7x	13.0x	9.3x	8.3x
	Hexagon	EUR	SEK	81.0	85.0	Hold	22.2	26.6x	23.5x	13.8x	12.7x
	Intuit	USD	USD	267.7	530.0	Buy	70.5	10.6x	9.2x	7.4x	6.1x
	Microsoft	USD	USD	373.0	550.0	Buy	2,784.2	19.0x	17.2x	11.9x	9.8x
	Nemetschek	EUR	EUR	52.6	NA	NA	7.1	19.6x	16.3x	13.6x	10.9x
	Oracle	USD	USD	148.5	300.0	Buy	432.8	18.9x	15.6x	11.6x	8.6x
	PTC	USD	USD	115.7	180.0	Buy	13.2	14.2x	12.9x	10.2x	9.3x
	Sage	GBP	GBp	817.6	900.0	Hold	10.7	14.4x	12.5x	10.4x	8.8x
	Salesforce	USD	USD	158.4	255.0	Buy	125.2	11.3x	10.3x	13.3x	7.8x
	SAP	EUR	EUR	135.9	200.0	Buy	173.9	17.0x	14.3x	10.2x	8.7x
	ServiceNow	USD	USD	98.3	135.0	Buy	102.9	24.2x	19.8x	12.8x	10.5x
	Synopsys	USD	USD	454.3	590.0	Buy	84.5	29.8x	25.1x	16.7x	15.1x
	Workday	USD	USD	124.2	180.0	Buy	33.3	11.7x	9.8x	9.6x	7.8x
Services Comps (SBC A	Accenture	USD	USD	129.0	140.0	Hold	80.3	7.6x	7.1x	4.9x	4.6x
	Capgemini	EUR	EUR	89.7	125.0	Buy	17.7	7.2x	6.5x	4.8x	4.4x
	CGI	CAD	CAD	91.3	NA	NA	14.2	9.7x	8.9x	6.7x	6.1x
	Cognizant	USD	USD	40.0	70.0	Buy	19.2	6.7x	6.1x	3.8x	3.1x
	HCL Technologies	INR	INR	1100.7	NA	NA	31.7	15.5x	14.2x	9.1x	8.3x
	IBM	USD	USD	271.6	NA	NA	254.5	19.2x	17.9x	13.8x	12.7x
	Infosys	INR	INR	1041.2	NA	NA	44.4	13.4x	12.6x	7.9x	7.3x
	TCS	INR	INR	2094.7	NA	NA	79.9	13.9x	12.9x	9.6x	9.2x
	Wipro	INR	INR	175.0	NA	NA	19.5	12.7x	12.1x	7.6x	7.1x

Market cap in EUR bn

Market cap in USD bn

Source: Deutsche Bank Research, Bloomberg Finance LP for non-covered companies. Updated June 29 2026.

Other related research:

- ❑ AI's tightest bottleneck: Memory chips
- ❑ Chips: The Biggest AI Bottleneck
- ❑ Investor Positioning and Flows: Bouncing Above Neutral
- ❑ What We Learned This Week: Stay in Rotation
- ❑ Bitcoin: Tinkerbell Blinks

Appendix 1

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Deutsche Bank
Research

What Q2's emerging themes mean for 2026 and beyond

1 Jul 2026

Luke Templeman | Thematic Strategist
Galina Pozdnyakova | Research Analyst

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
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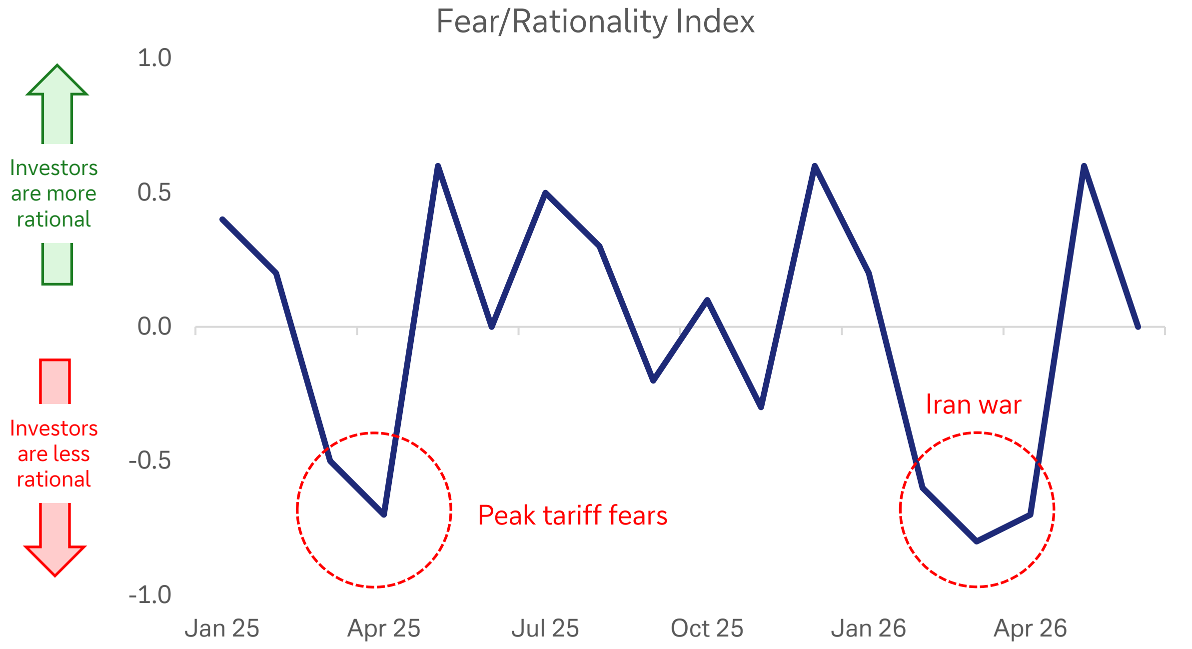
The themes driving markets in Q2 – based on AI analysis of our daily market commentaries

1. The AI revolution
2. US-Iran conflict & Strait of Hormuz blockade
3. Shifting central bank policies & inflation fears
4. Rise of "Middle Powers" & geopolitical realignment
5. US political turmoil & domestic division
6. China’s assertive foreign policy & economic resilience
7. Strains in Trans-Atlantic relations
8. Volatility in commodity markets (beyond oil)
9. Evolving role of Japan in global markets
10. Growing pains in emerging markets

Are investors rational?

As markets began to recover from their Iran war-low, the investor buying was very rational. Now, that equity markets are close to their peak again, 'rationality' has moved to neutral … but not into negative territory.

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



What does it mean to be 'rational'?

Risk-off periods:
The tariff fears in 2025 and then Iran war in 2026 were key buying opportunities in equity markets. Yet, the overwhelming fear exhibited by investors in these periods largely meant they sold off risk assets in an emotional and irrational response.

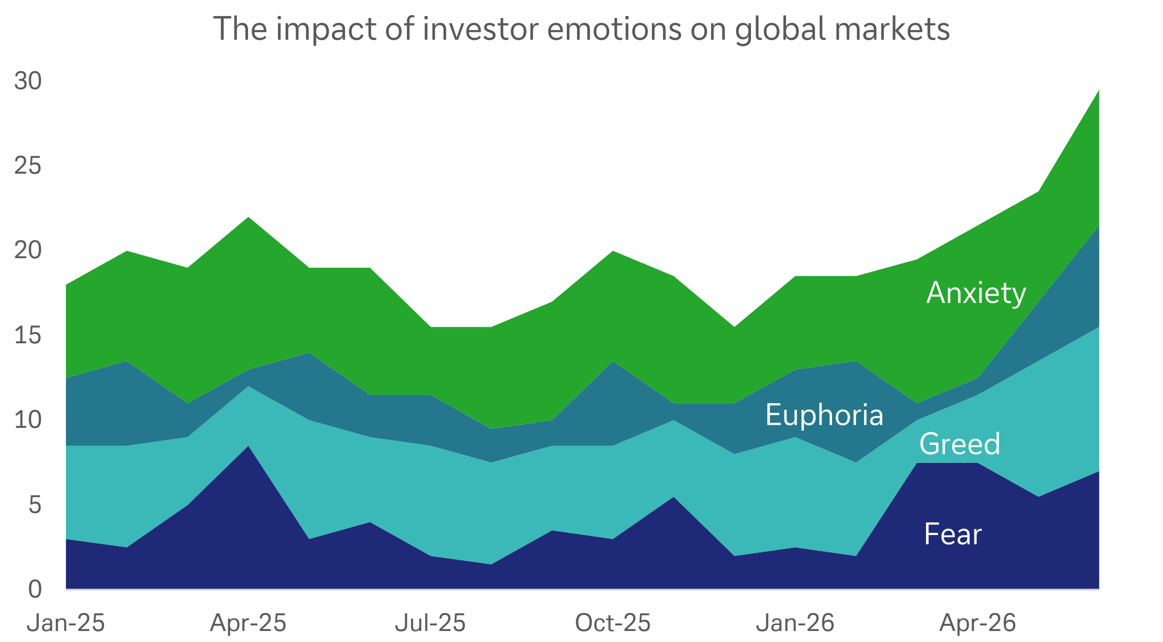
Fear Indicators:

Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted, strong emotional language (e.g., "roiled," "plunge," "blistering sell-off," "panic selling,").

The most ‘emotion’ we’ve ever seen

The four key market-moving emotions dominated trading in Q2. Hence the volatility in both directions.

Emotion was everywhere in Q2 – volatility followed



Status-quo bias is strong. That is creating more market volatility

Investors are loathe to consider the upside in bad times and the downside in good times. Thus, when sudden events change, market drops can be more aggressive than expected.

Some examples:

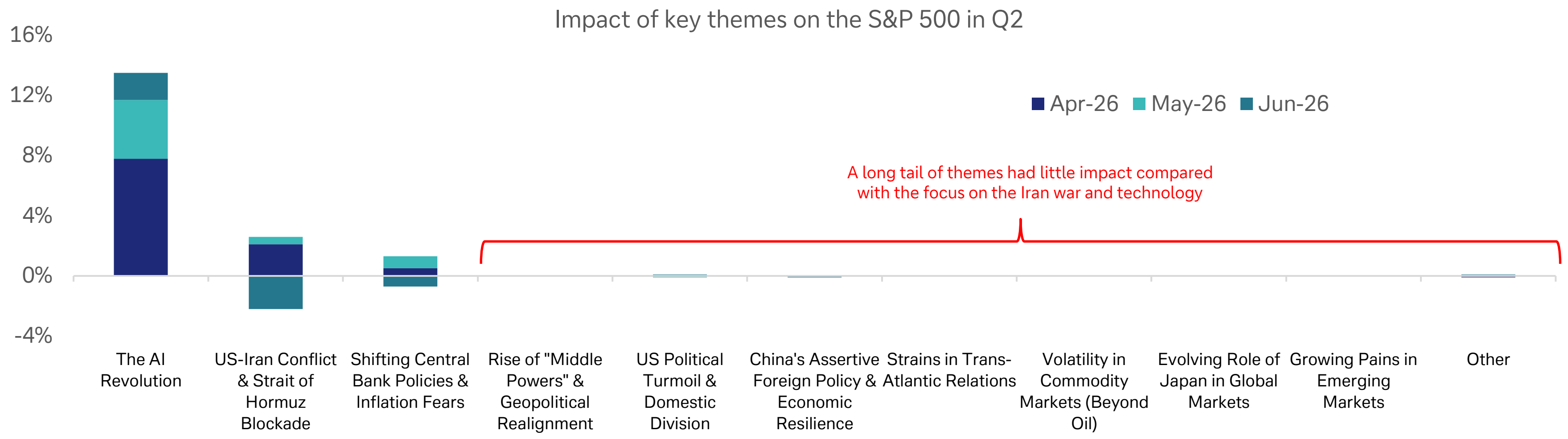
1. Rush to price in the downsides of tech disruption
2. Fear of considering sovereign risk
3. Fear of underperforming peers is greater than the desire to beat them
4. Unwillingness to price in the upside of iterative technology improvements
5. Narrow vision of what globalisation is and the fear that it has peaked

dbLumina's estimate of how each key theme in Q2 impacted the S&P 500

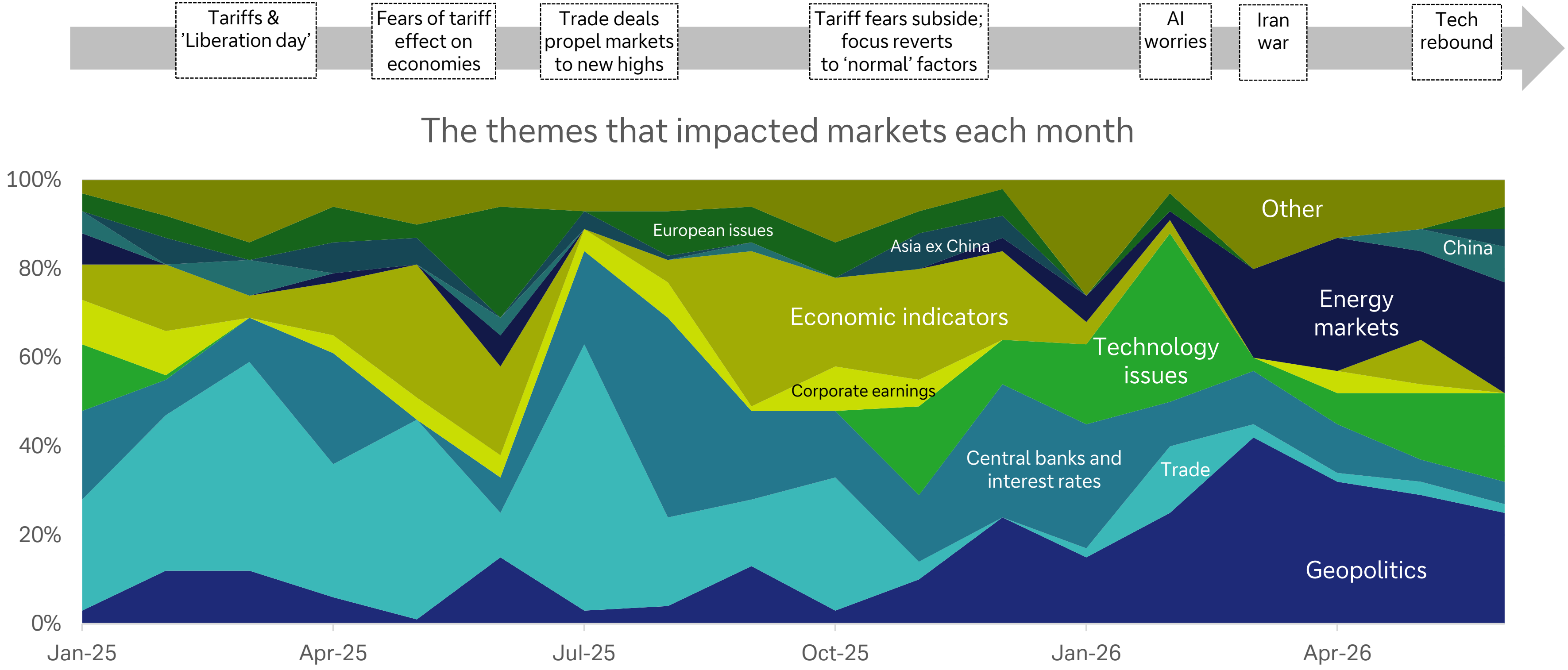
AI surge
Yet another strong rebound … after yet another period of AI-rally scepticism

Iran conflict
Markets settled after their initial shock

Fed policy
Uncertainty about both Warsh and the Fed's response to energy inflation



Our AI analysis shows how groups of themes have driven markets over time



The impact of today's themes 1 year ahead:

We asked our AI model, dbLumina, to analyse the impact of the themes for the quarter. The 'positive' and 'negative' indicators are relative to expectations at the beginning of the quarter. For example, a net negative score for the S&P 500 does not mean we expect the index to fall. Rather it means we see the rising probability that returns will be lower than previously expected.

	Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1	The AI Revolution	Strongly Positive	Strongly Positive	Positive (Upward)	Positive (Upward)
2	US-Iran Conflict & Strait of Hormuz Blockade	Negative	Negative / Volatile	Strongly Positive (Upward)	Strongly Positive (Upward)
3	Shifting Central Bank Policies & Inflation Fears	Negative	Negative	Strongly Positive (Upward)	Negative (Downward)
4	Rise of "Middle Powers" & Geopolitical Realignment	Neutral	Neutral	Neutral	Neutral
5	US Political Turmoil & Domestic Division	Slightly Negative	Negative / Volatile	Neutral	Neutral
6	China's Assertive Foreign Policy & Economic Resilience	Slightly Negative	Negative / Volatile	Neutral	Neutral
7	Strains in Trans-Atlantic Relations	Neutral	Slightly Negative	Neutral	Neutral
8	Volatility in Commodity Markets (Beyond Oil)	Neutral	Slightly Negative	Neutral	Neutral
9	Evolving Role of Japan in Global Markets	Neutral	Neutral	Neutral	Neutral
10	Growing Pains in Emerging Markets	Neutral	Neutral	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-2	-3	-5	-2

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

The impact of today's themes 3 years ahead:
The medium-term picture now broadly reflects the short-term picture – a big change from last quarter when several risks were weighing on economies and markets

	Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1	The AI Revolution	Very Strongly Positive	Positive	Mixed / Uncertain	Negative (Downward)
2	US-Iran Conflict & Strait of Hormuz Blockade	Neutral	Neutral	Neutral	Neutral
3	Shifting Central Bank Policies & Inflation Fears	Neutral	Neutral	Neutral / Stabilizing	Neutral / Stabilizing
4	Rise of "Middle Powers" & Geopolitical Realignment	Slightly Negative	Slightly Negative	Neutral	Neutral
5	US Political Turmoil & Domestic Division	Neutral	Neutral	Neutral	Neutral
6	China's Assertive Foreign Policy & Economic Resilience	Slightly Negative	Slightly Negative	Neutral	Neutral
7	Strains in Trans-Atlantic Relations	Slightly Negative	Slightly Negative	Neutral	Neutral
8	Volatility in Commodity Markets (Beyond Oil)	Neutral	Neutral	Neutral	Neutral
9	Evolving Role of Japan in Global Markets	Neutral	Neutral	Neutral	Neutral
10	Growing Pains in Emerging Markets	Neutral	Neutral	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-1	-1	Nil	1

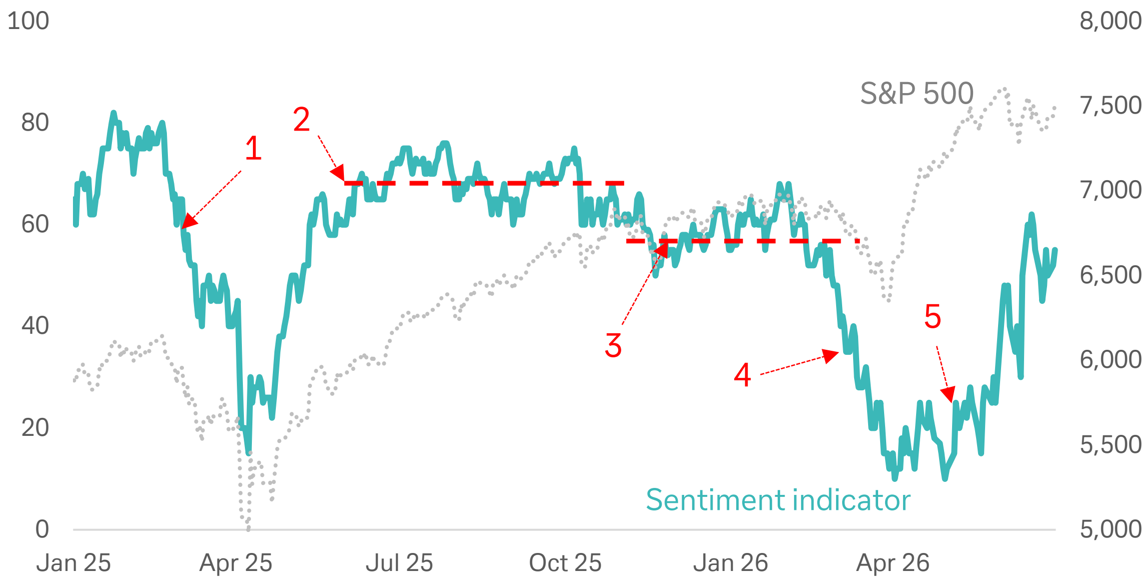
1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

Our gauge of investor sentiment uses AI to interpret macro, economic, and market signals

This is our response to the fact that so many confidence indices today fail to reflect technological and social megatrends that now affect markets.

It is early days, but our gauge appears to be developing some predictive qualities

dbLumina investor sentiment indicator v S&P 500



1. Indicator drops 20 points in Feb-25 before the most dramatic stock market declines relating to tariffs.

2. Consistent strong sentiment as markets recovered from the tariff-induced sell-off.

3. Lower sentiment as markets top-out on valuations and software/PC nerves grow. Yet, no catalyst for a large drop.

4. Our sentiment indicator commenced a steep drop on 3-Feb. It fell 20 points before the war in Iran kicked off.

5. The slower, more volatile recovery in our indicator foreshadowed the volatility in the S&P 500 during June

Emerging themes – part 1

We asked dbLumina to help us identify the key rising themes that have the potential to most impact markets and economies in the short term

1 The "sovereignty" premium in technology and AI

National governments are increasingly asserting control over the technology sector, extending beyond simple regulation to direct intervention in the name of "sovereignty."

This is evolving from a focus on supply chains into more direct control over AI models and data infrastructure. Examples include:

- The US asking OpenAI to restrict its latest model, GPT 5.6, to "government-approved entities" only.
- The US-led "Pax Silica" alliance expanding, aiming to secure AI supply chains, including critical minerals and energy
- China adding Japanese companies to an export control list
- Talks between Anthropic and the US Commerce Department over security fears related to its AI models

This trend reinforces the trend towards a future where access to advanced AI and technology is becoming fragmented along geopolitical lines, creating a "sovereignty" premium on nationally-controlled tech assets and disrupting the globalised tech ecosystem.

2 The rise of the populist left in Europe

While the rise of the populist right has been a long-running theme, a complementary trend is now gaining traction across Europe: the re-emergence of the populist left.

This is beginning to create new political uncertainties and potential fiscal challenges in key European economies. Examples include:

- Alexis Tsipras, Greece's former prime minister, announced his return to politics with a new party aimed at unifying the left.
- Across Europe's ten largest countries, support for populist left parties is now "rivalling that of strictly centrist parties".
- In the UK, Andy Burnham firmed as the favourite to become the next Prime Minister. He has previously supported greater government borrowing, causing some initial concern in gilt markets.
- Local elections in the UK showed a drive towards the populist left and right, putting pressure on centrist leadership.

The growing influence of the populist left could challenge the established fiscal consensus in Europe, potentially leading to increased government spending, greater sovereign debt issuance, and renewed focus on wealth distribution policies.

Emerging themes – part 2

A key learning from the ‘emerging theme’ analysis is the focus on some longer-term themes that have been pushed aside by the conflict in Iran and the focus on tech

3 The AI “picks and shovels" investment thesis

As the initial euphoria and fear around AI business models mature, a more pragmatic investment theme is emerging, focused on the underlying infrastructure and enabling technologies—the "picks and shovels" of the AI gold rush.

- The entry of three new firms—Vertiv, Lumentum, and Coherent—into the S&P 500, all of which specialise in AI supply chain components like power, cooling, and high-speed optical parts.
- Major plans from chipmakers Samsung and SK Hynix (potentially $1.3tn over ten years) for chipmaking plants and data centres
- Comments from TSMC that global chip supply will be insufficient to meet demand "for years to come" reinforcing the long-term demand for the foundational hardware of AI.

This theme suggests a broadening of the AI investment landscape, with increasing investor attention on the less glamorous but potentially more durable companies that provide the essential hardware and infrastructure for the AI revolution.

4 "Middle power" diplomacy and strategic hedging

The “middle powers" are navigating changes to US foreign and trade policy, the tense relationship with China, and other global conflicts by building new alliances. They aim to hedge their dependencies and carve out more independent foreign policy paths. Examples include:

- A meeting of Mercosur's leaders to discuss the implementation of their trade deal with the EU
- Canada inaugurating a sovereign wealth fund specifically to reduce its economic dependence on the US and its Prime Minister visiting President Xi in Beijing to reset relations.
- Colombia's election of a new hard-right president, signalling a domestic policy shift that has implications for its relationship with the US.

Frequently, geopolitical trends take a long time to play out. This trend is accelerating on a monthly basis. It indicates the oft-talked about multipolar world is quickly becoming a practical reality. This creates new opportunities and risks in international relations and trade.

Current evolving risks

Today's key concerns revolve around assumptions that key risks will turn out to be positive for markets. A big assumption.

1 Geopolitical Miscalculation in the Middle East

The fragile US-Iran ceasefire could unravel at a moment's notice.

Iran is constantly testing the truce, including attacks on shipping and back-and-forth military strikes

Markets appear far too sanguine about the risks of a reignition of a full-scale conflict.

2 AI-Driven Market Disruption and Valuation Risk

AI is a dual risk: the potential for disruptive technology to upend established business models and the risk of a valuation bubble in AI-related stocks.

Further AI sell-offs in non-tech sections also appear possible following similar instances in wealth management and private capital.

3 Sovereign Debt Fragility and Fiscal Strain

Global sovereign bond markets showed increasing signs of stress.

We do not expect a serious developed-market default, but several factors increase the risk of fear-based market sell-offs.

Japan and Europe remain clear risks. Secondary effects could press through other markets, particularly if oil prices flare up.

4 A US Consumer and Labor Market Slowdown

The economic indicators continue to show an inconsistent trend and investors may have become too used to the US labour market 'always finding a way through'.

Persistently low consumer sentiment risks turning into a broader downturn should another shock (such as a renewed energy crisis) hit the US economy.

Emerging risks underestimated by the market – part 1

Iran and tech dominate risk equations but several risks are underestimated

1 Social & political backlash against "Big Tech" and AI

An undercurrent of social and political resistance to the influence of technology and AI is steadily growing. While the market is focused on the economic disruption and investment opportunities of AI, a broader societal backlash is forming that could lead to significant regulatory, legal, and fiscal headwinds for the sector.

Examples:

- Legal & Regulatory: A California court ruled on tech companies creating addictive platforms for minors. Governments are also proposing legislation, such as Canada's bill to ban social media for children and similar measures being considered in the UK, France, and Spain.
- Societal & Fiscal: South Korea has proposed a "national dividend" funded by tax revenue from AI-related profits. Ireland has demanded that new data centres "Bring Your Own Power" (BYOP),

Potential Impact (1-3 Years):

- Inflation/Equity Prices: A wave of litigation and stringent regulations could impose massive costs on major technology firms
- Interest Rates/GDP Growth: New taxes on AI or technology, like the proposed "national dividend," could alter fiscal landscapes and influence government spending, potentially affecting GDP and central bank interest rate policy.

2 The politicisation of trade and infrastructure

Beyond standard tariffs, a more complex and unpredictable risk is emerging where critical trade routes and infrastructure are being used as leverage in geopolitical disputes.

The market has treated the Strait of Hormuz situation as a temporary, energy issue, yet it is a dangerous example of the politicisation of global chokepoints

Examples:

- Control over chokepoints: Iran has demonstrated its ability disrupt traffic in the Strait of Hormuz at a very low financial cost to establish a persistent form of leverage over the global economy.
- Targeting infrastructure: Continued Iranian attacks on neighbouring countries sets a dangerous precedent for how other conflicts may play out

Potential Impact (1-3 Years):

- Inflation/GDP Growth: The risk of similar disruptions in other global chokepoints could lead to recurring supply chain crises, driving sustained inflationary pressures and hampering global GDP growth.

- Equity Prices: Sectors reliant on global shipping, manufacturing, and stable energy supply may face recurring shocks to their earnings and valuations.

Emerging risks underestimated by the market – part 2

Iran and tech dominate risk equations but several risks are underestimated

3 The fragmentation of the EU's political centre

While the market has reacted to specific events like the UK leadership turmoil, it appears to be underestimating the broader, structural trend of political polarisation that could threaten the EU's cohesion and policy-making ability.

Examples:

- Rise of the Populist Left: Support for these parties now rivals that of strictly centrist parties. Just one example was the return of Alexis Tsipras in Greece.
- Internal EU Disagreement: Disagreements on core policies are becoming more pronounced. The summit on European competitiveness revealed a lack of alignment between France and Germany on joint eurobonds, leading to a suggestion that a smaller group of countries might push ahead with "enhanced cooperation".
- Weakening of Traditional Alliances: The UK's political instability has been a consistent theme, while the EU's response to US tariff threats and the US-Iran conflict has been disjointed

Potential Impact (1-3 Years):

- Interest Rates: A breakdown in fiscal cohesion could lead to diverging sovereign bond yields and increased risk premiums for fiscally weaker EU member states
- GDP Growth/Equity Prices: Political paralysis or a turn towards protectionist policies could stifle the single market, hindering the continent's growth and impacting the performance of European equities.

4 'Grey Zone' escalation and the limits of military deterrence

"Grey zone" tactics—actions that are hostile but fall below the threshold of conventional warfare—are growing. Iran's recent actions have demonstrated that a technologically inferior power can impose significant costs on a military superpower and its allies, a risk that the market has not fully priced in.

Examples:

- Asymmetric Warfare: Iran's ability to close the Strait of Hormuz along with its cheap rockets and drones that have been a severe burden on the far more expensive US and Israeli interceptors.
- Proxy Forces: Iran has proven adept at enlisting loyal groups like the Houthi rebels in Yemen and Hezbollah in Lebanon to exert pressure and escalate the conflict horizontally.
- Economic Disruption as a Weapon: Iran's strategy involved not just military threats but also creating regional instability and economic pain – tactics that could be replicated in other conflicts.

Potential Impact (1-3 Years):

- Inflation/GDP Growth: A proliferation of these tactics globally could lead to a state of perpetual, low-level conflict, disrupting trade, raising insurance costs, and creating a persistent drag on global growth and a floor for inflation.
- Equity Prices: Defence sector stocks may benefit, but the broader market would likely face a higher risk premium due to the constant threat of unpredictable disruptions.

Appendix 1

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FIFA World Cup: What economics sees that fans don't

June 30, 2026

This summer, two markets are running simultaneously: one financial, one sporting. While financial markets react daily to macroeconomics and geopolitics, the sporting market — currently represented by the football World Cup — unfolds across 16 venues in the US, Canada and Mexico, with 48 nations competing for a trophy and an expected 6bn global viewers.

These markets are more alike than most observers admit. Both:

— Thrive on scarcity and compelling narrative;
— Follow the same microeconomic laws of supply and demand, driving pricing and profit;
— Reprice in real-time based on new information;
— Exhibit "winner-takes-most" structures, where top performers outstrip the rest exponentially; and
— Exhibit herding behavior - retail investors and football fans alike chasing momentum, overpaying at peaks and panic-selling (or abandoning) at troughs.

This makes the FIFA World Cup an unparalleled "economics laboratory". Over 39 days, it puts core microeconomics on display — across tickets, host cities, tournament design, player pay, fan behavior, and even the business model of FIFA, football's international governing body, itself.

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1. The ticket market: Dynamic pricing meets the Tinkerbell problem

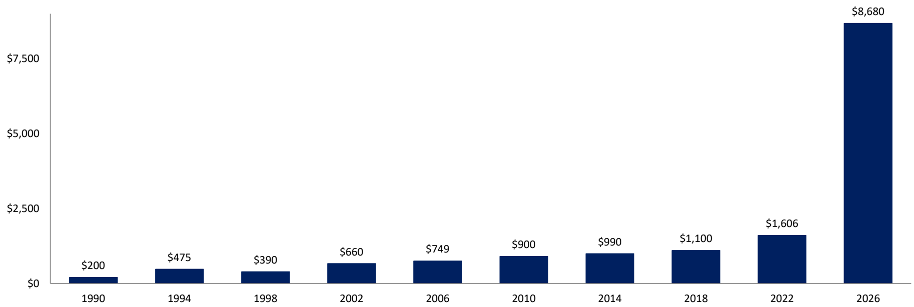
FIFA implemented algorithmic dynamic pricing for 2026 — a first for the World Cup. Think of it like Uber surge pricing: the fare changes depending on how many people want a ride at any given moment. The same mechanism now applies to football tickets, and the effects were immediate.

By February, the median resale ticket on SeatGeek had reached $1,291. The US opener against Paraguay saw prices exceed $3,200. Premium final seats at MetLife Stadium, with a face value of $6,730, were tripled by FIFA to $32,970 by early May. US President Donald Trump publicly said he would not pay those prices.

By May, demand softened — group-stage prices fell 28% to a median of $928. But once results started to matter, prices surged again. After the US beat Paraguay 4–1, tickets for the next US match climbed 65% in 48 hours. By mid-June, 84% of all matches had seen resale prices rise.

This is what we have previously called the "Tinkerbell Effect": while the product (a seat) does not change, the belief and narrative surrounding it do and that is enough to move the price. FIFA capitalized on this by creating an official resale marketplace, collecting a 15% fee from both buyer and seller. At the other extreme, Jordan vs Algeria and Curaçao vs Ivory Coast tickets were available for under $450.

Figure 1: World Cup final ticket prices



Source: The World Cup Guide Note: Category 1.

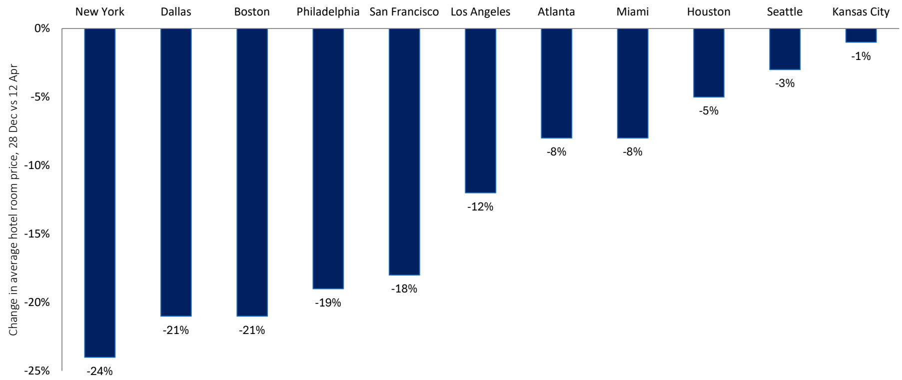
2. Host cities: Scarcity priced in, demand missing

The hotel market illustrates how quickly an anticipated demand shock can reverse. Following the December group-stage draw, room rates in some host cities surged by up to 300% overnight. By April, 80% of US hotel operators were below their initial forecasts. New York was down 65% vs expectations; Seattle, 80%. Game-day rates had fallen roughly one-third from their December peaks. Mexico was the exception: tighter local supply kept prices elevated.

This shortfall reflects more price-sensitive demand than many expected. And it is not without precedent. The World Cup in France in 1998 offers the closest parallel: organizers predicted 500,000 extra tourists, yet fewer visitors came than in a normal June. Non-football fans simply avoided the country; a pattern economists call the "crowding-out effect." Overnight hotel stays by non-residents fell 13% year-on-year. The same dynamic is playing out in 2026, compounded by visa and entry frictions that reduced cross-border travel — a factor cited by 70% of hotel operators as a primary drag on international demand. Several high-profile cases reinforced this constraint, including denied entry to officials and ticket restrictions affecting entire national allocations.

The overall macroeconomic impact is limited. The US GDP uplift is estimated at about 0.05% — negligible relative to the headline projections. External assessments consistently find realized gains materially below advertised figures, in line with research showing that 12 of the last 14 World Cups produced net economic losses for host regions. With Canada's hosting costs at ~C$1.07bn (~C$82mn per match), the event is less about economic stimulus and more about prestige spending.

Figure 2: World Cup hotel rates (%)



Source: Business Traveller, citing Lighthouse Intelligence. Comparison of average hotel room prices booked on 28 December vs. 12 April.

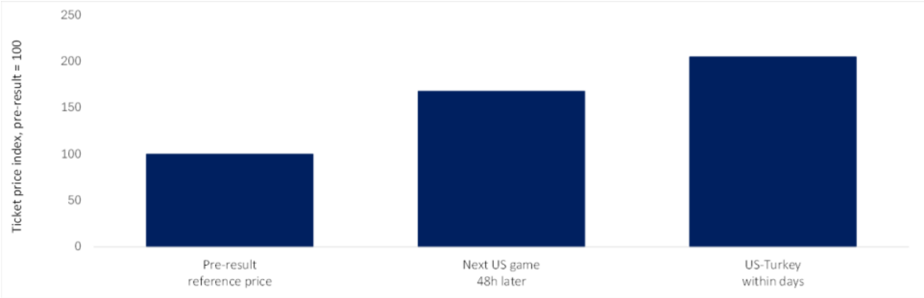
3. Fans: "The ball is round", behavior isn't

Fans price emotion, not assets. After the US defeated Paraguay 4–1, tickets for the next game surged 68% in 48 hours; US v Turkey prices climbed 105% in days. The product remains unchanged; only the narrative shifted. As Nobel Prize-winning economist Daniel Kahneman showed, people's valuations are reference-dependent and heavily influenced by recent information. In this case, a single victory fundamentally altered fans' willingness to pay.

Fan behavior also reflects Nobel Prize winner Richard Thaler's classic biases. Fans holding tickets for matches their team will no longer play face a sunk-cost dilemma. Many attend rather than sell at a loss, as loss aversion dominates: avoiding a realized loss matters more than optimizing value. What appears to be demand is partly behavioral inertia leading to predictable mispricings. Seats are filled not because their marginal value has increased, but because past spending anchors decisions. Markets clear, but not efficiently.

The contrast is broadcasting. While stadium access is scarce and volatile, global viewership scales at near-zero marginal cost. FIFA's ~$4.2bn in broadcasting rights captures value from the marginal viewer, not the marginal ticket buyer — a reminder that, in this market, price reflects psychology as much as scarcity.

Figure 3: How winning changes ticket prices



Source: seatgeek.com

4. Tournament design: All about game theory

Rules shape behavior, sometimes more than talent does. Tournament design significantly influences outcomes. The 1982 match between West Germany and Austria is the classic illustration: both teams knew that a 1–0 win for West Germany would send them both through at Algeria's expense, and that's what happened. Given the rules, the outcome was entirely rational. FIFA's response — requiring final group-stage matches to kick off simultaneously — was an institutional fix to a game-theory problem.

The 2026 format partly recreates these conditions. With 12 four-team groups and eight best third-place finishers advancing to the round of 32, there are again situations where teams gain more from managing a result than from maximizing performance. Cape Verde illustrated the new incentive structure: three draws were enough to qualify (the first time since World Cup 1998). The early phase of the tournament rewards not losing as much as it rewards winning.

This played out on the pitch. Against England, Ghana showed little interest in pressing for a win; having already won their earlier match, a draw was the rational outcome to fully secure qualification. England's possession — however dominant — could not change that calculus. Norway, already through and needing only a draw to top the group, fielded a rotated side against France, rationally conserving energy for the knockout rounds.

Yet, 2026 also exposes the limits of purely incentive-based models. Turkey, despite being already eliminated and with nothing at stake, beat the US 3–2 in stoppage time in a match with no bearing on standings. No prize-money model — with payouts ranging from $9mn for a group-stage exit to $50mn for the champion — would have predicted that result. Pride, national honor, and players auditioning for future contracts generated an intensity that financial incentives alone cannot explain.

Incentives matter, but they do not explain everything.

Figure 4: Ranking of the third best

Ranking	Country	Points	+/-
1	DR Congo	4	+1
2	Sweden	4	0
3	Ghana	4	0
4	Ecuador	4	0
5	Bosnia-Herzegovina	4	-1
6	Algeria	4	-2
7	Paraguay	4	-2
8	Senegal	3	+2
	Eliminated		
9	Iran	3	0
10	South Korea	3	-1
11	Scotland	3	-3
12	Uruguay	2	-1

Source: FIFA; https://www.skysports.com.

5. Penalties: Behavioral economics under pressure

As tournament rounds advance, quality differentials compress. By the semi-final stage, the marginal gap between sides is small enough that outcome is determined not by aggregate talent but by individual decision-making under extreme pressure. Penalty shootouts are that mechanism institutionalized.

The incidence is significant. Four of the last 10 World Cup champions survived at least one shootout en route to the trophy — in 2022, Argentina survived two against the Netherlands and France. A material share of tournament outcomes are therefore decided not by 90-minute performance differentials but by a discrete, high-variance event with well-documented behavioral distortions.

Shootouts perfectly illustrate loss aversion, mixed-strategy equilibrium, and irreversible decision-making. Kahneman and Tversky's framework explain how players, weighing the asymmetric reputational cost of missing against scoring, systematically favor "safe" shots even when an unpredictable strategy is superior. The fear of missing is not irrational; it is a rational response to asymmetric social payoffs.

The data are unambiguous. World Cup shootout conversion stands at ~70%, against ~80% for penalties in normal and extra time — a 10pp gap attributable to psychological load, not technique or distance. Premier League conversion runs at 80% in lower-stakes contexts. The distance is unchanged; what changes is the psychological weight attached to each kick.

The differential in success rate may be explained by goalkeepers systematizing their preparation for shootouts with more transparent data. Pickford's water bottle at Euro 2024 — carrying printed tendency data on each Swiss taker, direction by direction — captured this precisely: "AKANJI: DIVE LEFT." He dived left and saved it. England used the same approach in the 2018 World Cup round-of-16 against Colombia. Instinct has been replaced by a signal-processing problem. As goalkeepers industrialize their reads, kickers randomize further — producing exactly the mixed-strategy equilibrium game theory predicts, with neither side able to sustain a systematic edge.

The underlying condition remains fixed: a professional athlete, optimally prepared, missing from 12 yards not because of technique but because tournament stakes have recalibrated every cognitive signal. A penalty is ultimately a test of psychological resilience, not technical skill.

Figure 5: 10 last World Cup winners' penalty shoutouts

Winner	Match won on penalties
Argentina 2022	Netherlands (1/4); France (final)
France 2018	—
Germany 2014	—
Spain 2010	—
Italy 2006	France (final)
Brazil 2002	—
France 1998	Italy (1/4)
Brazil 1994	Italy (final)
Germany 1990	England (1/2)
Argentina 1986	—

Source: FIFA; https://www.skysports.com.

6. Players: The GOATs take it all

In 1981, economist Sherwin Rosen argued that small differences in talent, amplified by technology that lets the best performers reach unlimited audiences, produce extreme income inequality. The 2026 squad lists are the most striking illustration of that thesis in the sport's history.

Cristiano Ronaldo of Portugal enters his sixth World Cup at the age of 41 with annual earnings of ~$300mn — a base salary of ~$235mn at Al-Nassr (tax-free) plus endorsements including a lifetime Nike deal — making him the first-ever soccer player to cross $1bn in net worth. Argentina's Lionel Messi earns ~$140mn annually; his Inter Miami contract includes a revenue-sharing clause tied to Apple TV's MLS Season Pass subscriptions, giving him equity-like upside in platform growth. France's Kylian Mbappé, at ~$100mn per year, is already building a venture portfolio before age 28.

Below this elite tier, the drop-off is steep. Mid-squad players earn 10–20% of their income from endorsements. New Zealand's Tim Payne worked in construction between international caps. If football's income distribution were mapped as a Gini coefficient — the standard measure of inequality used for whole economies — it would likely rank among the most unequal in the world.

The Saudi Arabian influence makes this more extreme still, with sovereign wealth capital deployed as soft power, pricing players at levels no normal commercial model would typically justify. This is structurally identical to US hyperscalers committing over $700bn to AI infrastructure in 2026: strategic positioning, not near-term return.

Figure 6: World Cup 2026: Top 5 vs. bottom 5 squad market values (€mn)

Top 5	Market value
France	€1,520mn
England	€1,360mn
Spain	€1,220mn
Portugal	€1,010mn
Germany	€947mn

Bottom 5	Market value
Qatar	€20mn
Jordan	€20mn
Iraq	€21mn
Curaçao	€26mn
Iran	€32mn

Source: Transfermarkt; Deutsche Bank Research.

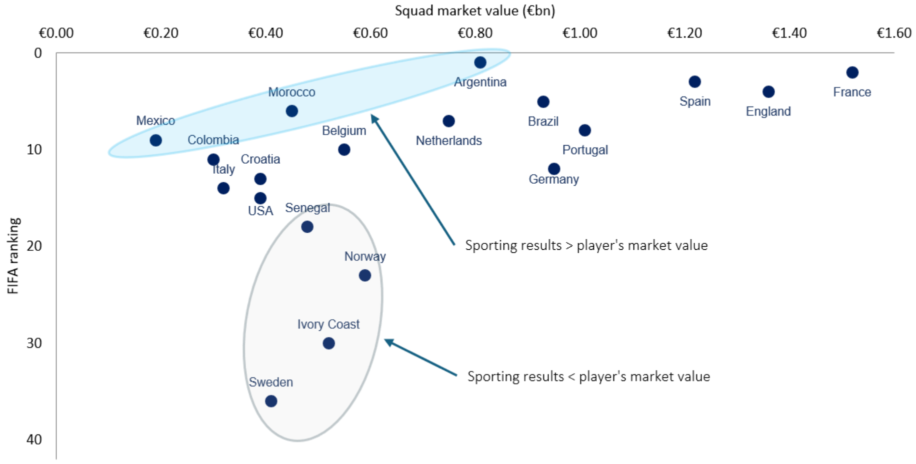
7. Teams: Better than the sum of their GOATs

Moneyball — Michael Lewis's 2003 book about the Oakland Athletics — showed how statistical analysis could greatly assist in building a winning team. It works particularly well in baseball because the sport decomposes into a large number of relatively homogenous individual duels, simplistically: pitcher against batter, each with a measurable contribution that can be compared with many other similar events. The whole approximates the sum of its parts.

Football is more complicated as it has a low number of repeatable scoring 'duels'. Rather, a player's value is frequently derived from hard-to-measure intangibles that necessitates the use of complex 'non-linear dynamic system models'. These are only starting to understand how players interact with each other in a way that can be shaped by training and chemistry built over months. Alchemy matters in a way that spreadsheets find it difficult to capture. The World Cup bears this out. Italy's winning team in 2006 had five players that knew each other well from Juventus (Buffon, Cannavaro, Zambrotta, Camoranesi, and Del Piero); Spain in 2010 had seven from Barcelona (Valdés, Puyol, Piqué, Xavi, Iniesta, Busquets, and Pedro); Germany in 2014 had six from Bayern (Neuer, Lahm, Boateng, Schweinsteiger, Kroos, and Müller). In 2026, the pattern holds among the favorites. Germany is again built around a Bayern spine of six — Neuer, Kimmich, Tah, Musiala, Goretzka, and Pavlović — alongside four from Dortmund and three from Stuttgart. Spain again leans on Barcelona, with Cubarsí, Pedri, Lamine Yamal, and potentially Ferran Torres and Gavi. France has six Paris St Germain players in the mix.

The counterexamples are equally instructive. Cohesion without quality produces coordination without results; the winning formula requires both. Saudi Arabia fielded six or seven Al-Hilal players; Czech Republic 10 from Slavia Prague. Both are eliminated.

Figure 7: Correlation between FIFA ranking and team market value (€ bn)



Source: Transfermarkt, FIFA, Deutsche Bank Research.

8. "Football is a simple game", scaling isn't

FIFA is best understood not as a sports organizer but as a platform — like Apple or Uber — sitting in the middle between two groups (teams on one side; broadcasters, sponsors and fans on the other), capturing value from both. The more users on each side, the more valuable the platform becomes.

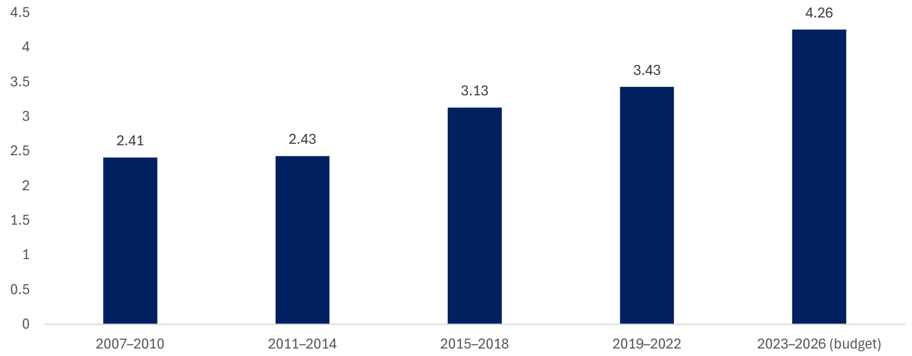
This year's move of expanding the tournament from 32 to 48 teams increased the number of matches from 64 to 104. More matches, combined with two new mandatory 3-minute stoppage breaks per game, mean more advertising slots, more sponsorship activations, and more data generated. The revenue arithmetic is direct: broadcast revenue above $4.2bn (US rights up ~94% vs 2022), sponsorship at ~$2.4bn (+37%), and total cycle revenue approaching $13bn, up from $7.6bn in Qatar. Scale, not scarcity, is now the dominant strategy.

FIFA is also moving further into the value chain. Its official resale marketplace mirrors Apple's App Store: a controlled secondary market framed as consumer protection, charging 15% fees on every transaction. This is vertical integration — capturing more of the value the platform creates.

The next frontier is data. With 150mn data points generated per match and 104 matches played, the 2026 tournament represents the largest structured sports dataset ever assembled. Data, not just matches, is becoming a monetizable asset.

The trade-off is familiar to every platform that has ever grown. More content increases reach but risks diluting quality at the margin. Curaçao vs Tunisia in a half-empty stadium is not Brazil vs Argentina, whatever the algorithm prices it at. The balance between scale and quality is now FIFA's central strategic question.

Figure 8: FIFA World Cup broadcast rights revenue has nearly doubled over the past two decades ($ bn)



Source: FIFA Annual Reports (2010, 2014, 2018, 2022, 2024). Figures refer to four-year commercial cycles; 2023–26 is FIFA's budgeted revenue.

9. Football market as capital markets: A broader ecosystem

The maturation of capital markets produced more than trading profits. It generated an entire industrial infrastructure: data terminals (Bloomberg, Refinitiv), analytics vendors, and retail execution platforms that democratized market access. Football's commercial ecosystem is replicating this structure, driven by the same dynamic: more data, more participants, more liquidity.

The data layer is expanding fastest. FIFA is tracking ~150mn data points per match in 2026, with the match ball's internal sensor logging 500 movements per second. Every player is 3D-scanned and tracked 50 times per second, generating a continuous biometric and performance data stream across all 104 matches. FIFA's AI Pro system, distributed to all 48 teams, provides identical pre- and post-match analytical capabilities via AI agents. The analogy to Bloomberg terminals holds: what those terminals did for price discovery in the 1980s, real-time tracking infrastructure is doing for football performance and wagering today.

The wagering market has scaled accordingly. Global wagers were $35bn at Qatar 2022; 2026 is projected to surge by 43% to exceed $50bn driven by the expanded format, broader regulatory access, and North American time zones. Prediction markets represent a new layer: Kalshi and Polymarket alone surpassed Super Bowl prediction-market volume within 48 hours of tournament kickoff.

Brazil is the sharpest case study in what happens when deep football fandom meets regulatory liberalization. Online betting was legalized in 2018 but unregulated until January 2025. The results were immediate: licensed operators generated $7bn in gaming revenue in the first year, with BRL 10bn collected in tax revenue, making Brazil one of the world's largest regulated betting markets in its first year of operation. The driver is simple: roughly one in four of Sofascore's Brazilian users are active bettors, and 18 of 20 Série A clubs carried betting sponsors in 2025 — though that figure has since pulled back to 12–13 as the market matures. The fandom is the distribution network.

The capital markets analogy holds at the ecosystem level. Retail participation, data infrastructure, and institutional liquidity are scaling simultaneously, each reinforcing the others. The sports analytics market alone — $2.29bn in 2025, forecast at $4.75bn by 2030 — is the infrastructure layer that makes the wagering layer possible, precisely as Bloomberg and Refinitiv underpinned the growth of retail trading.

The economic product has shifted in the process: the World Cup used to monetize attention through tickets, sponsorship, and broadcast rights. The same 90 minutes now generate a continuous data inventory — odds, expected goals (xG), corners, cards, player props, micro-markets — and what is being priced is real-time uncertainty, not a match result.

10. Prediction markets as capital markets: Repricing in real time

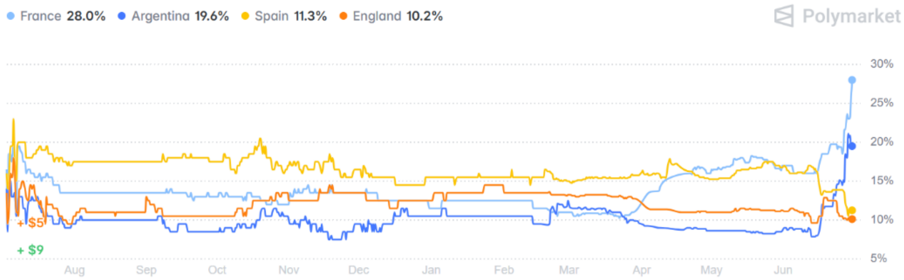
Financial markets price expectations, not outcomes. When tensions flare around the Strait of Hormuz, oil prices move immediately: the market is revising its probability distribution over future states. Prediction markets apply the same framework to football: each contract is a continuously updated implied probability, and every red card, substitution, or missed penalty is a signal that triggers instant repricing.

The market is no longer marginal. Peak betting traffic is expected to average 100,000 wagers per minute around key matches in the 2026 World Cup. US sportsbooks (FanDuel, DraftKings, BetMGM) run full tournament slates. But the more structurally interesting development is the rise of prediction markets proper: Kalshi and Polymarket together went from $2.2bn to $4.8bn in trading volume on the opening day alone, surpassing Super Bowl prediction-market volume within 48 hours. Unlike sportsbooks, these platforms allow participants to enter and exit positions continuously, acting as liquid secondary markets in probabilistic outcomes.

Argentina's tournament offers the cleanest illustration. The defending champions entered at ~9% implied probability — unusually low for a title holder — reflecting market uncertainty about Messi at the age of 38 and a congested field. A perfect group stage (three wins, no goals conceded) pushed that to ~13%. Then came the more revealing repricing: after the group stage draw, Argentina's probability surged to ~21% without kicking another ball. Portugal's failure to top its group had handed Argentina an easier path to the semi-finals with no heavyweight opponent until that stage. The market priced the bracket, not a result. No new information about Argentina's squad quality had changed; what changed was the probabilistic map of their path to the trophy.

That is the efficient market hypothesis operating in compressed time. The tournament is, in this reading, a six-week price discovery mechanism: 104 matches, $50bn in open interest, and a continuous information flow that no participant can fully anticipate.

Figure 9: World Cup Winner



Source: Polymarket.com.

Conclusion: Can America make football great again?

The 2026 World Cup has been a live economics seminar mostly confirming the theory. Dynamic pricing has extracted value but triggered a backlash. The anticipated demand boom in host cities has softened under geopolitical pressure. Game-theory distortions have emerged on cue. Superstar pay has widened income disparity further. Fans have overpaid at peak and held tickets at a loss. And FIFA, like any dominant platform, continues to capture a larger share of the value it enables.

The final whistle blows on July 19 at MetLife Stadium in New Jersey. Regardless of the score, the economic outcome is already visible: higher prices, strong revenues, and a tournament that has solidified football's role as both a global spectacle and a significant business engine. Andrew Giuliani — executive director of the White House's World Cup task force — has already suggested expanding the tournament to 64 teams and proposed a US bid for 2038. The arithmetic is simple: more matches, more markets, more revenue.

Appendix 1

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Washington's Post-Iran Playbook: Risks, Deals and Pressure Points

June 29, 2026

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Executive Summary

— Markets should expect more policy volatility, not less, after Iran. Washington's willingness to consider using force means geopolitical tail risks will likely remain difficult to price.

— The midterms may cap some near-term volatility but increase incentives for visible foreign policy wins. President Trump has reason to avoid market-disruptive escalation before November, but pressure for quick wins could accelerate action on Cuba and Europe.

— After November, foreign policy could become an even bigger focus. If domestic legislating becomes harder, the administration is likely to lean more heavily on executive tools, sanctions, tariffs, security deals and international resource coalitions.

Context

— The first half of 2026 has demonstrated that the Trump administration has a higher tolerance for market volatility than many anticipated, including decisions that risk disrupting trade, commodity flows and supply chains.

— Markets will remain sensitive to signalling from the Trump administration on international security. Trump's willingness to threaten force to advance talks drove repeated bouts of volatility in H1 – but the signals will remain hard to price as the continued ambiguity may lead markets to misinterpret geopolitically driven tail risks over the next two and a half years.

— The Iran ceasefire partially frees Trump to refocus on longstanding policy priorities in the second half of 2026. But domestic political pressures will likely weigh more heavily on the Trump administration's foreign policy post-Iran. Public support for further military campaigns – especially lengthy ones or those involving ground forces – is low. But investors should not overestimate how far this constraint will shape Trump's international strategy over the coming two and a half years.

— The administration has shown it is willing to use, or threaten to use, air and naval forces to resolve international disputes where negotiations stall. The interventionism of H1 2026 defied almost every expectation set in 2025 for Trump's second term. While it is not our base case, investors should consider tail-risk scenarios that hit markets through energy supply, freedom of navigation in key waterways, or spillover into tariffs and export controls.

From ceasefire to stress test

— Even after the Memorandum of Understanding with Iran, the US faces serious diplomatic and security challenges in the Middle East. The Phase 2 negotiations with Iran that began on 21 June (focusing on the nuclear programme and deal implementation) will be complex and almost certainly run past the initial 60 days. Reopening the Strait of Hormuz to full pre-war

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traffic could take months and require a major military commitment to fully de-mine its central corridor and reassure shipping firms.

— Washington is consequently pushing European allies and other regional actors to shoulder more of the burden after the initial re-opening. Recent strains in US-Israel relations could complicate the administration's efforts to maintain delicate ceasefires in Lebanon and Gaza, and have already affected the truce with Iran and the navigability of the Strait of Hormuz. Alongside US efforts to manage post-war security assurances for Gulf Arab countries (which we discussed in our recent note), the Middle East will likely remain a focus for the rest of Trump's term.

NATO's burden-sharing bill comes due

— We expect pressure on European allies to carry more of the security burden will intensify as US stockpile and resource constraints bite after the Iran war. Trump's demands will likely headline the NATO Leaders Summit in Ankara on 7-8 July. It is slated to assess the credibility of national defence spending plans and advance the emerging "NATO 3.0" agenda under which European NATO members take on more responsibility for their own conventional defence.

— Ahead of the meeting, the US has announced some military pullbacks from Europe and reportedly notified allies of further planned reductions to its NATO Force Model contributions. To slow the withdrawal of hard-to-replace US capabilities (such as air defence, long-range weaponry and strategic airlift), European leaders will likely stress their commitment to NATO's new 3.5%-of-GDP target for "core" defence spending and their readiness to help clear the Strait of Hormuz.

Tariff pressure remains a dealmaking tool

— The July expiry of Section 122 tariffs and their replacement by Section 301 levies should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. While EU-US trade relations are more stable now than in early 2025, there remains a tail risk that a transatlantic security or sovereignty dispute (renewed American claims on Greenland, for instance) spills over into the economic relationship.

Ukraine diplomacy sees renewed momentum

— Trump will likely renew his push for a ceasefire in the Russia-Ukraine war. The G7 Leaders' 16 June commitment to boost support for Ukraine and increase pressure on Russia struck European capitals as an encouraging sign that the US was converging on their approach to the conflict. After meeting Volodymyr Zelensky the same day, Trump signalled he could reinstate Russian oil export sanctions to press Moscow towards a settlement. On 17 June, the US Treasury's most recent waiver of Russian oil sanctions expired. But US-Europe rhetorical alignment could reverse quickly – for instance if US-Russia-Ukraine negotiations resume in a format that largely excludes the EU/UK. While a ceasefire this year is far from guaranteed, any deal brokered by the Trump administration could offer Russia sanctions relief and access to frozen assets.

Asia policy balances deterrence with détente

- The US strategy for the Asia Pacific is crystallising around pragmatic balancing with China and more burden-sharing with allies. In a 30 May speech to the Shangri-la Dialogue, Defence Secretary Pete Hegseth reaffirmed the 2026 US National Defence Strategy's aim of building up forces to strengthen deterrence and denial along the First Island Chain and in the Taiwan Strait. But with resources stretched and stockpiles to rebuild post-Iran, Washington will likely pair this build-up with mounting demands that allies spend more on their own defence, using incentives such as favourable access to weapons programmes, intelligence sharing and industrial collaboration.
- We expect the administration to prioritise stabilising relations with Beijing in the lead-up to the Trump-Xi summit in Washington, D.C. on 24 September, where the 2025 Busan trade and critical minerals truce may be extended. Even so, a steadier US-China relationship will not undo deepening long-term competition on military, economic and technology issues. Strong bipartisan alignment in Congress on a more hawkish China policy should also keep pressure on the administration to limit any rapprochement with Beijing.

The Americas become a legacy play

- Extending the "Donroe Doctrine" in the Americas should offer Trump a legacy-building opportunity. The US stands to gain from a rightward shift in regional governments (now including Colombia and Peru), letting the administration expand its Shield of the Americas initiative to counter drug trafficking and curb Chinese involvement in critical infrastructure. US pressure on Cuba is mounting, which may lead to more brinksmanship or a partial negotiated solution with Havana in the next few months.
- We expect Trump to also push to widen American involvement in anti-cartel efforts in Mexico, which the Sheinbaum government will likely accommodate through already-close US-Mexico security cooperation. The US will likely also keep leading the rapid re-development of Venezuela's oil and minerals sector and collaborating with the Rodriguez government on anti-gang measures. The drive for a democratic transition in Caracas is expected to persist but may take longer to materialise as the administration focuses on near-term stability.

The impact of midterm elections

- The 3 November midterms will likely bring domestic politics to the fore and shape the Trump administration's international agenda. The thesis that the midterms will steady US policy is partly credible: Trump has a strong incentive not to shock equity and bond markets or stoke inflation ahead of the vote. But the midterms will cut both ways. Some foreign policy issues should benefit from more stability before November – namely US-China ties and the security situation in the Gulf. Yet an administration looking to demonstrate strength through quick wins may pursue other issues more aggressively. Investors should watch rising US pressure on Cuba and the political relationship with Europe as likelier sources of nearer-term movement.
- What happens after the midterms actually matters more. Recent polling and historical precedent – governing parties usually shed seats at midterms – point to Republicans losing control of at least the House of Representatives. Even so, this would not decisively restrict the administration's foreign policy in its final two years: the US executive holds broad foreign affairs and military powers. While a Democratic-controlled House may attempt to constrain

certain Trump initiatives through spending and oversight powers, the administration has already proved willing to work around congressional limitations.

— With little room to pass major domestic legislation, Trump is likely to become a “foreign policy president”, focused on cementing his legacy. This should include: embedding existing tariff and investment deals with key US partners; deepening US leadership in the Americas; creating the conditions for a pull-back of American resources from Europe (including brokering a Russia-Ukraine ceasefire); and broadening US efforts to reduce rare earths and critical minerals dependence on China, including through international coalitions such as the US-led Pax Silica initiative.

Appendix 1

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Research Institute

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Research

Q2 2026 Global Markets Survey

June 2026

Jim Reid | Camilla Siazon

With special thanks to Anthony Chaimowitz from the dbDataInsights team

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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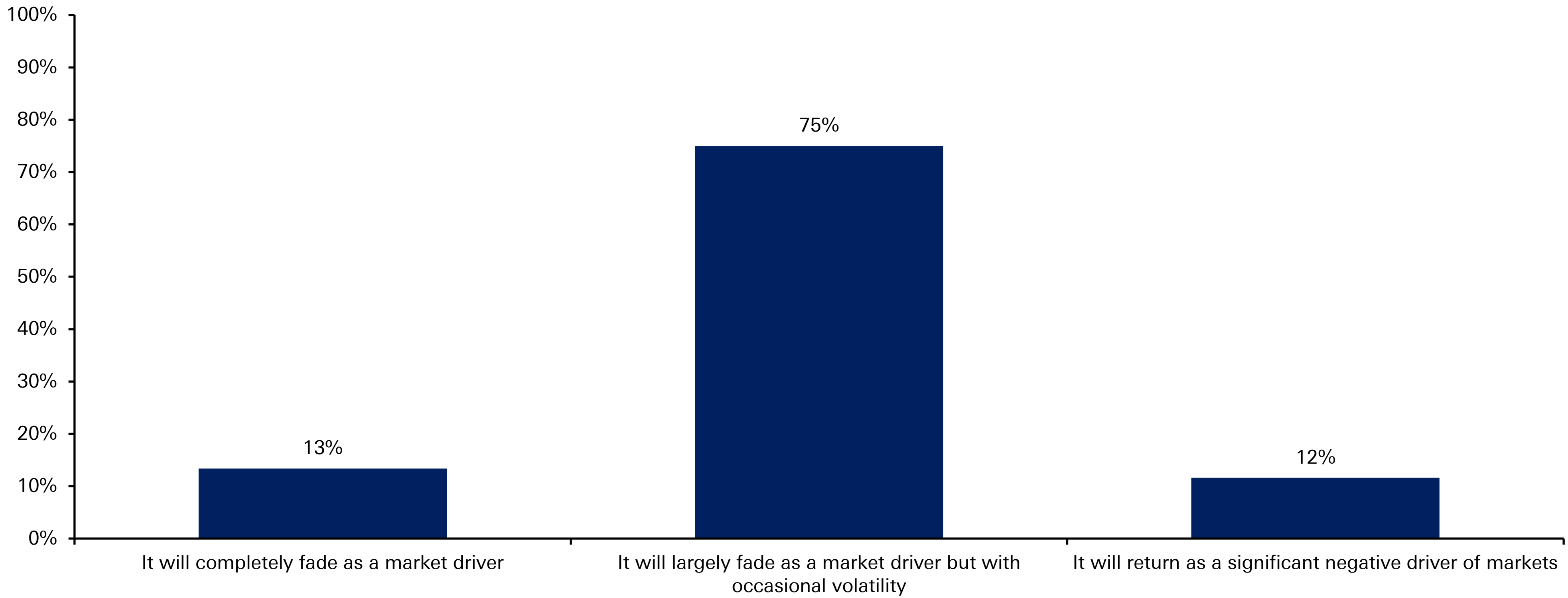
Our Q2 2026 global financial market survey had 275 responses from around the world, conducted between June 22nd and June 26th.

An overwhelming 75% believe the Iran conflict will now largely fade as a market driver into year-end, albeit with occasional volatility. With the 13% who think it will completely fade as a factor that makes 88% who are reasonably sanguine about events in the Middle East. Just 12% think it will return as a big negative driver.

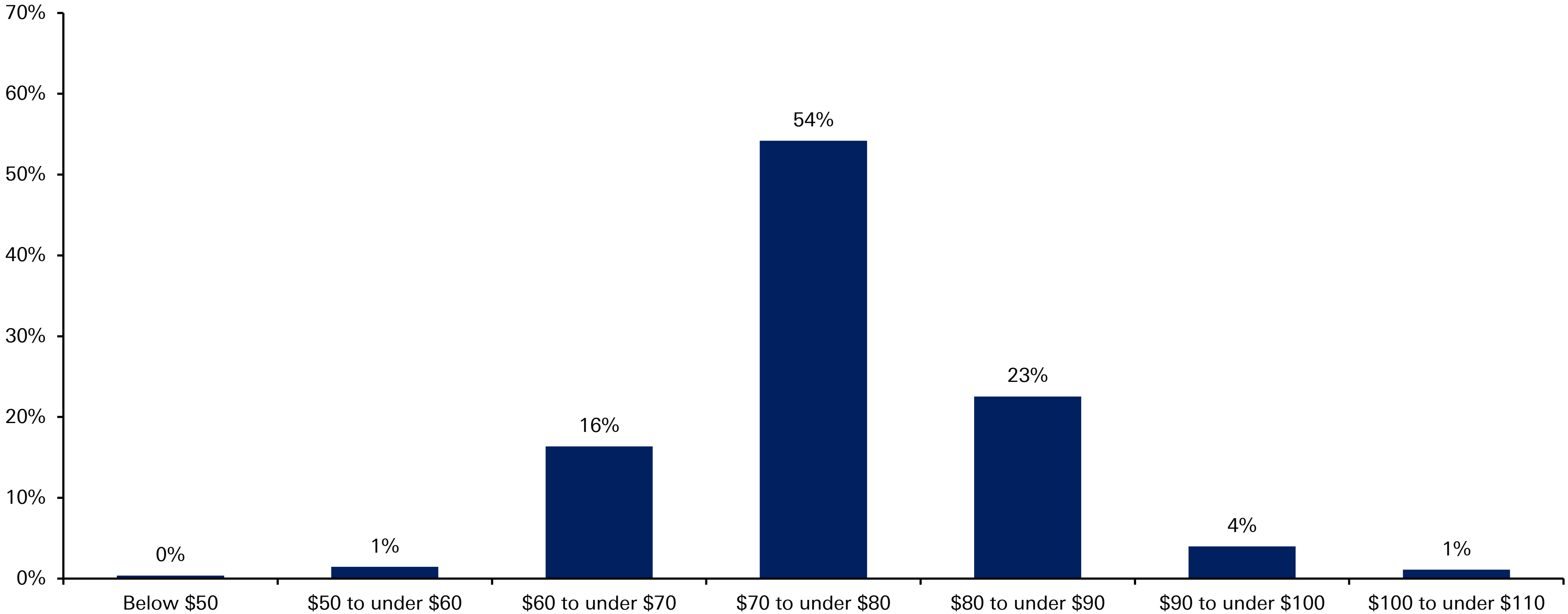
When thinking about the Iran conflict, which of the following best reflects your view of its impact on financial markets for the rest of 2026?



Source: dbDataInsights Survey, Deutsche Bank Research

The more sanguine view on Iran has also led investors to dial down expectations of higher oil prices. A majority of respondents now think Brent will stay between $70-80/bbl for the remainder of the year. Compare that to April, when Brent peaked at ~$118/bbl.

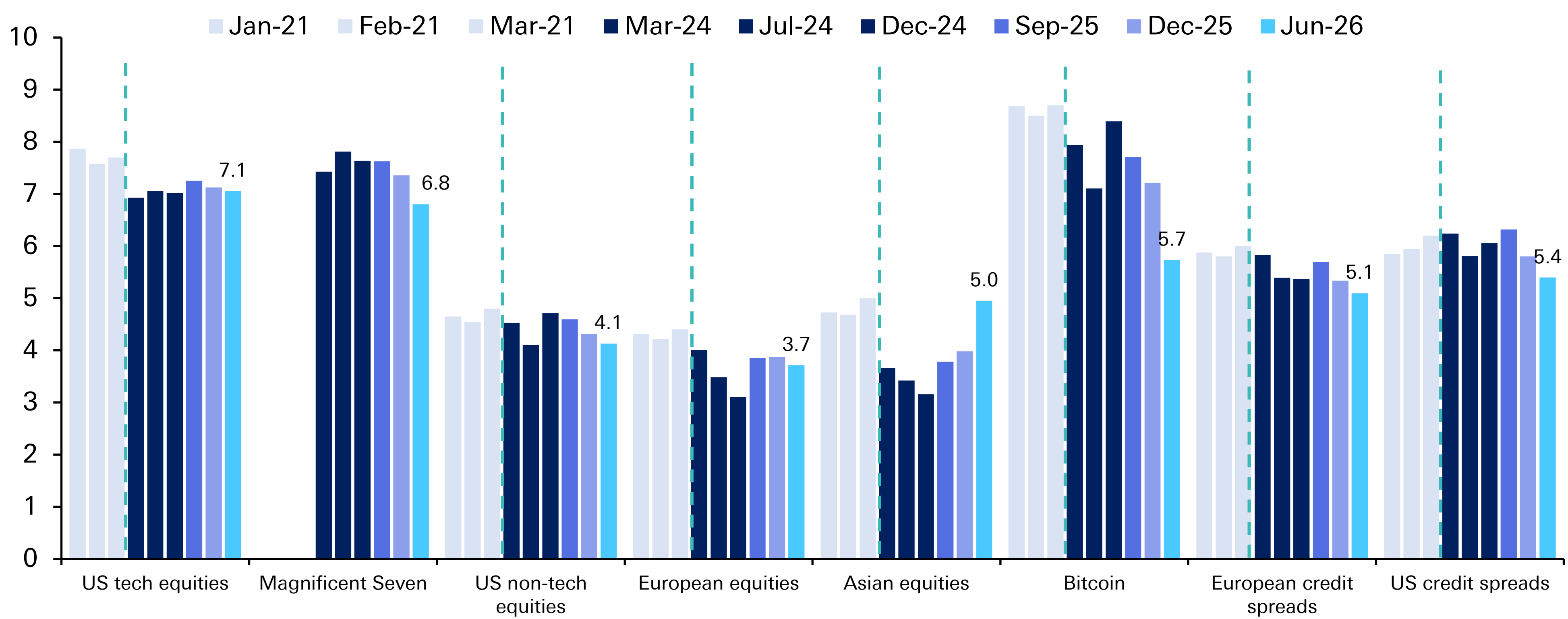
Brent crude oil prices started the year at around $60, rose to approximately $70 prior to the Iran conflict, peaked near $120, and have since declined to around $80. With this in mind, where do you expect Brent crude prices to average for the remainder of the year?



Source: dbDataInsights Survey, Deutsche Bank Research

In our regular bubble question that we first asked in 2021, and then after a gap, in most quarters since 2024, we've seen the lowest perceived bubble risk for the Mag-7. Both US tech and Mag-7 still score around 7 out of 10 on our bubble risk scale, though. Note that Bitcoin bubble risk has collapsed to 5.7 after the near halving since last October.

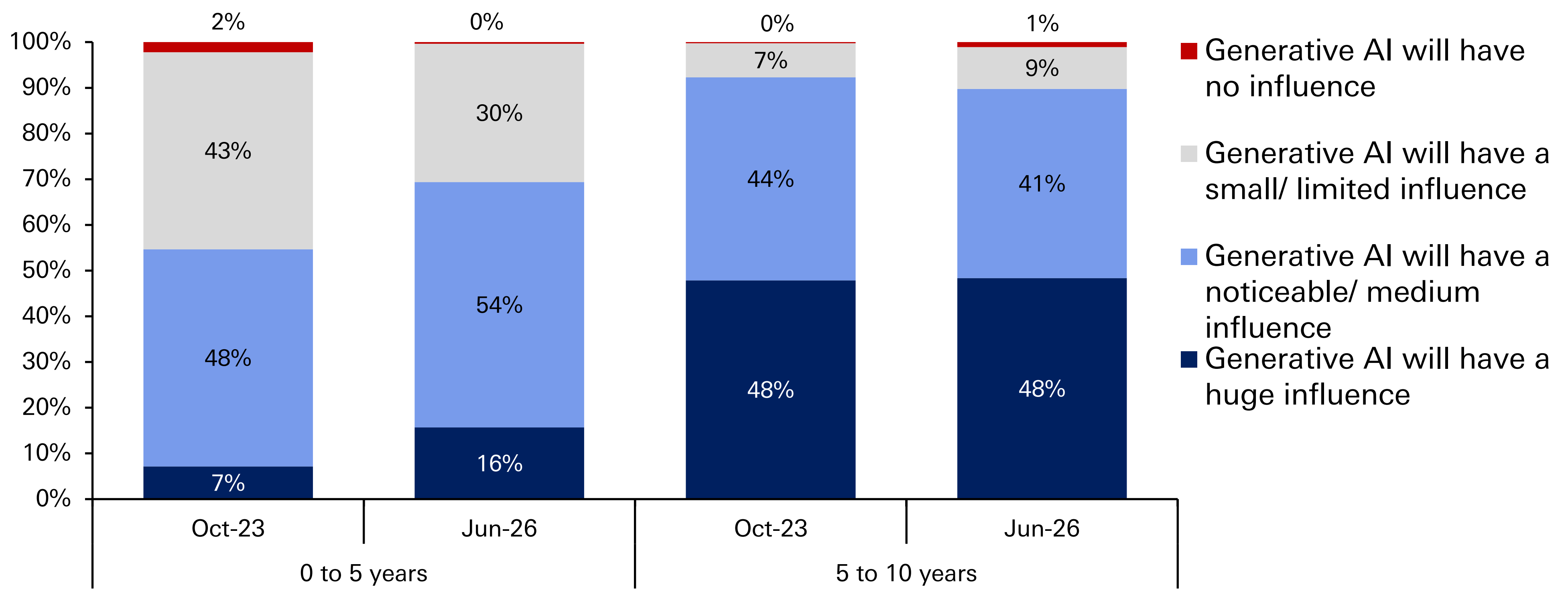
Looking at the following assets and using a scale of 0 to 10, where 0 means 'No Bubble' and 10 means 'Extreme Bubble', please tell us where you think these assets currently are.



Source: dbDataInsights Survey, Deutsche Bank Research

We last asked about Gen AI's potential for productivity growth back in October 2023. Its expected near-term positive impact has increased since then – although some of that likely reflects the nearly 3 years of extra time since we last asked. Surprisingly, the anticipated 5- to 10-year impact has hardly changed, even if just under half still think it will have a huge impact on productivity.

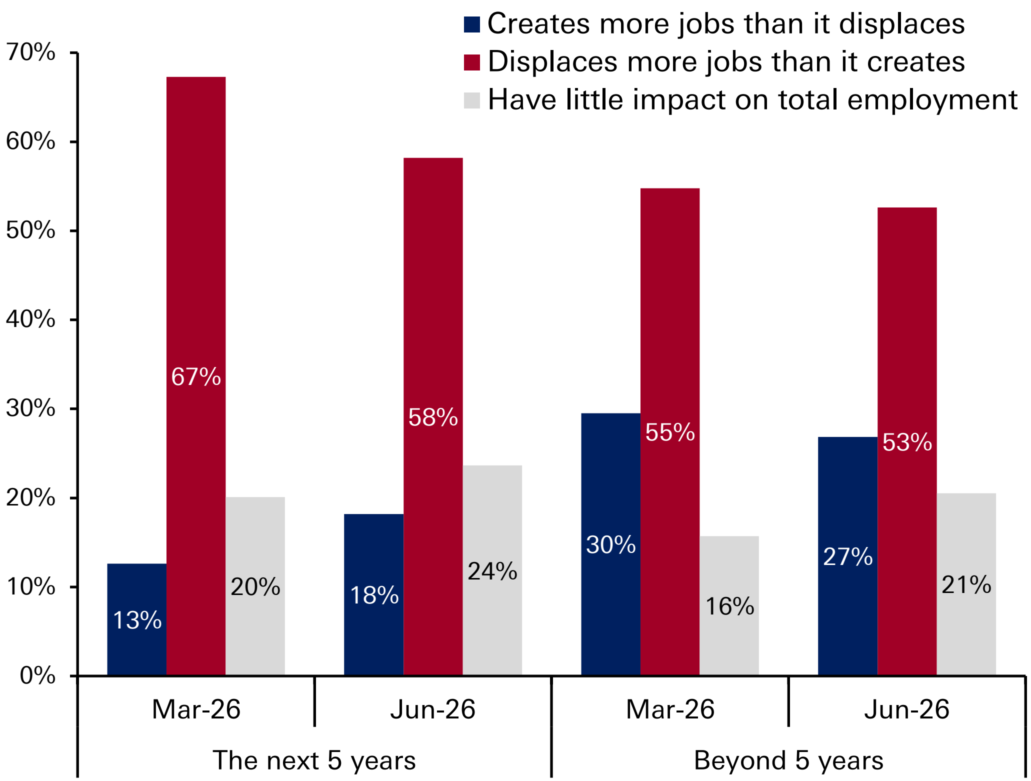
What influence, if any, do you think future iterations of Generative AI will have on productivity growth within the following time frames?

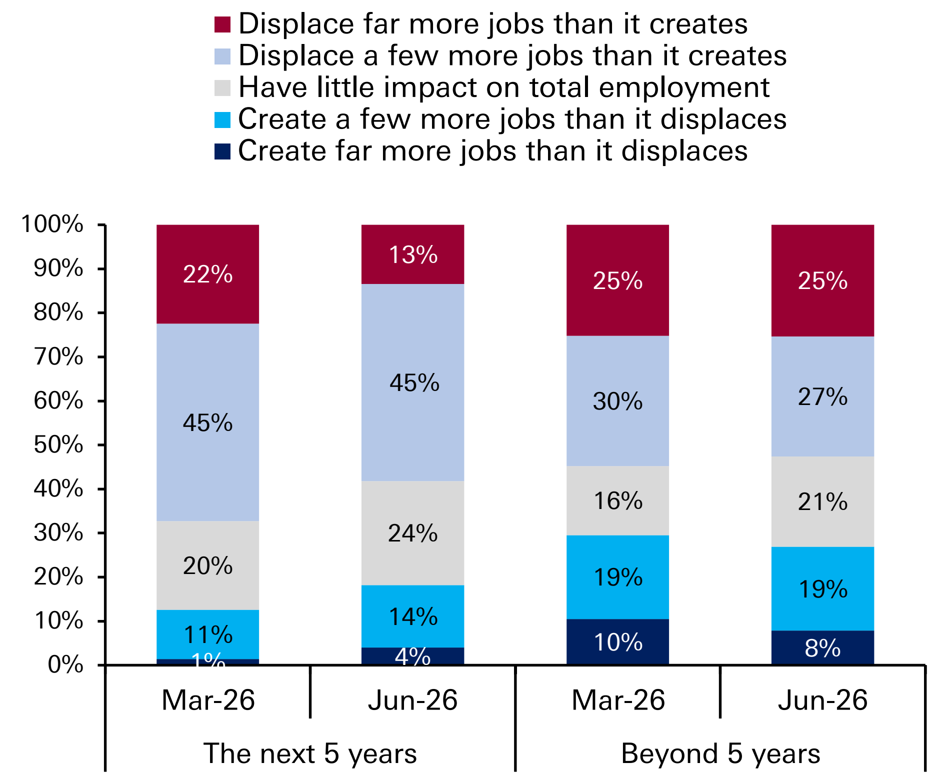


Source: dbDataInsights Survey, Deutsche Bank Research

Market sentiment on AI's impact on jobs remains pessimistic, but we've seen a small improvement since we asked in March. The first chart aggregates the data into binary positive or negative, and the second breaks it down as to how positive or negative.

For the following time horizons, which best describes your expectation for AI's impact on employment? Assume a scenario where AI adoption stays on its current trajectory.

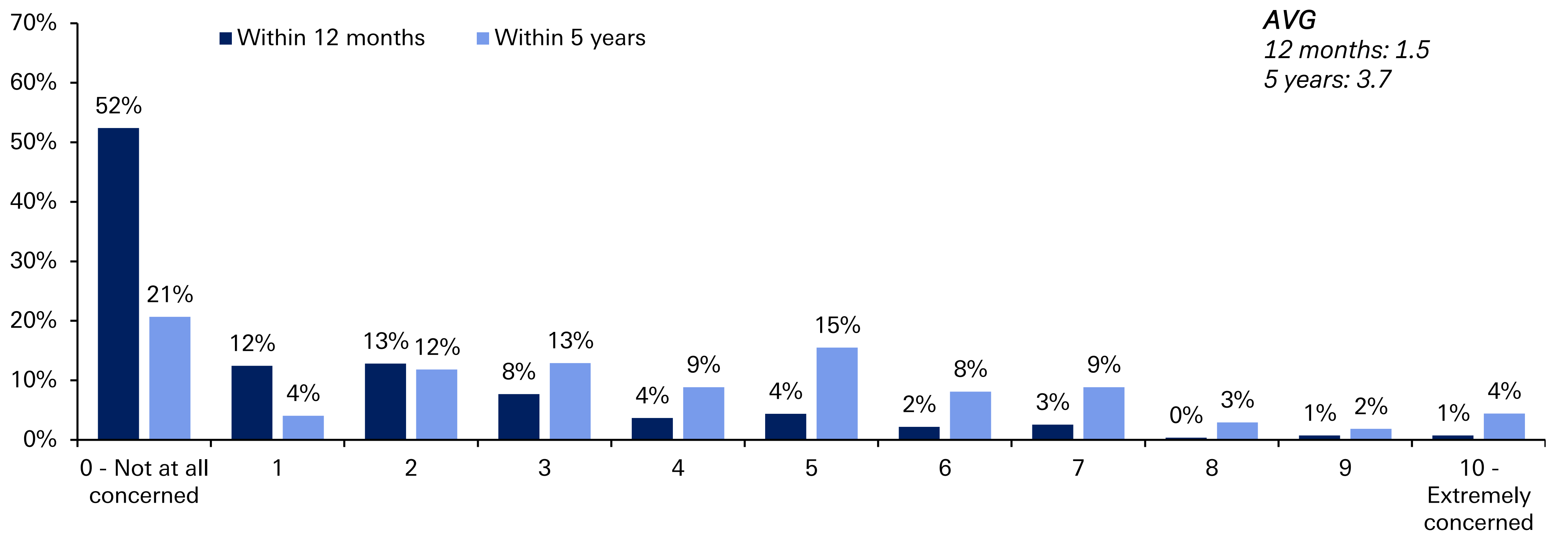




Source: dbDataInsights Survey, Deutsche Bank Research

52% say they are not concerned at all about their own jobs because of AI within the next 12 months. This drops to 21% within the next 5 years. The average scores (scale of 1-10) for these periods are 1.5 and 3.7, respectively. So, combining the last two charts suggests that people are more concerned about other people's jobs than their own due to AI.

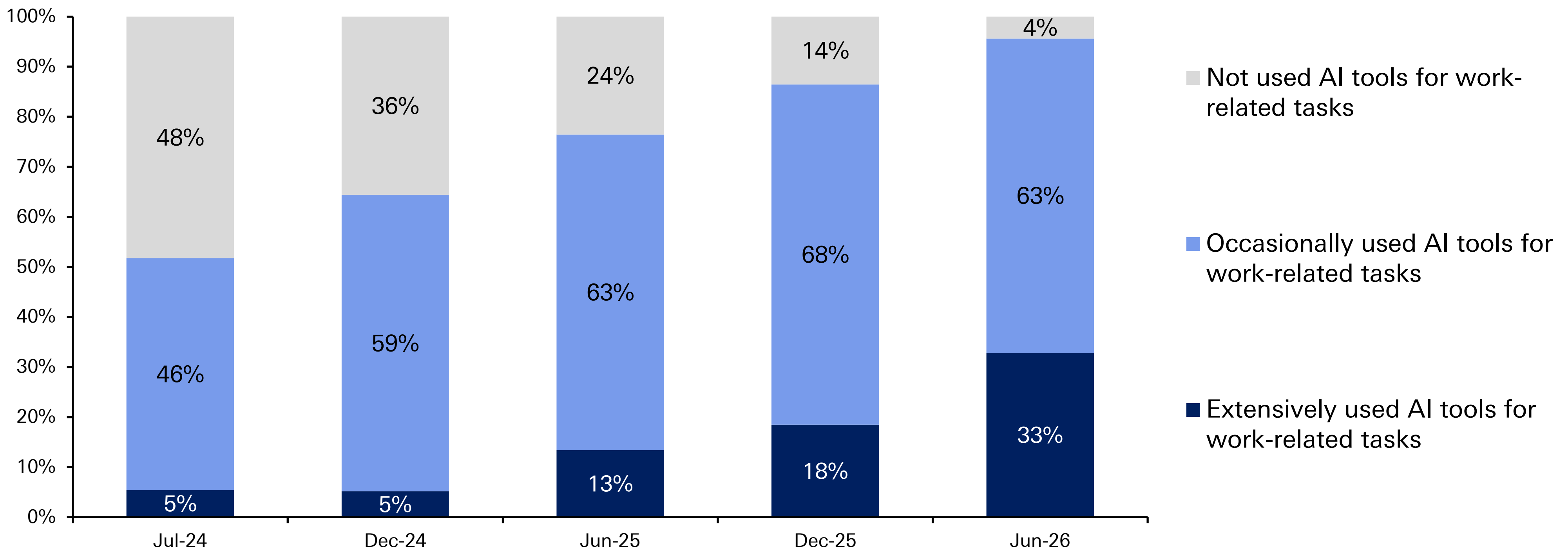
Using a scale of 0 to 10, where 0 is 'Not at all concerned' and 10 is 'Extremely concerned', how concerned are you that you will need to look for a new job within the following time frames because of AI?



Source: dbDataInsights Survey, Deutsche Bank Research

Since March, AI adoption at work continues to accelerate as it has done steadily since we first asked 2 years ago. Now only 4% have not used AI for work, compared to 14% last December and 48% 2 years ago. "Extensively used" has also risen from 18% to 33% now since December – a pretty big increase.

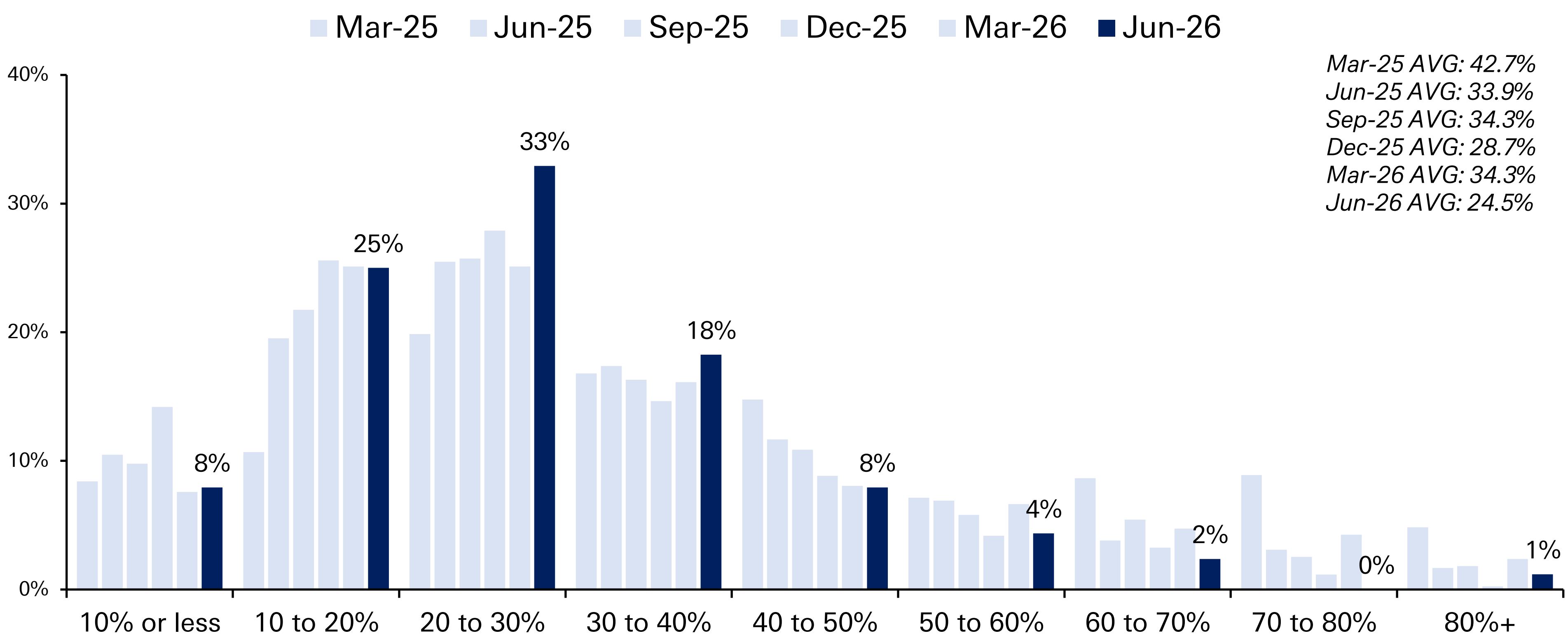
Which of the following best describes your use of AI tools for work-related tasks within the last three months?



Source: dbDataInsights Survey, Deutsche Bank Research

Average perception of 12-month US recession risk is back down to 24.5%, its lowest score since we asked the question in this format last March. Back then it was 42.7% – and even this March, when the Iran War was in its early stages, it was 34.3%.

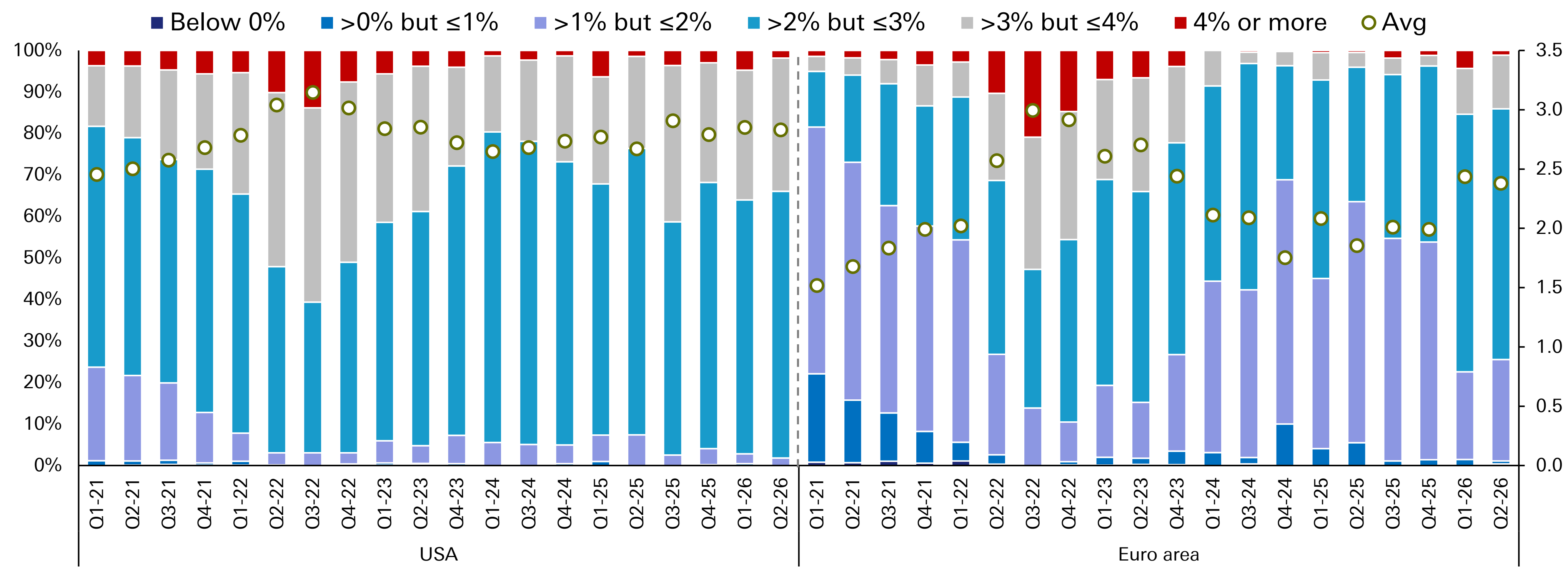
What do you think the probability of a US recession in the next 12 months is?



Source: dbDataInsights Survey, Deutsche Bank Research

Europe inflation expectations over the next 5 years are unchanged from March, at 2.4%. 5-year US inflation expectations have edged back down to 2.8% from 2.9% in March.

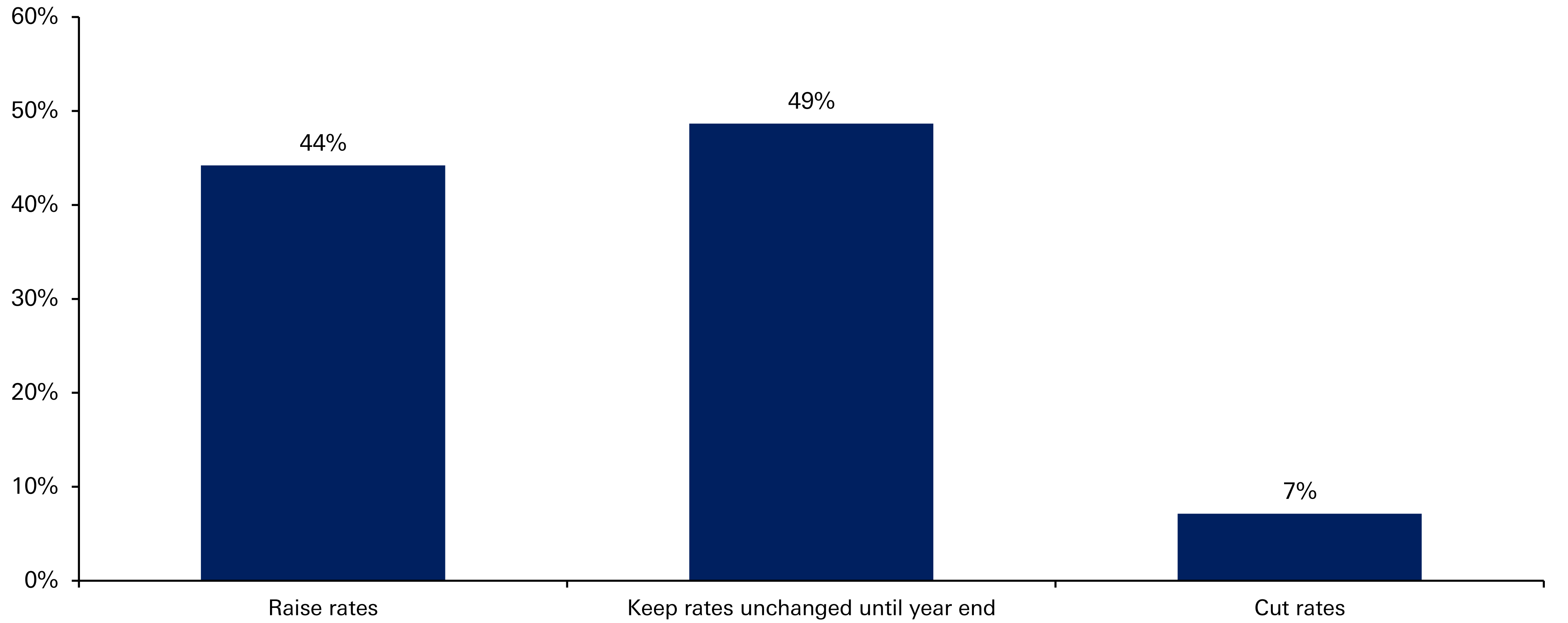
Looking at the below, what do you think inflation will average in the following regions over the next 5 years?



Source: dbDataInsights Survey, Deutsche Bank Research

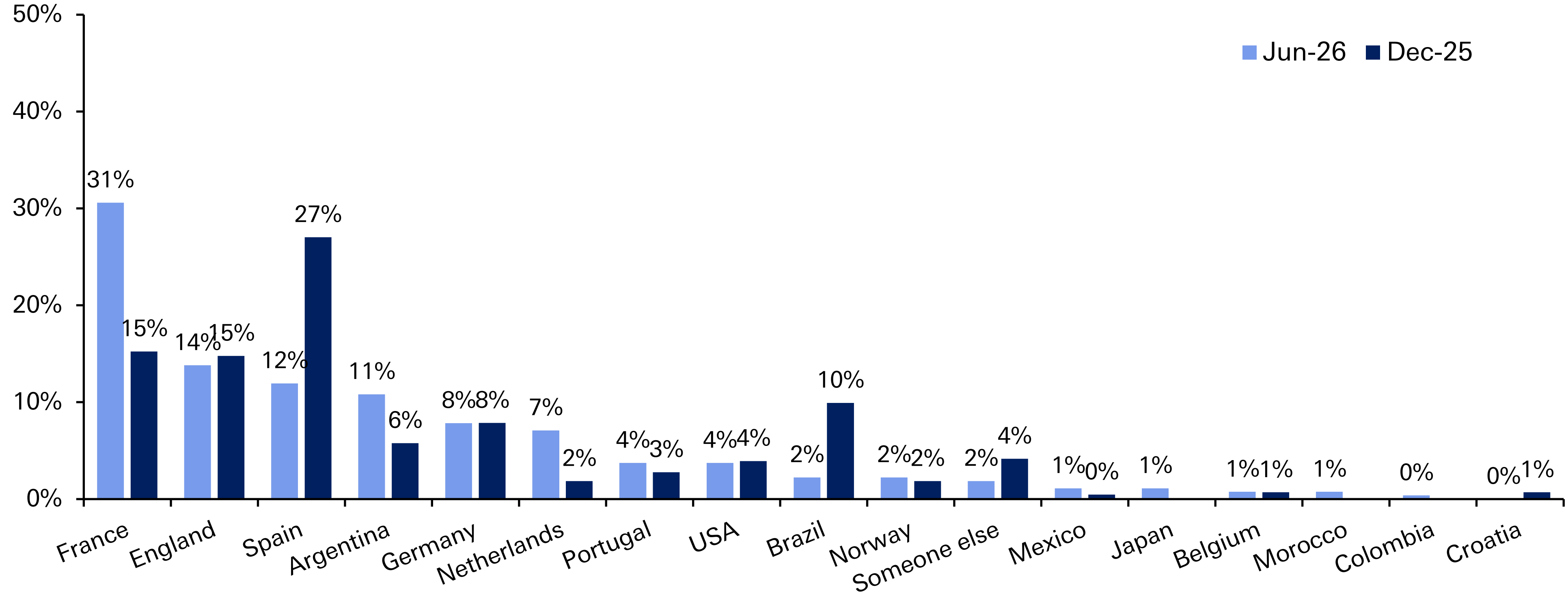
Markets are pricing in almost one-and-a-half hikes by December. Yet perhaps the receding threat of an oil-driven shock has made our respondents lean slightly more dovish, with 49% thinking the Fed will stay on hold this year and 7% thinking cuts are still on the agenda. 44% think a hike is coming.

What do you think the Fed's next move will be in 2026?



And finally, a majority (31%) think France will win the World Cup this year, followed by England (14%). Spain and Brazil have seen the biggest shifts in opinion since December.

Who do you think will win the 2026 football World Cup?



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US at 250 – Why has the US been so successful and can it continue?

June 24, 2026

US at 250

To mark the 250 year anniversary of the US Declaration of Independence, this paper looks at how the US emerged as a global superpower, and why it's likely to remain one despite new challenges.

First, we consider how the US went from being a comparatively small country to the world's pre-eminent global power. These reasons range from the US' natural advantages, like favourable geography, to factors like its institutional stability and risk-tolerant capital markets.

We then consider the challenges that threaten US outperformance. China's rapid growth means the US faces its biggest rival since its own emergence as the world's dominant power. China's rise also coincides with several other issues: the rules-based international system that the US helped create is under immense strain; the reserve currency status of the US Dollar is under pressure; and the country's public debt-to-GDP ratio is set to hit new records in the coming years.

Finally, we look at why the US is likely to sustain its outperformance, thanks to a collection of reinforcing advantages. In fact, the US has consistently emerged from challenging periods successfully, be that after the Great Depression, the malaise of the 1970s, and again around the GFC and its aftermath. We conclude with a few lessons for investors and for the rest of the world.

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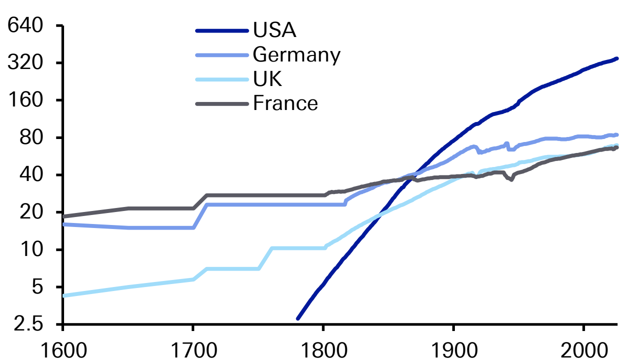
1. The Lessons from History: What underpins America's Success?

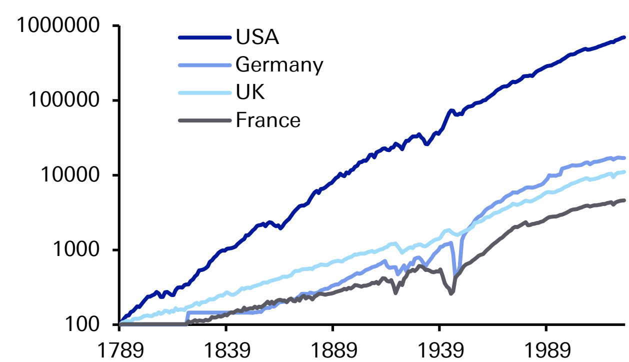
Before delving into the drivers of US success, let us bring with a few highlights of its historical outperformance. Since the United States' founding, the country has achieved remarkable economic success on a whole range of metrics. For one, the US saw rapid growth in population. It started as a comparatively small entity at its founding, but it quickly surged to overtake the large European countries in the mid-19th century and has continued to outpace them since. And with more people, it was little surprise that US GDP rapidly outpaced other countries too.

Figure 1: Population on a log scale (millions)



Source : Finaeon, Deutsche Bank

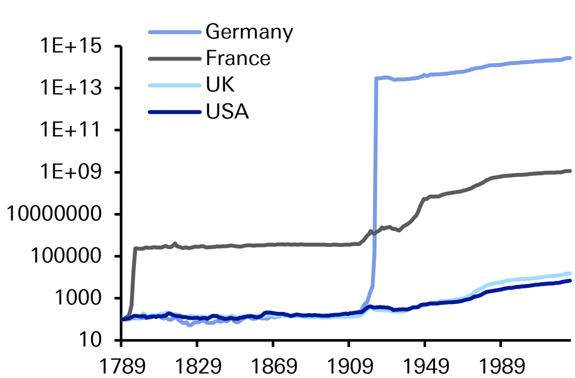
Figure 2: Real GDP on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

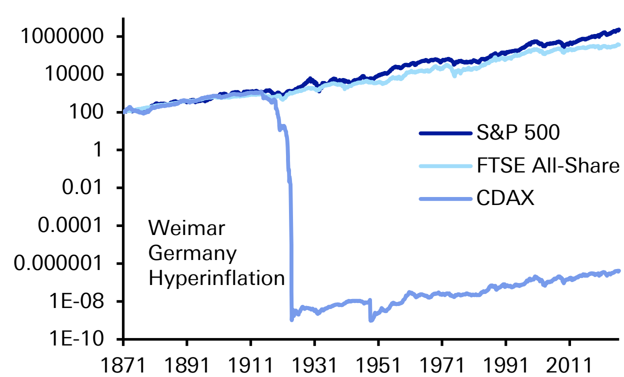
This success extends beyond GDP growth. Unlike France or Germany, the US has never had a bout of hyperinflation in its history, with its currency maintaining its value comparatively well. Meanwhile, the US' long-term equity performance has also been very strong. In the time since equity data is available from the late-19th century, real returns have outpaced the UK and Germany by substantial margins.

Figure 3: Consumer Price Index on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

Figure 4: Real equity returns on a log scale (Jan 1871 = 100)



Source : Finaeon, Deutsche Bank

So what has caused this incredible success?

The first reason, and one of the most critical historically, has been the US' political and institutional stability. Clearly, there have been moments of intense turmoil, most notably with the Civil War in the 1860s. Its historical path also should not be over-romanticised, with US territorial expansion during the 19th century having many similarities to that of the Old World colonial empires. Nonetheless, the US is incredibly rare in that its political system is recognisably the same over the last 200 years. This relative stability and early development of property rights provided a fertile environment for long-term investments, which in turn has aided economic growth over the centuries.

The second reason is its geographic advantages. The US possesses vast arable land, navigable rivers, large coastlines, and access to two oceans. It therefore found itself more insulated from the destructive effects of the world wars, with productive capacity not impacted in the same way. Moreover, the country borders Mexico and Canada, who in both population and GDP terms are much smaller than the United States, meaning it didn't face the security risks that many European powers faced over the 19th and early-20th centuries.

The third reason has been its abundance of energy resources, particularly relative to Europe. In large part a corollary of its geographic strengths, this energy abundance has given the US several advantages. First of all, lowering costs for households and industry, which has helped the economy be more resilient against geopolitical shocks. Moreover, with the US becoming a net energy exporter in recent years, it also strengthens the external position.

The fourth reason for the US' relative success was that its main competitors in Europe were deeply affected by the world wars and wider political turmoil. Their productive and financial capacity was severely degraded, with the destruction causing huge loss of life as well. Even among those who survived, many scientists, engineers and entrepreneurs left for the US in the first half of the 20th century. Whilst not a US "success" as such, this meant that on a relative basis, the US' divergence widened considerably. Indeed, in the first half of the 20th century, the US economy grew by 5.7 times, Germany by 3.4 times, the UK by 2.0 times, and France by 1.8 times.

The fifth reason for the US' outperformance is scale. It has a large domestic market of over 300m people, with high average incomes, a common language, and low internal barriers to trade. This has given firms the ability to reach a large-scale domestically before expanding abroad. Indeed, it is notable that 8 of the world's 10 largest firms are based in the US, whereas none are in Europe.

The sixth reason is the structural advantage of the US Dollar, which remains dominant in global trade and FX reserves. This dollar demand matters because it lowers borrowing costs and raises demand for US Treasuries. In turn, it leaves the US with an exceptional capacity to run bigger fiscal and external deficits without facing a funding crisis, and expands the geopolitical leverage that the US possesses. This has been dubbed the "exorbitant privilege", and has been a major advantage in recent decades.

The seventh reason is financial depth. The US has a large banking system, but also a wide range of non-bank financing, which means startups have access to other sources of capital. In fact, in the decade from 2013 to 2023, annual venture capital financing was 0.7% of GDP in the US, compared to just 0.2% in the EU. This financial depth is important because innovative firms can often be loss-making for lengthy periods, so countries with bigger pools of patient risk capital are better placed to commercialise new technologies.

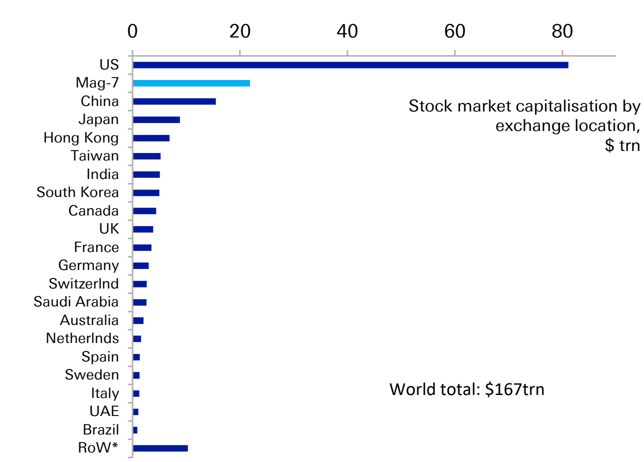
The eighth reason is its self-compounding advantages in education and research. The US has many of the world's strongest research universities, including 7 of the top 10 in the Times Higher Education World University Rankings for 2026. Moreover, this research also has an economic angle, as scientific discoveries feed into startups, workforce training and industrial scale-ups. In turn, this becomes self-compounding, because top global talent is attracted to the US, and ensures it remains at the forefront of new sectors such as artificial intelligence.

The ninth reason is its pro-business architecture, including a greater tolerance of business failure than many European systems. For instance, Chapter 11 reorganisation is designed to preserve and restructure viable firms rather than liquidate them. This more positive approach to business failure ties into the empirical literature, which suggests that more debtor-friendly and efficient insolvency frameworks are associated with greater entrepreneurship and innovation.

The tenth reason is adaptability. The cultural acceptance of failure and capacity to reinvent itself have boosted US ability to adapt to the changing world. This has allowed it to navigate repeat boom-and-bust cycles, as well swings of the policy pendulum – between openness and isolationism, between protectionism and free trade – without threatening overall the institutional stability we highlighted above. And while capitalism has been a key underlying driving force, it is pragmatism rather than ideology that have driven continued success over time.

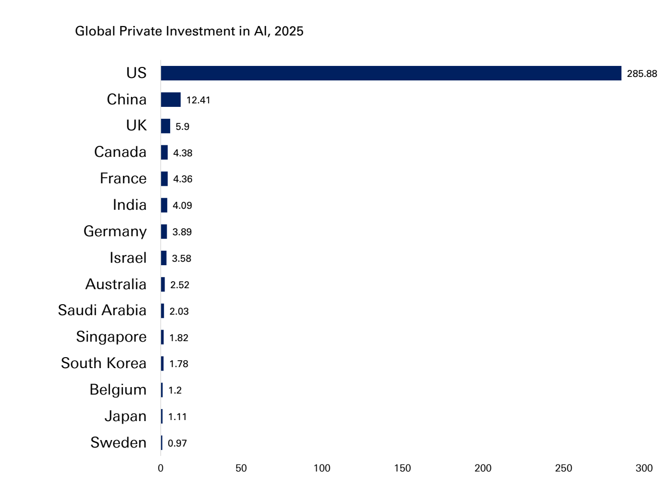
It is also important to note that these advantages do not play out individually, but are mutually reinforcing, with the interaction between them allowing the US to benefit from network and externality effects that few if any countries can match.

Figure 5: The US accounts for nearly half of the world's public stock market capitalisation



Source : Bloomberg Finance LP, Deutsche Bank Research; Data as of 22 June 2026
** Rest of World comprises of a further 62 country exchanges.*

Figure 6: US private AI investment reached $285bn in 2025, more 20x as much as in China



Source : Stanford University 2026 Artificial Intelligence Index Report, Deutsche Bank Research

US history has been marked by challenges and recoveries

That said, this success story was far from a straight line. **US outperformance has frequently been questioned, yet it has repeatedly defied the sceptics, including at numerous points in the last century.**

Right from the country's founding, questions were raised about its long-term potential, and in the Civil War of the 1860s, the nation's survival was at stake after less than a century.

Even over the past 100 years, which has ostensibly been a period of US pre-eminence, there have frequently been doubts. For example, after the Wall Street Crash of 1929, the **Great Depression** saw unemployment peak above 25% and remain above 10% for an entire decade. The associated stock market collapse saw the S&P 500 fall by -86% from peak-to-trough, not reaching its 1929 peak again until 1954. This was a global shock, but it undermined faith in the US system of free-market capitalism.

However, the country eventually pulled out of the slump. That started with Franklin Roosevelt's New Deal program, which helped to restore confidence and economic activity. Then the start of WWII saw economic activity surge even further, particularly as there wasn't fighting on US soil. By the end of WWII, the US had never been in a stronger position relative to its peers, not least with Europe having to rebuild after the war.

Similar doubts were clear as the US grappled with the various **crises of the 1960s and 70s**. This period saw major political turmoil and multiple assassinations, including President Kennedy in 1963, his brother and presidential candidate Robert Kennedy in 1968, and civil rights leader Martin Luther King Jr. in 1968. A few years later, Richard Nixon became the first president to resign from office amidst the Watergate scandal. US military power was also facing questions at this time, as the Vietnam War turned into a protracted conflict that also faced substantial domestic opposition.

Then on the economic front, inflation started to gather pace from the late-1960s, surging further after the first oil shock of 1973, leading to a 16-month recession. Measures were imposed to conserve energy, including the National Maximum Speed Limit, which wasn't repealed until 1995. Then in 1979, a second oil shock drove inflation up again.

This turmoil raised questions about whether policymakers could effectively tackle the crises of the day. In 1979, President Jimmy Carter delivered what was widely referred to as the "malaise" speech, although it was actually called "A crisis of confidence". He said there was "growing doubt about the meaning of our own lives and in the loss of a unity of purpose for our nation." He spoke of a "growing disrespect" for institutions, and said how the public often "see paralysis and stagnation and drift." Even the US president was acknowledging that the country was facing huge issues.

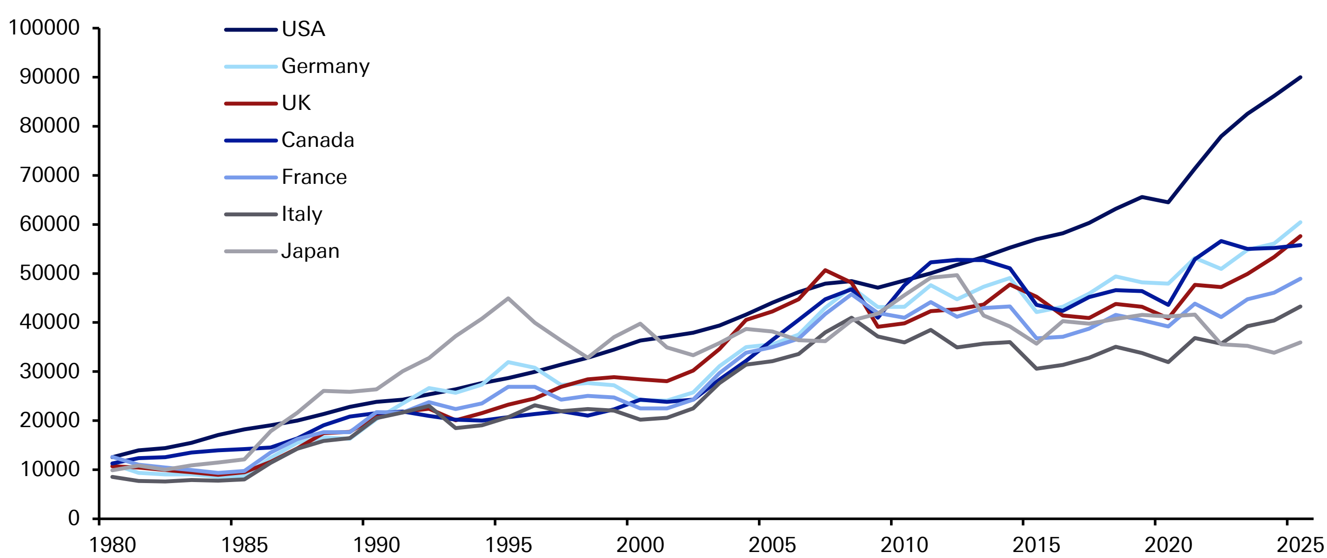
Nevertheless, the US then recovered strongly into the 1980s and 90s. Inflation was tamed, albeit with drastic monetary tightening, and the economy saw rapid growth once it recovered from the early 1980s recession. In 1991, the dissolution of the Soviet Union marked an end to the Cold War, and the 1990s were widely considered a unipolar moment where the US was left as the unmatched global superpower. So once again, the US came through a period of doubt over its success.

But the unipolar moment proved brief, as the **global financial crisis** cast fresh doubt over US prestige. Unemployment surged above 10%, and the economy entered an 18-month recession, the longest since the Great Depression. Moreover, the subsequent recovery was very sluggish, with growth rates well below those of previous decades. Economists resurrected the idea of "secular stagnation", originally put forward by Alvin Hansen in the 1930s, suggesting the US faced an era of permanently slower growth.

These questions about US pre-eminence came as China was growing rapidly at the same time. In 2005, China's GDP (in USD terms) was still only 18% of the US total, but that surged to 62% by 2015. That was an incredible shift in relative economic weight in the space of a decade.

But for all the doubts of the 2010s, the secular stagnation narrative has since disappeared in the 2020s. The US economy bounced back remarkably quickly from the pandemic, aided by huge quantities of stimulus that were far larger than 2008. Moreover, recent years have seen the US at the forefront of AI developments, raising hopes of a new productivity surge, with its lead extending over the other advanced economies. So yet again today, the US has emerged from a rockier period around the financial crisis and its aftermath, into a relatively strong position that includes the highest GDP per capita of any G7 country.

Figure 7: GDP per capita (USD terms)



Source : Haver Analytics, Deutsche Bank

2. The Challenges Ahead: Why the US now faces a crucial decade

The above discussion highlights how the US has repeatedly overcome periods of doubt, with predictions of American decline proving premature each time. However, in several respects today's challenges feel more acute. The US is not as dominant a geopolitical power as it used to be, given China's emergence as a serious rival. Moreover, its debt-to-GDP ratio is on the verge of new records, meaning its fiscal space is far more constrained to deal with new challenges. As such, the next decade will offer a further test of US adaptability, and whether the political system can respond to these interacting pressures.

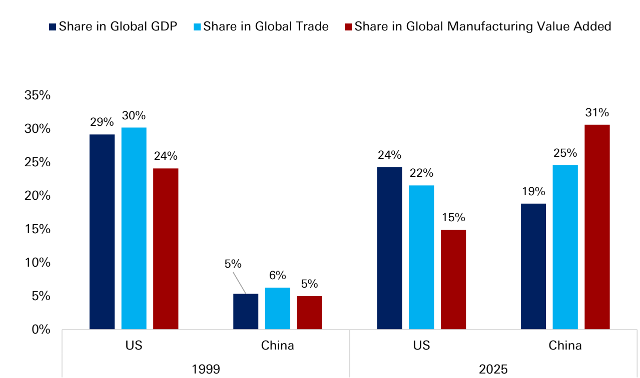
What are the biggest near-term challenges for the United States?

Since the end of the Cold War, the most significant adjustment has been the end of the unipolar moment and China's emergence as a genuine peer competitor, which is different in scale and capability from any rival the US has faced since the late-19th century. China has already overtaken the US on manufacturing output, merchandise trade, and GDP (on a PPP basis), and is rapidly closing the gap in advanced technologies and semiconductors. The US retains decisive leads in nominal GDP, capital markets depth, the dollar system, and frontier innovation, but the gap has narrowed substantially.

The military balance tells a similar story. US defence spending remains roughly three times China's at market exchange rates, but the PPP-adjusted gap is far smaller. More broadly, the gap between US military spending and other major economies has shrunk over the past 15 years. That comes as the proliferation of low-cost precision and autonomous systems is eroding the historic advantages of expeditionary power projection, raising the cost of underwriting the global security order the US has provided since 1945.

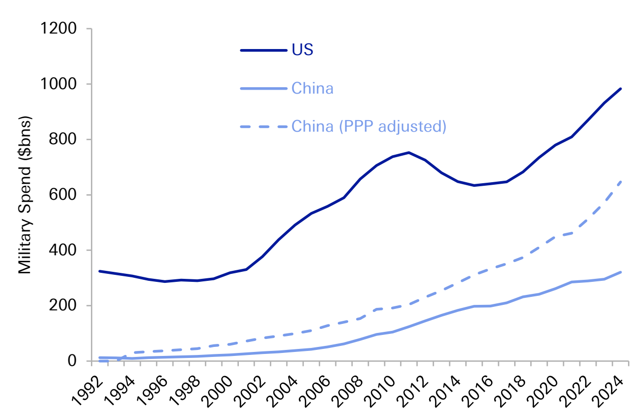
We can also see the emergence of Chinese leadership in a whole range of areas. Among others, China accounts for a third of all global manufacturing activity, with a sector as large as the US, Japan, Germany and South Korea combined. The cost of electricity there is a fraction of what people pay in Europe or the US. And just as the US dollar's role has come under pressure, China is strengthening the global influence of the RMB, in part because of its vast domestic savings surplus.

Figure 8: China has overtaken the US as the world's largest trade and manufacturing hub, though the US is still ahead on nominal GDP



Source : Deutsche Bank Research, UN Industrial Development Organization, Haver. All in Constant 2020 USD, Data for 2025

Figure 9: US lead on military spending has narrowed over the past 15 years



Source : World Development Indicators, World Bank Data, Deutsche Bank Research

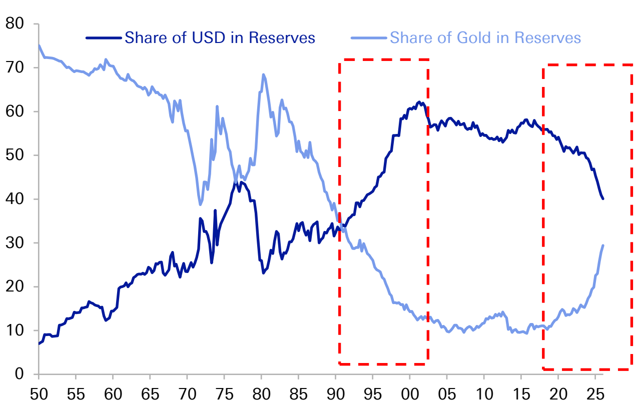
The emergence of China comes as the rules-based international system designed and led by the US for eight decades is under strain from multiple directions. Admittedly, these strains have not just emerged. Even before the Trump administration, emerging markets had grown in relative size and become less willing to accept Western-designed institutions. Meanwhile, the political support for globalisation has frayed across advanced economies.

But recent US policy choices have raised profound questions about the future global order. In 2025, the US effective tariff rate rose to its highest since the 1930s, representing the largest peacetime trade barrier increase in a century. Strategically, the US alliance network has been an enormous economic asset, underwriting dollar dominance, securing trade routes, and providing the soft-power foundation for global rule-setting. Yet US membership of NATO has become an openly debated question, and tariffs have been raised on US allies as well as rivals.

This uncertainty has contributed to growing pressure on the US' status as the global reserve currency. The dollar's share of global reserves has fallen from roughly 72% to 58% over two decades — a gradual decline rather than a collapse, but a clear trend nevertheless. The reduction has accrued to a basket of "non-traditional" reserves and to gold, which has seen the largest central bank buying programme in over half a century. In addition to the overall transition away from a 1990s unipolar world, specific factors have added to the challenges that dollar dominance faces. This includes the weaponisation of the dollar through sanctions — notably the freezing of Russian central bank reserves in 2022, which gave non-Western-aligned countries a strategic incentive to develop alternatives even at material cost — as well as the structural decline of the petrodollar arrangement as energy trade has fragmented and the US has become a net oil exporter.

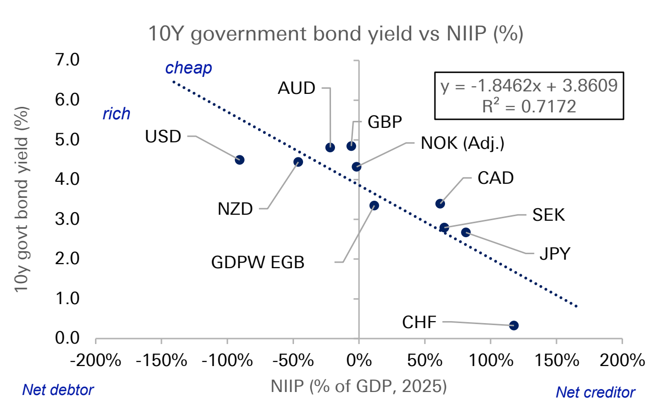
Irrespective of the cause, were the US "exorbitant privilege" to erode further, this would bring major costs to the US and allow other currencies to benefit (see also the DBRI note last year on the potential benefits for the euro). As mentioned earlier, the structural subsidy that the US receives from its reserve currency status includes lower borrowing costs, the ability to run persistent external deficits without the discipline imposed on other economies and the advantages of the dollar being the unit of account for global trade, commodities and cross-border lending.

Figure 10: The "end of history" in the 1990s saw a sharp rise in the USD and decline in gold's share of reserves



Source : Deutsche Bank, Bloomberg Finance LP, Eichengreen, Chitu and Mehl (2014), IMF International Liquidity.

Figure 11: 10yr yields vs Net international investment Position – An illustration of US "exorbitant privilege"



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP, IMF
Updated 22 June 2026

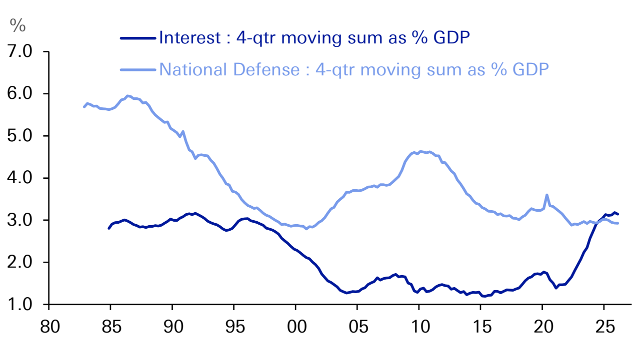
None of this implies imminent displacement. No alternative is remotely ready to assume the dollar's role. Network effects and inertia in trade invoicing and central bank balance sheets all favour continued dollar dominance. Indeed, the pound sterling held a prominent role for many decades after the US had emerged as the world's largest economy. So the realistic risk for the next decade is not collapse but gradual erosion, with a slow loss of the privilege margin that has boosted US economic performance.

The US fiscal trajectory is the most plausible catalyst to accelerate that erosion, and for institutional investors is the single most concrete macroeconomic risk facing the United States. Over the last four years since 2022, the federal deficit has been consistently running at around 5-6% of GDP. Those are the highest peacetime deficits in US history outside of a major recession, and this is happening in a full-employment economy. Debt held by the public is set to surpass 100% of GDP this year, and the CBO projections point to an unsustainable trajectory. Interest payments on the federal debt now exceed defence spending and are the fastest-growing line item in the budget.

Most significantly, the binding constraints are now set to arrive within the next decade. The latest estimates project that the Social Security trust fund will be depleted by late 2032, which would trigger an automatic benefit cut absent legislative action. The Medicare hospital insurance trust fund faces a similar reckoning shortly after. So while the unsustainability of the US public debt trajectory has been in the headlines for many years, these events are now more imminent – ones that the next US administration that takes office after the 2028 election will have to face.

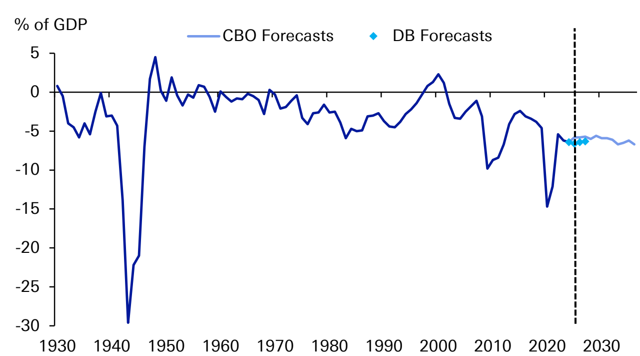
To be fair, the fiscal challenge is far from arithmetically catastrophic – a combination of AI-driven productivity growth and moderate fiscal consolidation could lead to significant improvements in the long-term debt profile. The difficulty is whether the political system can deliver such an outcome. Past fiscal consolidations required bipartisan compromise. Yet the current political environment offers little prospect of similar cooperation, with polarisation a mounting issue. Attempts in recent decades (like the Simpson-Bowles Commission in 2010) have failed to gain sufficient support.

Figure 12: The Crossover: Rising Federal Interest Payments have Overtaken Defense Spending



Source :CBO, OMB, Haver Analytics, Deutsche Bank Research

Figure 13: A Century of Deficits: US Federal Budget Balance from the Great Depression to the 2030s

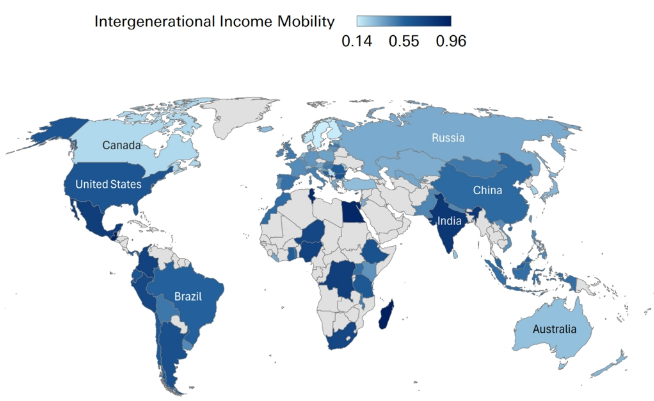


Source : CBO, OMB, Haver Analytics, Deutsche Bank Research

In the meantime, demographic trends are compounding both the fiscal and innovation challenges. The US fertility rate has fallen to 1.6 and the dependency ratio is set to deteriorate sharply as the baby boomer cohort fully enters retirement. Even though the population is comparatively younger than many other developed countries, it is still ageing. For example, the UN projects that the share of the US population aged 65 and over will rise from 19% in 2026 to 22% by 2036, rising further to 28% by the end of the century.

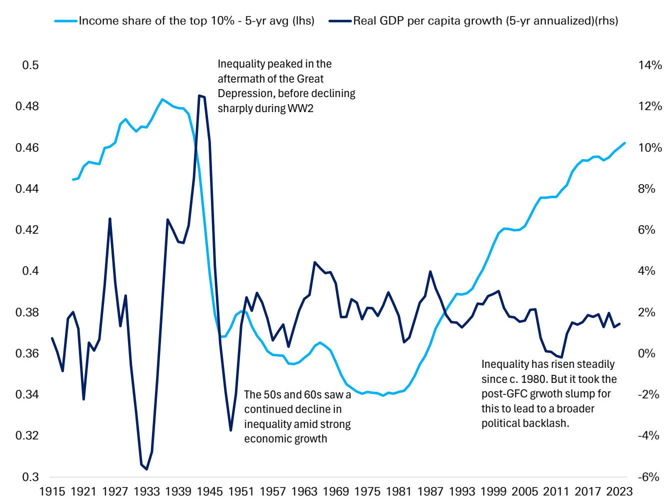
An important connecting driver across fiscal, political and demographic challenges has been a rise in inequality. Its steady rise since the 1980s has gone hand-in-hand with a rise in political polarisation, especially once combined with the economic shock that followed the GFC and the sluggish recovery in its aftermath. A corollary of this is that the traditional "American dream" of social mobility has found itself under mounting pressure. The US now has lower intergenerational income mobility than almost all DM economies, which risks corroding the social contract on which open markets and creative destruction depend. It has also led to negative social effects, notably a widening of the life expectancy gap between Europe and the US since the 1980s.

Figure 14: The US has lower intergenerational mobility than most DM economies
(lower score = higher mobility)



Source : World Bank Data, Deutsche Bank Research

Figure 15: The rise in inequality since 1980 added to the political challenges facing the US, especially after growth slowed post-GFC



Source : Finaeon, World Inequality Report, Deutsche Bank Research

3. Why the US Should Remain a Long-Term Success Story

The challenges facing the US are real, but the weight of evidence still suggests it will remain the world's leading economy for the foreseeable future. Its collective structural advantages remain difficult to replicate, and it retains the same adaptability that's helped it recover from previous difficulties.

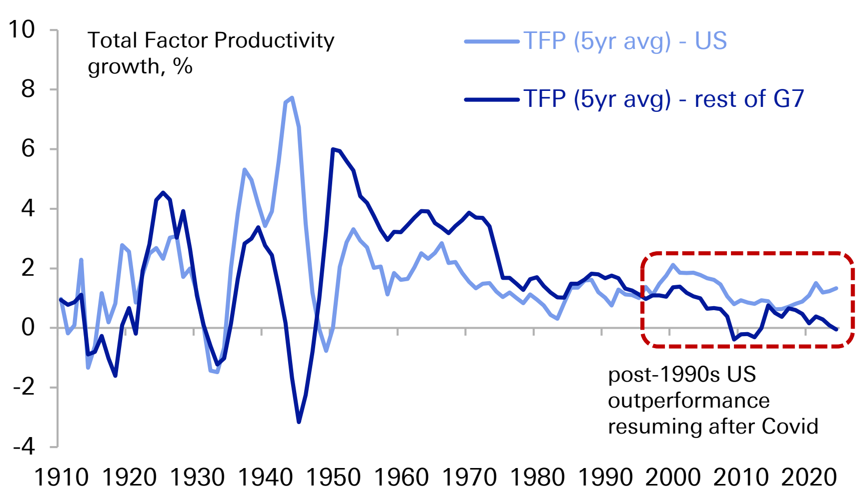
The starting point is that the US should remain the world's largest economy well beyond the critical decade coming up. A global demographic slowdown means that previous assumptions of rapid population and economic growth in emerging markets no longer look inevitable. Conversely, US economic growth has generally surprised on the upside since the pandemic, with the "secular stagnation" narrative of the 2010s quietly fading. Indeed, the US economy has successfully weathered multiple shocks in recent years, ranging from rapid Fed rate hikes, to tariffs, to the Iran conflict. Moreover, there is further upside potential to productivity growth (and hence economic growth) from the US' significant lead on AI.

This AI boom is the latest expression of the US capacity for reinvention that stretches back through its history: a willingness to embrace disruptive, capital-intensive technology and absorb the dislocation it brings. AI is one of many new technologies that have often led to economic volatility (and boom-bust cycles on occasion), but whose benefits have delivered broad, economy-wide productivity gains over time. That was clear historically with canals, railroads, and electrification, and also with more recent innovations like the internet.

Thus, the US is well-positioned to lead the AI cycle for several mutually reinforcing reasons: an innovation-friendly ecosystem of research universities and venture capital; a cultural and institutional acceptance of the boom-bust investment cycles that frontier technology requires; the deepest capital markets in the world to fund it; and abundant domestic energy. The shale revolution has transformed the US into a net energy exporter, and cheap, plentiful power is now a decisive input advantage in the race to build and run data centres. Few competitors can match this combination.

The AI boost comes at a time when the US has already re-established its productivity outperformane in recent years. Since the pandemic, US productivity growth has risen above its post-1970 average, which in turn is boosting GDP and the revenue base. The potential fiscal arithmetic is powerful. According to CBO estimates, if productivity growth were just half a point above their baseline in the coming decade, then debt held by the public would only rise to 110% rather than 120% by 2036 – see our colleagues' note on the US debt outlook last autumn for more. However, productivity gains alone are unlikely to resolve the fiscal issue, in part because the political incentive is to spend the fiscal dividend rather than save it and in part given the uncertainty of how the gains would be distributed — a theme we return to below.

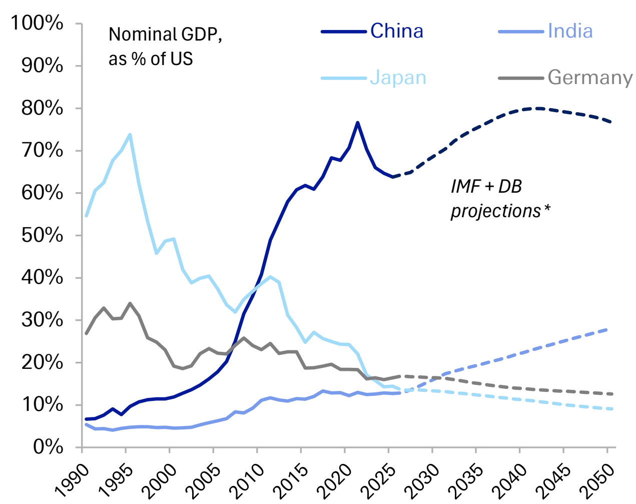
Figure 16: US has resumed its productivity outperformance in recent years



Source : Deutsche Bank Research, www.longtermproductivity.com
Rest of G7 is the simple average of Canada, France, Germany, Italy, Japan and the UK.

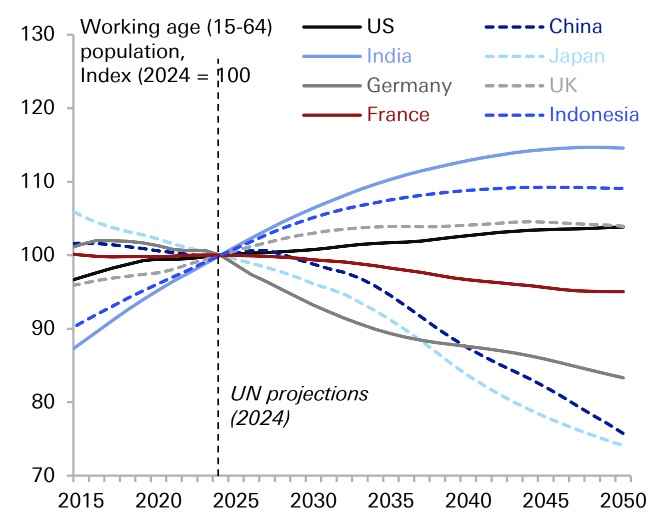
On demographics, the US also enjoys a clear advantage over its principal rivals. China's working-age population will begin a pronounced contraction in the 2030s, with Europe in aggregate also seeing a decline, whereas the US will see moderate growth under a normalised migration trend[1]. With a growing working-age population, the US would avoid this drag from a shrinking labour force. Moreover, the US has had a consistent capacity to absorb immigration at levels few other developed economies have managed. And with the demographic slowdown now reaching emerging markets, few major economies now have clearly better long-term demographic potential than the US. For instance, while India will maintain positive demographic momentum over the next couple of decades, with its fertility rate now below replacement rate, its advantage will shrink by the middle of the century.

Figure 17: US likely to remain the world's largest economy for the foreseeable future



Source : IMF World Economic Outlook, Deutsche Bank Research
** IMF projections to 2031; DB estimate to 2050 based on (i) UN population projections (2024), (ii) stable productivity growth among DM economies, (iii) gradually slowing labour productivity growth for China (to 2.0% by 2050) and India (3.5% by 2050), (iv) stable REER exchange rates.*

Figure 18: The US has better demographics than most DM countries and China, while India's advantage is set to peter out over time



Source : UN Population Database (2024), Deutsche Bank Research
Recent research suggests the 2024 UN demographic projections are likely too optimistic (e.g. www.mercatus.org/macro-musings/jesus-fernandez-villaverde-quandary-global-demographic-decline). However, this downside is largely global and unlikely to see relative US underperformance under normalised migrations trends.

Even on the financial side, whilst the dollar's role has come under pressure, it still retains a dominant and unrivalled position. The network effects that sustain it are exceptionally strong, and there is no obvious successor in the wings, unlike when the dollar overtook sterling in the interwar period. The depth and liquidity of US capital markets, inseparable from the dollar's role, are themselves a structural draw that even US rivals rely upon. Indeed, it is notable that the asset seeing a major increase in official reserves is gold, rather than another currency. And while new payment technologies could bring new risks to dollar dominance, US dominance in the stablecoin space will help limit these risks for the foreseeable future.

1 The UN population projections assume net migration into the US of c. 1.2m annually in the coming years. This would be roughly in line with the average migration rate since 1960 (0.3pp of population annually), while being well below the c.2.5m peak reached in 2023 but well above the c.300k latest Census projection for 2026.

The Conditions for Continued Success: Pitfalls to Avoid

Yet much as there are credible reasons for US outperformance to continue, this cannot be taken for granted.

The first key challenge is to ensure that the gains from future technological growth are broad-based, as excessive concentration risks adding to political and institutional dysfunction. We know from history that technological progress does not automatically deliver broadly shared prosperity. For instance, during the early Industrial Revolution, aggregate output rose for decades as wages for ordinary workers stagnated, a period that has been dubbed “Engels’ pause”. This offers a cautionary lesson on AI, and with the labour share of income in the US already near historic lows today, a further deterioration risks intensifying the social and political backlash building against the technology. It’s important to avoid that, as a backlash against technological advances would in turn negate its gains. A related risk is that if AI gains accrue to capital through profit margins while labour is shed, this would negate the fiscal benefits from higher productivity given the focus of the US tax system on labour taxation.

This brings us to the second challenge – managing the fiscal adjustment and the related timing risk. The US is currently running budget deficits at levels previously unprecedented outside of a major recession or war. The ideal scenario would be some sort of bipartisan consensus, as in the early 1990s, to deal with the fiscal challenges ahead in a pre-emptive manner. However, political incentives to reduce deficits appear limited, raising the prospect that it will be up to markets to ultimately force consolidation. The risk is that this could come at a moment of heightened vulnerability, such as a wider economic crisis or recession, resulting in a harsher correction. A degree of financial repression, especially if combined with genuine consolidation, could end up playing a role in managing the debt burden. But a full embrace of that route — in an extreme case, via debt monetisation by the Fed — would merely accelerate the erosion of the dollar's special status.

The third challenge is protecting the US’ institutional foundations. Institutional stability (and the consistent rule of law and secure property rights underpinning it) have been the bedrock on which the capital-markets and dollar advantages ultimately rest. If that eroded over the coming decades, it would create a more uncertain climate for long-term investments and dampen the international appeal that has attracted top talent and financial capital to the United States.

The fourth challenge is a global one, in that the US has benefited immensely from the post-war system of US leadership. However, that settlement has come under increasing strain in recent years, particularly as emerging markets that have grown in size seek greater influence in global affairs. After all, the US only makes up around 26% of global GDP in USD terms. But with its allies, the influence of the US increases substantially. For instance, the G7 as a whole makes up 44% of global GDP, and if you include further US allies such as Australia and South Korea, its effective alliance network still comprises more than 50% of global GDP in dollar terms. With the US standing as the clear and unsurpassed leader of this group, maintaining this network has enormous benefits.

What will the US’ ongoing success imply, and what are the key takeaways?

To wrap up, let us briefly consider two concluding questions. First, what does the above analysis mean for investors looking at the US over the long term. And second, what lessons can the rest of the world learn from the success of the US model over time.

For those allocating capital, the case for the US as the world's pre-eminent investment destination remains intact. An investor into the US buys a bundle that is hard to replicate – productivity leadership, the deepest and most liquid capital markets in the world, the rule of law that underpins them, and the flexibility conferred by the dollar's reserve role. That combination should persist. However, the risk premium around the US has risen, as fiscal strains and a wider geopolitical reshuffling point to a wider distribution of outcomes and bigger tail risks.

In practice, this could play out in several ways. First, the dollar's "exorbitant privilege" is unlikely to disappear within the next decade, but gradual diversification of both reserves and payments is real and persistent. Already, the dollar's global importance has ebbed in recent years. These growing question marks around the dollar’s status are the first risk. Second, US equity leadership is now heavily concentrated in a handful of AI-exposed names, so the bull case increasingly rests on a small group of companies, again widening the tail risk outcomes. And third, there is a tail risk of a correlated scenario where a fiscal reckoning coincides with a correction in the AI investment cycle, pressuring equities, duration and the dollar simultaneously, and undermining the diversification that normally cushions portfolios. This is a scenario worth planning for, even if it is not the central case.

What can the rest of the world learn from the US’ success? In part, it is simply down to luck, including geographic advantages and an abundance of resources, such as energy, that cannot be replicated elsewhere. However, there are several features that could be reproduced elsewhere.

The first is its resilient institutional architecture, with a legal system and property rights that have provided a favourable environment for long-term investment. Although the US has faced severe crises in the last 250 years, including a civil war, its system of government has been recognisably the same throughout that time.

Second, it has a well developed financial system, and a tolerance for risk that’s enabled the US to embrace new technologies and emerge quicker from downturns, with resources shifting quickly to new and productive sectors. US history demonstrates how suppressing volatility and preventing resource re-allocation can be counterproductive over the long term.

Third, a key component of US success has been its adaptability to different contexts, and its ability to leverage existing strengths when under pressure. These range across its energy resources, capital markets, universities, military power, reserve currency status, and technological leadership. Even when one point has appeared under threat, others have been able to compensate.

Appendix 1

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Remembering Alan Greenspan

June 22, 2026

Authors

Peter Hooper
Vice-Chair of Research

First Impressions

Alan Greenspan, who passed away today at the age of 100, was one of the most consequential figures in the history of American economic policy—and someone I had the privilege of working with for more than two decades, first as a colleague at the Federal Reserve and later as a friend and adviser at Deutsche Bank. His passion for economic analysis, his respect for data, and his willingness to challenge conventional wisdom profoundly influenced my own approach to economics and policy.

I joined the Federal Reserve staff in September 1973 as a newly minted PhD in Economics from the University of Michigan—four years after my new Fed colleague Don Kohn—and served through the chairmanships of Arthur Burns, G. William Miller, and Paul Volcker. When Greenspan arrived in August 1987, I was in mid-level management in the Board's International Finance economic research staff. I still remember the day he arrived. He toured the building with my colleague Ted Truman, meeting many members of the staff, and I had the opportunity to shake his hand and welcome him personally. It was a small gesture, but a characteristic one: Greenspan appreciated the value of his staff and moved quickly to engage them.

A Remarkable User of Expertise

That engagement was immediately substantive. Greenspan had ambitious ideas about how to use the Fed's deep reservoir of talent. He was fascinated by data—not only the headline numbers but obscure series and unusual relationships that others overlooked.

Much of my early Fed career predated the Internet. Yet the institution possessed an extraordinary asset: specialists in virtually every corner of the economy. One could simply pick up the phone and speak with an expert on housing, productivity, corporate profits, or almost any topic imaginable. It was, in effect, a pre-Internet web of distributed expertise. Greenspan understood this instinctively and made exceptional use of it.

Don Kohn, in his remembrance published today at Brookings, recalled that Greenspan arrived "like a kid in a candy store," eager to tap the knowledge embedded throughout the organization. That description captured him perfectly.

Models, Markets, and Instinct

Greenspan brought macroeconomic instincts somewhat different from those of most Fed staff. I had been trained in the neo-Keynesian tradition, while Alan placed greater faith in markets and the informational efficiency of prices.

Having spent much of my own career modeling trade flows and global economic interactions, I came to appreciate both the power and limitations of formal econometric models. Greenspan shared my strong support for free trade, but he was less inclined than many Fed economists to rely heavily on model-based

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forecasts. Instead, he searched for revealing relationships in data and drew heavily on information gathered from his extensive network of business leaders and industry contacts.

Over the years I occasionally worked on speeches for him, mostly in an editorial capacity, since he inevitably rewrote drafts into precisely what he wanted to say.

"Irrational Exuberance"

One memorable episode involved the now-famous phrase "irrational exuberance" in a speech prepared for the American Enterprise Institute in December 1996. Several staff members, myself included, were skeptical. Greenspan insisted. History vindicated him.

Years later I asked him the secret of his remarkable longevity and productivity. His answer surprised me: soaking in a hot tub. The same ritual that, as he later recalled, had inspired the concept of "irrational exuberance" sustained him through his tenth decade.

The Productivity Revolution

The 1990s were, in many ways, a magical time for the Federal Reserve. Greenspan earned his reputation as the "Maestro"—the title Bob Woodward gave him in his 2000 book—in large part because of his success in steering the economy through the long expansion of that decade.

By the mid-1990s, many policymakers worried that growth was running beyond the economy's capacity and that inflation pressures were building. Greenspan disagreed. He became convinced that information technology and the computerization of business processes were producing a genuine acceleration in productivity growth, allowing the economy to sustain faster expansion without inflation.

Greenspan's recognition of the productivity surge before most others demonstrated his ability to identify anomalies that did not fit conventional wisdom and pursue them to their logical conclusion. He was right. The economy grew faster, and inflation remained subdued. My good friend Steve Landefeld, former Director of the Bureau of Economic Analysis, well remembers how Alan, one of their most ardent supporters, frequently challenged him to expand their measures of sustainable GDP growth.

Communication and Transparency

Greenspan believed that some ambiguity served monetary policy well. He thought the central bank retained greater flexibility when markets focused on incoming economic data rather than endlessly parsing every word uttered by policymakers.

Nevertheless, under his leadership the Fed began its gradual move toward greater transparency, introducing post-meeting statements and the policy bias. Ben Bernanke later accelerated that transition through press conferences, forward guidance, and eventually the dot plot.

Bringing Greenspan to Deutsche Bank

Greenspan was more than a mentor to me; he was an inspiration who demonstrated the importance of rigorous analysis and intellectual curiosity.

He was gracious enough to acknowledge our relationship in his own way. When he inscribed a copy of his memoir, The Age of Turbulence, he referred to me as a mentor—a generous inversion that I treasure, even though I would reverse the description.

After leaving the Fed in 1999 and joining Deutsche Bank, I became convinced that Alan would make a valuable adviser to the firm. With David Folkerts-Landau's blessing, I approached him after his retirement in January 2006, but he wanted first to finish his book. Months later, after learning he had accepted a position with PIMCO, I reminded him of our earlier conversation. He agreed to join Deutsche Bank as a senior adviser for two years, beginning in 2007.

Around the World with the Maestro

Alan saw clients globally for Deutsche Bank, and the attraction was exactly what the firm had hoped when he joined as senior adviser: access to his strategic insight and his extraordinary perspective on markets, monetary policy, and risk. Clients everywhere relished hearing the Maestro's latest thinking—not simply because he had once chaired the Federal Reserve, but because he could translate complex economic developments into practical judgments that sophisticated market participants could use.

Intellectual Honesty

Alan inspired me in many ways, though we had our differences. I was uncomfortable with his early confidence that private institutions could regulate themselves more effectively than government authorities—a view that events eventually proved overly optimistic.

Yet those disagreements never diminished my admiration for him. What mattered most was that Alan continued to interrogate his own views when events challenged them. Testifying before Congress in October 2008, Greenspan acknowledged that he had "made a mistake" in presuming that the self-interest of banks and other institutions would be sufficient to protect their shareholders. That willingness to confront error openly, rather than evade it or rationalize it away, was one of the clearest expressions of his intellectual honesty—and one of the qualities I most respected in him.

His Living Legacy

What living legacy does Greenspan leave at the Federal Reserve?

Part of the answer may be found in Kevin Warsh, who was sworn in as Fed chairman just one month ago. At his White House ceremony, Warsh paid explicit tribute: "Chairman Greenspan was the first to tell me and show me what this role demands."

Warsh's instincts resemble Greenspan's in two important respects. First, just as Greenspan resisted premature tightening in the 1990s because he believed information technology was driving a productivity revolution, Warsh has argued that artificial intelligence could prove a significant disinflationary force. In both cases, the judgment rested on perceiving structural change before the data fully confirmed it.

Second, Greenspan's preference for brevity when discussing the outlook for monetary policy appears to have found an echo in Warsh's early approach. In his first press conference, he reduced the Fed's post-meeting statement from 341 words to 132. As my Deutsche Bank colleague Matthew Luzzetti observed, that move reverses a decades-long trend toward ever greater transparency.

Whether Warsh's views on AI prove as well founded as Greenspan's insights about information technology will be closely watched. Inflation today is stickier, geopolitical risks are greater, and the enormous investment required to build AI infrastructure may itself prove inflationary in the near term.

“We Had Fun”

Don Kohn concluded his remembrance by recalling Greenspan summing up their years together with a simple observation: “We had fun, didn’t we, Don?”

Yes, Alan—we had fun.

And that spirit of curiosity, intellectual excitement, and joy in understanding how the world works enriched economic policymaking in the United States and around the globe. It certainly enriched my own life and career, and strengthened my desire to remain active in professional economics. For that, and for much else, I will always be grateful.

Appendix 1

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